

**GREEN  
CLIMATE  
FUND**

**Meeting of the Board**  
26– 28 February 2019  
Songdo, Incheon, Republic of Korea  
Provisional agenda item 34

**GCF/B.22/10/Add.09**

**1 February 2019**

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# Consideration of funding proposals – Addendum IX

## Funding proposal package for FP105

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### **Summary**

This addendum contains the following six parts:

- a) A funding proposal summary titled “BOAD Climate Finance Facility to Scale Up Solar Energy Investments in Francophone West Africa LDCs” submitted by the Banque Ouest Africaine de Développement;
- b) No-objection letters issued by the national designated authority(ies) or focal point(s);
- c) Environmental and social report(s) disclosure;
- d) Independent Technical Advisory Panel’s assessment;
- e) Response from the accredited entity to the independent Technical Advisory Panel’s assessment; and
- f) Gender documentation of the funding proposal.

These documents are presented as submitted by the accredited entity and the national designated authority(ies) or focal point(s), respectively. Pursuant to the Comprehensive Information Disclosure Policy of the Fund, the funding proposal titled “BOAD Climate Finance Facility to Scale Up Solar Energy Investments in Francophone West Africa LDCs” submitted by the Banque Ouest Africaine de Développement is being circulated on a limited distribution basis only to Board Members and Alternate Board Members to ensure confidentiality of certain proprietary, legally privileged or commercially sensitive information of the entity.

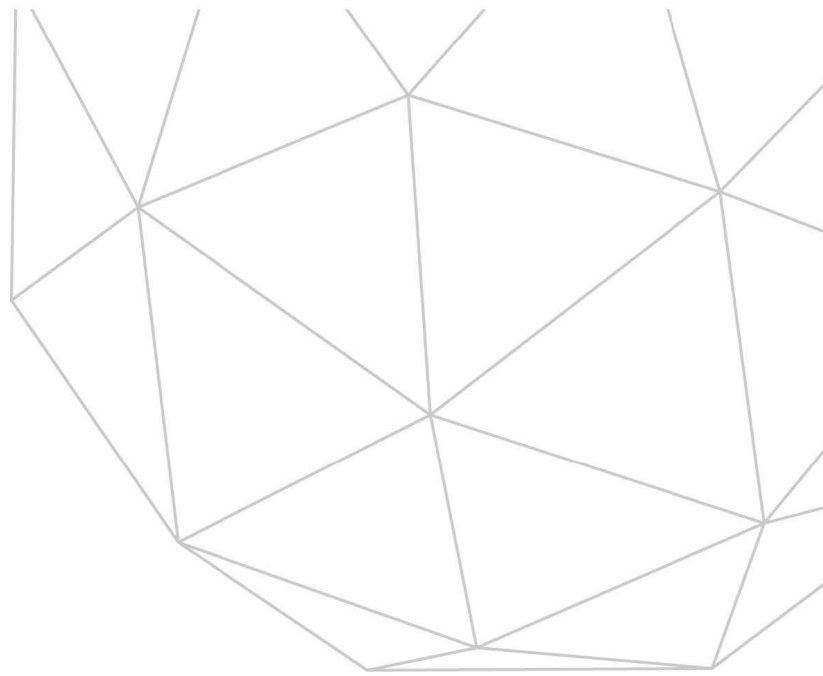
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GREEN  
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FUND



# Funding Proposal

Version 1.1

**The Green Climate Fund (GCF) is seeking high-quality funding proposals.**

Accredited entities are expected to develop their funding proposals, in close consultation with the relevant national designated authority, with due consideration of the GCF's Investment Framework and Results Management Framework. The funding proposals should demonstrate how the proposed projects or programmes will perform against the investment criteria and achieve part or all of the strategic impact results.

Project/Programme Title: BOAD Climate Finance Facility to Scale Up Solar Energy Investments in Francophone West Africa LDCs

Country/Region: Benin, Burkina Faso, Guinea-Bissau, Mali, Niger and Togo

Accredited Entity: West African Development Bank (BOAD)

Date of Submission: 20/07/2018

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### *Note to accredited entities on the use of the funding proposal template*

- Sections **A, B, D, E** and **H** of the funding proposal require detailed inputs from the accredited entity. For all other sections, including the Appraisal Summary in section F, accredited entities have discretion in how they wish to present the information. Accredited entities can either directly incorporate information into this proposal, or provide summary information in the proposal with cross-reference to other project documents such as project appraisal document.
- The total number of pages for the funding proposal (excluding annexes) is expected not to exceed 50.

**Please submit the completed form to:**

[fundingproposal@gcfund.org](mailto:fundingproposal@gcfund.org)

Please use the following name convention for the file name:

"[FP]-[Agency Short Name]-[Date]-[Serial Number]"

## List of Abbreviations

|           |                                                   |
|-----------|---------------------------------------------------|
| ABREC     | African Biofuel and Renewable Energy Company      |
| ACCF      | Africa Climate Change Fund                        |
| AE        | Accredited Entity                                 |
| AECF      | Africa Enterprise Challenge Fund                  |
| AEMP      | Africa Energy Market Place                        |
| AFD       | French Development Agency                         |
| AfDB      | African Development Bank                          |
| AMA       | Accreditation Master Agreement                    |
| AREF      | Africa Renewable Energy Fund                      |
| BAU       | Business as Usual                                 |
| BCEAO     | Central Bank of West African States               |
| BOAD      | West African Development Bank                     |
| CCAC      | Climate & Clean Air Coalition                     |
| CDM       | Clean Development Mechanism                       |
| CEB       | Communauté Electrique du Bénin                    |
| COP       | Conference of the Parties                         |
| CSP       | Concentrated Solar Power                          |
| DAC       | Development Assistance Committee                  |
| DAO       | Tender Dossiers                                   |
| DBSA      | Development Bank of South Africa                  |
| DFI       | Development Finance Institution                   |
| ECOWAS    | Economic Community of West African States         |
| ECREEE    | Centre for Renewable Energy and Energy Efficiency |
| EDM       | Energie du Mali                                   |
| EE        | Energy Efficiency                                 |
| EE        | Executing Entity                                  |
| EF        | Emission Factor                                   |
| EIB       | European Investment Bank                          |
| ElectriFi | Electrification Finance Initiative                |
| EREI      | Renewable Energy Investment Initiative            |
| EREP      | Renewable Energy Policy                           |
| ESAP      | Environmental and Social Action Plan              |
| ESIA      | Environmental and Social Impact Assessment        |
| EU        | European Union                                    |
| EU-AITF   | EU-Africa Infrastructure Trust Fund               |
| FAA       | Funded Activity Agreement                         |
| FDE       | Energy Development Fund                           |
| GCF       | Green Climate Fund                                |
| GDP       | Gross Domestic Product                            |
| GEF       | Global Environment Facility                       |
| GHG       | Greenhouse Gas                                    |
| IFC       | International Finance Corporation                 |
| IMF       | International Monetary Fund                       |
| IPP       | Independent Power                                 |
| IRED      | Regional Initiative for Sustainable Energy        |
| IRENA     | International Renewable Energy Agency             |
| KfW       | German Development Bank                           |

|            |                                                             |
|------------|-------------------------------------------------------------|
| kWh        | kilowatt per hour                                           |
| LDC        | Least Developed Country                                     |
| MAI        | Notice of Expression of Interest                            |
| MDGs       | Millenium Development Goals                                 |
| MRV        | Monitoring, Reporting and Verification                      |
| MW         | MegaWatt                                                    |
| NAMA       | Nationally Appropriate Mitigation Action                    |
| NDA        | National Designated                                         |
| NDC        | Nationally Determined Contributions                         |
| NEAP       | National Environmental Action Plan                          |
| NEMP       | National Environmental Management Plan                      |
| NEPAD-IPPF | New Partnership for Africa Development –                    |
| NFI        | National Financial Institutions                             |
| NGO        | Non-Governmental Organisation                               |
| NREAP      | National Renewable Energy Action Plan                       |
| NREPs      | National Renewable Energy Policy                            |
| NSDP       | National Sustainable Development Policy                     |
| ODA        | Official Development Assistance                             |
| OFID       | OPEC Fund for International Development                     |
| PDA        | Detailed Project Study                                      |
| PIDG       | Private Infrastructure Development Group                    |
| PPA        | Power Purchase Agreement                                    |
| PPIAF      | Public-Private Infrastructure Advisory Facility             |
| PRODERE    | Regional Program for the Development of RE and EE           |
| PV         | PhotoVoltaic                                                |
| RCC        | Regional Collaboration Centre                               |
| RE         | Renewable Energy                                            |
| REF        | Rural Energy Fund                                           |
| SAGSD      | Strategy for Accelerated Growth and Sustainable Development |
| SDGs       | Sustainable Development Goals                               |
| SE4ALL     | Sustainable Energy for All                                  |
| SEFA       | Sustainable Energy Fund for Africa                          |
| SF-SLM     | Strategic Framework for Sustainable Land Management         |
| SOGEM      | Société de Gestion de l’Energie de Manantali                |
| TA         | Technical Assistance                                        |
| TWh        | Terawatt per hour                                           |
| UNFCCC     | United Nations Framework Convention on Climate Change       |
| USADF      | United-States African Development Foundation                |
| VAT        | Value-Added Tax                                             |
| WAEMU      | West African Economic and Monetary Union                    |
| WAPP       | West African Power Pool                                     |
| XOF        | CFA Franc                                                   |

| A.1. Brief Project / Programme Information                   |                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------------------------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A.1.1. Project / programme title                             |                          | BOAD Climate Finance Facility to Scale Up Solar Energy Investments in Francophone West Africa LDCs                                                                                                                                                                                                                                                                                                                                                                                 |
| A.1.2. Project or programme                                  |                          | programme                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| A.1.3. Country (ies) / region                                |                          | Benin, Burkina Faso, Guinea-Bissau, Mali, Niger and Togo                                                                                                                                                                                                                                                                                                                                                                                                                           |
| A.1.4. National designated authority (ies)                   |                          | <p>Benin: Ministry of Living Environment and Sustainable Development</p> <p>Burkina Faso: Prime Minister's office</p> <p>Guinea-Bissau: Secretariat of State of the Environment/General Direction of the Environment</p> <p>Mali: The Environment and Sustainable Development Agency</p> <p>Niger: National Council of the Environment for Sustainable Development</p> <p>Togo: Directorate of Environment, Ministry of Environment and Forest Resources and Forests Resources</p> |
| A.1.5. Accredited entity                                     |                          | <b>West African Development Bank (BOAD)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| A.1.5.a. Access modality                                     |                          | <input checked="" type="checkbox"/> Direct <input type="checkbox"/> International                                                                                                                                                                                                                                                                                                                                                                                                  |
| A.1.6. Executing entity / beneficiary                        |                          | <p>Executing Entity: West African Development Bank (BOAD)</p> <p>Beneficiary: Project developers, primarily from the six beneficiary countries, local public institutions and BOAD</p>                                                                                                                                                                                                                                                                                             |
| A.1.7. Project size category (Total investment, million USD) |                          | <input type="checkbox"/> Micro ( $\leq 10$ ) <input type="checkbox"/> Small ( $10 < x \leq 50$ )<br><input checked="" type="checkbox"/> Medium ( $50 < x \leq 250$ ) <input type="checkbox"/> Large ( $> 250$ )                                                                                                                                                                                                                                                                    |
| A.1.8. Mitigation / adaptation focus                         |                          | <input checked="" type="checkbox"/> Mitigation <input type="checkbox"/> Adaptation <input type="checkbox"/> Cross-cutting                                                                                                                                                                                                                                                                                                                                                          |
| A.1.9. Date of submission                                    |                          | 20 July 2018                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| A.1.10. Project contact details                              | Contact person, position | Yacoubou BIO-SAWÉ, Director for Environment and Climate Finance<br>Ibrahim Traore, Head of Climate Finance Division                                                                                                                                                                                                                                                                                                                                                                |
|                                                              | Organization             | BOAD                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                                              | Email address            | <a href="mailto:itraore@boad.org">itraore@boad.org</a> <a href="mailto:ybiosawe@boad.org">ybiosawe@boad.org</a>                                                                                                                                                                                                                                                                                                                                                                    |
|                                                              | Telephone number         | +228 22 23 26 92                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                                              | Mailing address          | 68 Avenue de la Libération<br>1172 Lomé<br>Togo                                                                                                                                                                                                                                                                                                                                                                                                                                    |

|                                                    |                                                                                                            |
|----------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| A.1.11. Results areas <i>(mark all that apply)</i> |                                                                                                            |
| Reduced emissions from:                            |                                                                                                            |
| <input checked="" type="checkbox"/>                | Energy access and power generation<br>(E.g. on-grid, micro-grid or off-grid solar, wind, geothermal, etc.) |
| <input type="checkbox"/>                           | Low emission transport<br>(E.g. high-speed rail, rapid bus system, etc.)                                   |

- ☐ **Buildings, cities and industries and appliances**  
(E.g. new and retrofitted energy-efficient buildings, energy-efficient equipment for companies and supply chain management, etc.)
- ☐ **Forestry and land use**  
(E.g. forest conservation and management, agroforestry, agricultural irrigation, water treatment and management, etc.)

**Increased resilience of:**

- ☐ **Most vulnerable people and communities**  
(E.g. mitigation of operational risk associated with climate change – diversification of supply sources and supply chain management, relocation of manufacturing facilities and warehouses, etc.)
- ☐ **Health and well-being, and food and water security**  
(E.g. climate-resilient crops, efficient irrigation systems, etc.)
- ☐ **Infrastructure and built environment**  
(E.g. sea walls, resilient road networks, etc.)
- ☐ **Ecosystem and ecosystem services**  
(E.g. ecosystem conservation and management, ecotourism, etc.)



## A.2. Project / Programme Executive Summary (max 300 words)

*Please provide a brief description of the proposed project/programme, including the objectives and primary measurable benefits (see [investment criteria in section E](#)). The detailed description can be elaborated in [section C](#).*

About half of the population in the West Africa Region do not have access to electricity. Countries from this region face interrelated challenges of energy access, energy security and high cost of electricity. This is attributed to the high dependence on fossil fuels, the insufficient availability of public funds to invest in energy infrastructure to meet the growing electricity demand, the inability to attract private capital in power generation at scale, and the inefficiencies along the value chain. This situation hampers the social, economic, and industrial development of the region.

The region's energy mix mostly comprises diesel and heavy fuel oil (76%). Hydropower and solar represent respectively 19% and 5%. Given the continuous decline in the cost of solar electricity generation<sup>1</sup>, the region would benefit from investing massively in solar technologies to reduce its dependence on expensive fossil fuels for power generation, while supporting a shift towards cheaper, cleaner and reliable sources of electricity generation. With large financing needs over the sector's entire value chain, the solar sector increasingly relies on the private sector to deploy grid-connected solar power generation projects. While the region is endowed with high solar irradiation especially in the Sahel<sup>2</sup>, the total installed solar PV capacity in the countries targeted by the proposed programme was only around 70 MW at the end of 2017<sup>3</sup>, which is far below the national plans and NDC targets of 1,192 MW solar energy capacity by 2030.

The rationale of the proposed GCF programme "BOAD Climate Finance Facility to Scale Up Solar Energy Investments in Francophone West Africa LDCs" (hereafter the "programme") is to decarbonise the energy mix of the Francophone West Africa Region and to improve access to clean energy by scaling up cost-competitive solar technologies through private capital leveraging. The programme aims to accelerate private sector investments in the medium/large scale on grid-connected solar sector and to create a market for investments in solar technologies (PV and CSP) in the Francophone West African region. Initially, the programme will focus on the six Least Developed Countries (LDC) of this region, namely Benin, Burkina Faso, Guinea-Bissau, Mali, Niger and Togo.

BOAD will play a catalytic role with a blended finance approach to increase solar investment by providing long term debt financing to qualifying private sector companies for solar projects that are commercially viable but that cannot currently attract affordable financing at scale. The programme will contribute to a paradigm shift to low-emission development pathways through the mobilisation at scale of private climate finance to add 215 MW of new solar energy capacity; equivalent to a 400% increase of the regional solar capacity in the targeted six LDCs.

The total programme cost is EUR 122 million, of which EUR 61 million is GCF financing and EUR 61 million is co-financing from BOAD. The proposed programme envelope is expected to leverage another EUR 125 million from project developers and other lenders (private and public sector banks in the region). The total programme capital cost is expected to result in the reduction or avoidance of approximately 4.8 million tCO<sub>2e</sub> during the lifetime of the programme. This increased investment will help fill the current investment shortfall to achieve West Africa's Nationally Determined Contributions (NDCs).

## A.3. Project/Programme Milestone

Expected approval from accredited entity's Board (if applicable)

25/02/2019

<sup>1</sup> Solar PV module prices dropped by more than 80 percent since 2009 (IRENA, 2018. Renewable Power Generation Costs in 2017).

<sup>2</sup> The average solar radiation is of 5-6 kWh/m<sup>2</sup> per day throughout the year (ECOWAS renewable energy policy, 2013)

<sup>3</sup> IRENA, Renewable Energy Statistics, 2018

|                                             |                                                    |
|---------------------------------------------|----------------------------------------------------|
| Expected financial close (if applicable)    |                                                    |
| Estimated implementation start and end date | Start: <u>01/06/2019</u><br>End: <u>01/06/2026</u> |
| Project/programme lifespan                  | 20 years                                           |

### B.1. Description of Financial Elements of the Project / Programme

The proposed programme envelope aims to provide EUR 117 million of senior debt and standby loans (tenor extension instrument) to selected solar projects in Least Developed Countries (LDCs) from the Francophone West Africa region and EUR 5 million of technical assistance to local stakeholders (from private and public sectors).

The GCF contributions will account for 50% of the total funding of EUR 122m required to deliver the Programme. The GCF financing will include a loan to BOAD for on-lending to selected projects in solar PV and CSP technologies. BOAD will provide 50% co-financing and blend GCF funds with its existing funds.

The programme is designed to maximize total investment, using limited public funds to leverage greater private investment. The total commitment is expected to leverage a substantial amount of financing from project developers and private sector banks.

*Table 1: Financial structure of the proposed programme*

| Components                                                                                                               | GCF<br>Financing<br>(EUR) | BOAD Co-<br>financing<br>(EUR) | Total<br>co-<br>financing<br>amount<br>(EUR) | Private sector leverage                             |                                                          |
|--------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------------|----------------------------------------------|-----------------------------------------------------|----------------------------------------------------------|
|                                                                                                                          |                           |                                |                                              | Equity from<br>Private<br>Sector<br>Actors<br>(EUR) | Loans from<br>Commercial<br>and Public<br>Banks<br>(EUR) |
| Component 1: To scale up commercial and sustainable financing for solar investments - Direct lending (loan component)    |                           |                                |                                              |                                                     |                                                          |
| Component 1.1: Senior Loan Facility                                                                                      | 45 000 000                | 60 000 000                     | 105 000 000                                  | 50 000 000                                          | 78 100 000                                               |
| Component 1.2: Standby Loan Facility<br>(credit enhancement in the form of tenor extension)                              | 12 000 000                |                                | 12 000 000                                   |                                                     |                                                          |
| Component 2: Technical assistance (grant component)                                                                      |                           |                                |                                              |                                                     |                                                          |
| Component 2.1 Technical assistance to<br>BOAD and local project developers                                               | 2 000 000                 | 250 000                        | 2 250 000                                    |                                                     |                                                          |
| Component 2.2 Technical assistance to<br>national institutions to address legal and<br>regulatory barriers in the market | 2 000 000                 | 250 000                        | 2 250 000                                    |                                                     |                                                          |
| Component 2.3 Marketing and Awareness<br>Raising                                                                         |                           | 500 000                        | 500 000                                      |                                                     |                                                          |
| TOTAL (EUR)                                                                                                              | 61 000 000                | 61 000 000                     | 122 000 000                                  | 50 000 000                                          | 78 100 000                                               |

Financial barriers are a significant obstacle to the mobilization and scaling up of climate finance (i.e. inadequate pricing, inflexible grace periods and loan tenors that are not adapted to the characteristics of renewable energy investments). The lack of sufficient private financing and the short tenor (5-7 years from local commercial investors) have been identified as key barriers for renewable energy (RE) project financing in the West Africa region.



|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |               |                 |              |                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----------------|--------------|----------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p>* BOAD's grant will be in-kind contribution.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |               |                 |              |                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p>* Please provide a confirmation letter or a letter of commitment in section I issued by the co-financing institution.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |               |                 |              |                |
| <p>(d) Financial terms between GCF and AE (if applicable)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p><i>In cases where the accredited entity (AE) deploys the GCF financing directly to the recipient, (i.e. the GCF financing passes directly from the GCF to the recipient through the AE) or if the AE is the recipient itself, in the proposed financial instrument and terms as described in part (b), this subsection can be skipped.</i></p> <p><i>If there is a financial arrangement between the GCF and the AE, which entails a financial instrument and/or financial terms separate from the ones described in part (b), please fill out the table below to specify the proposed instrument and terms between the GCF and the AE.</i></p> |               |                 |              |                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Financial instrument</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Amount</b> | <b>Currency</b> | <b>Tenor</b> | <b>Pricing</b> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | .....         | <u>Options</u>  | ( ) years    | ( ) %          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p><i>Please provide a justification for the difference in the financial instrument and/or terms between what is provided by the AE to the recipient and what is requested from the GCF to the AE.</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                         |               |                 |              |                |
| <b>B.3. Financial Markets Overview (if applicable)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |               |                 |              |                |
| <p>BOAD is the development finance institution of the member countries of the West African Economic and Monetary Union (WAEMU) which includes Benin, Burkina-Faso, Côte d'Ivoire, Guinea Bissau, Mali, Niger, Senegal and Togo.</p> <p>The Central Bank of West African States (Banque Centrale des États de l'Afrique de l'Ouest - BCEAO) is the central bank serving the eight west African countries (Benin, Burkina Faso, Ivory Coast, Guinea-Bissau, Mali, Niger, Senegal and Togo) which share the common West African CFA franc currency and comprise the WAEMU.</p> <p>In addition to centralizing the cash reserves of the Union, the main mandates of the Central Bank are: to issue currency, to manage monetary policy, to organize and monitor banking activities and to provide assistance for the WAEMU member States.</p> <p><b>Overview of commercial market interest rates</b></p> <p>Commercial market interest rates in the region are high, and in most cases steady as can be seen in Table 2 below. In terms of average lending rates according to the type of borrower, rates remained unchanged for all types of borrowers in Benin, Burkina Faso, Côte d'Ivoire, Guinea-Bissau, Mali, Senegal and Togo, with the exception being Niger from November to December 2017.</p> <p><i>Table 2: Average lending rates according to the type of borrower (%) – for tenors mainly below 5 years.</i></p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |               |                 |              |                |

|                      | State and similar organizations |        | Individuals |        | State-owned corporations and Pes |        | Private businesses in the production sector |        | Individual businesses |        | Other borrowers |        | All    |        |
|----------------------|---------------------------------|--------|-------------|--------|----------------------------------|--------|---------------------------------------------|--------|-----------------------|--------|-----------------|--------|--------|--------|
|                      | Nov.17                          | Dec.17 | Nov.17      | Dec.17 | Nov.17                           | Dec.17 | Nov.17                                      | Dec.17 | Nov.17                | Dec.17 | Nov.17          | Dec.17 | Nov.17 | Dec.17 |
| <b>Bénin</b>         | 9,80                            | 9,80   | 9,07        | 9,07   | 9,29                             | 9,29   | 8,44                                        | 8,44   | 11,62                 | 11,62  | 5,16            | 5,16   | 8,76   | 8,76   |
| <b>Burkina</b>       |                                 |        | 7,56        | 7,56   | 6,50                             | 6,50   | 7,34                                        | 7,34   | 11,04                 | 11,04  | 6,63            | 6,63   | 7,51   | 7,51   |
| <b>Côte d'Ivoire</b> | 6,02                            | 6,02   | 7,59        | 7,59   | 9,00                             | 9,00   | 7,42                                        | 7,42   | 5,14                  | 5,14   | 5,19            | 5,19   | 6,68   | 6,68   |
| <b>Guinée-Bissau</b> |                                 |        | 9,13        | 9,13   |                                  |        | 11,91                                       | 11,91  |                       |        | 5,66            | 5,66   | 9,70   | 9,70   |
| <b>Mali</b>          | 8,00                            | 8,00   | 9,10        | 9,10   | 10,13                            | 10,13  | 7,49                                        | 7,49   | 10,04                 | 10,04  | 5,59            | 5,59   | 7,65   | 7,65   |
| <b>Niger</b>         | 11,00                           | 10,41  | 10,87       | 10,35  | 8,00                             | 5,50   | 10,21                                       | 8,47   | 10,19                 | 8,64   | 8,25            | 8,81   | 10,30  | 8,57   |
| <b>Sénégal</b>       | 6,10                            | 6,10   | 9,16        | 9,16   | 11,00                            | 11,00  | 5,27                                        | 5,27   | 10,40                 | 10,40  | 7,10            | 7,10   | 5,90   | 5,90   |
| <b>Togo</b>          |                                 |        | 8,64        | 8,64   | 6,58                             | 6,58   | 8,57                                        | 8,57   | 10,14                 | 10,14  | 3,22            | 3,22   | 8,68   | 8,68   |

Source : BCEAO.

Overall lending in West Africa remains difficult for most banks due to particularly challenging conditions and internal vulnerabilities. Credit conditions are reflected in high interest rates from the commercial banks. The region is characterized by inadequate pricing, inflexible grace periods and loan tenors that are not adapted to the characteristics of a solar investment. The local commercial banks are unable to extend loans with tenors of no more than 5-7 years due to the short-term maturity profile of their deposit and the absence of deep capital markets to source long-term financing. The limited borrowing capacity and the lack of available and accessible commercial finance by local financial institutions hinders the ability of businesses to invest in renewable energy, and in particular solar energy. These financial barriers are a significant barrier to the mobilization and scaling up of climate finance.

Given that the current market conditions in the target region are not favorable and that commercial banks are risk adverse, the GCF's participation will be critical to bring private investors onboard. The programme is designed to address these financial barriers by providing affordable and long-term financing to scale up solar energy in the LDC countries of the region:

- To overcome the barrier of high capital cost, external climate finance support is necessary, combined with private financing and public funding. The access to concessional financing from the GCF and BOAD will crowd in private sector capital to develop and implement solar investment. By providing concessional loan to BOAD at an attractive interest rate, the blended cost of borrowing for BOAD will be reduced compared to the cost of borrowing of BOAD on capital markets. Therefore, BOAD will be able to lend to projects at long tenors and more attractive rate than the rate applied without GCF support.
- To overcome the barrier of the length of tenor which is a key limitation encountered by project developers seeking local financing, the GCF will incentivise the participation of local commercial banks in the Programme by providing up to EUR 12 million of Standby Loans to BOAD which will be on-lent to sub-projects for extension of the commercial banks loan tenor. A standby loan facility will be provided when commercial banks cannot extend the tenor beyond the year 5-7, as a tenor extension instrument. The repayment of the outstanding commercial bank loan at the end of its tenor shall either be rescheduled by the commercial lenders/banks or repaid as scheduled by drawing down on the standby loan facility. Standby loan facility will not cover the risk of a commercial bank tranche during their initial tenor. Commercial banks will provide senior debt, backed by the standby loan facility as a tenor extension instrument provided by BOAD. The standby loans will automatically provide a credit enhancement to the underlying sub-project by extending the tenor of the outstanding commercial bank loan at maturity to match the tenor of BOAD's loan to the sub-project.

Please fill out applicable sub-sections and provide additional information if necessary, as these requirements may vary depending on the nature of the project / programme.

### C.1. Strategic Context

The Paris Agreement, the Marrakech agreements (COP 22) and the SE4ALL initiative have all emphasized the urgency to restore balance in the energy mix globally and in the region by scaling-up Renewable Energy (RE) projects, in particular through Public-Private-Partnerships and private investment. The international community as a whole now pushes for private-led green growth, as public and national financial institutions cannot bear the investment costs alone.

In West Africa, there is an urgent need to increase electricity supply and access for the population as almost half of it are without access to electricity. In recent years, countries from West Africa have renewed their commitment to the promotion of clean energy as well as regional cooperation in order to meet their Nationally Determined Contributions (NDCs). In particular the governments from West Africa have demonstrated a growing interest to large-scale, grid-connected solar technology which constitutes the best alternative to fossil fuels considering the region's potential<sup>4</sup>.

The ECOWAS region is made up fifteen member countries that are located in the Western African region: Benin, Burkina Faso, Cabo Verde, Ivory Coast, Gambia, Ghana, Guinea, Guinea Bissau, Liberia, Mali, Niger, Nigeria, Senegal, Sierra Leone and Togo. These countries have both cultural and geopolitical ties and shared common economic interest. The proposed programme focuses on six LDCs of West Africa region, namely: Benin, Burkina Faso, Guinea-Bissau, Mali, Niger and Togo.

The ECOWAS region has developed its RE strategy, which complements the ECOWAS White Paper (2006) to promote "access to energy services for populations in rural and peri-urban areas" and improve "the business environment for a competitive private sector". The EREP (ECOWAS Renewable Energy Policy) strategy (2013) came following the realization that the demographic transition and climate change were significant threats to African economies and aims at overcoming the various market barriers by translating regional pledges into national initiatives and targets. EREP ambitious objective is to achieve nearly 30% of installed RE capacity in the energy mix – totally 7,606MW – by 2030<sup>5</sup>.

Based on the EREP baseline, the ECOWAS Centre for Renewable Energy and Energy Efficiency (ECREEE) has provided national governments with guidelines to develop a RE strategy encompassing legal, technical and financial aspects fitting their national needs and challenges. In addition, national governments were strongly encouraged to take steps towards the alleviation of market barriers for private investment through the simplification of the regulatory framework and additional fiscal incentives<sup>6</sup>.

As indicated in Tables 3 and 4, the regional and national targets of the beneficiary countries are to increase significantly the share RE in the energy mix by 2020 and by 2030. Table 6 highlights the specific targets for solar energy by 2030 and 2030 and is a clear evidence of the strong commitment of the target countries to develop the market for solar energy.

*Table 3: EREP Regional objectives for grid-connected renewable energies (Guiding NREPs in UEMOA/ECOWAS countries)*

| Type of target                                                                       | Expected result                        |
|--------------------------------------------------------------------------------------|----------------------------------------|
| <b>EREP RE options in % of peak load</b>                                             | 2020: 10 %<br>2030: 19 %               |
| <b>RE share in the electricity mix (including medium and large-scale hydro)</b>      | 2020: 35 %<br>2030: 48 %               |
| <b>RE share in the energy mix (excluding large hydro)</b>                            | 2020: 10%<br>2030: 19%                 |
| <b>Added installed RE capacity from wind, solar, bioenergy and small-scale hydro</b> | By 2020: 2,424 MW<br>By 2030: 7,606 MW |

<sup>4</sup> Rapport d'Étape sur les Énergies Renouvelables et l'Efficacité Énergétique de la CEDEAO, 2014, p. 12

<sup>5</sup> ECREEE.org

<sup>6</sup> ECREEE, ECOWAS Renewable Energy Policy (EREP), 2015, p. 47-49



|                                                                     |                                                                                               |
|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Possible contribution of each RE technology to the targets</b>   | Wind: 13%<br>Solar: 28%<br>Hydro: 33%<br>Biomass: 26%                                         |
| <b>Production using RE</b>                                          | 2020: 8.350 GWh<br>2030: 29.229 GWh                                                           |
| <b>RE share of energy demand</b>                                    | 2020: 5 %<br>2030: 12 %                                                                       |
| <b>Total RE production (including medium and large-scale hydro)</b> | 2020: 23 %<br>2030: 31 %                                                                      |
| <b>Target for regional manufacture of RE equipment</b>              | 2020: 7% of installed RE equipment, by value<br>2030: 20% of installed RE equipment, by value |

Table 4: National targets for grid-connected RE in the target countries

| Country                          | Share of RE in total installed capacity (including medium and large-scale hydro) |       | Share of RE in electricity mix (including medium and large-scale hydro) |      |
|----------------------------------|----------------------------------------------------------------------------------|-------|-------------------------------------------------------------------------|------|
|                                  | 2020                                                                             | 2030  | 2020                                                                    | 2030 |
| <b>Benin<sup>7</sup></b>         | 10.5%                                                                            | 18.8% | 64.7%                                                                   | 35%  |
| <b>Burkina Faso<sup>8</sup></b>  | 24%                                                                              | 36%   | 17%                                                                     | 27%  |
| <b>Guinea-Bissau<sup>9</sup></b> | 26%                                                                              | 52%   | 24%                                                                     | 75%  |
| <b>Mali<sup>10</sup></b>         | 59.5%                                                                            | 58.3% | 49.3%                                                                   | 37%  |
| <b>Niger<sup>11</sup></b>        | 27%                                                                              | 28%   | 40%                                                                     | 57%  |
| <b>Togo<sup>12</sup></b>         | 41.9%                                                                            | 45.3% | 17%                                                                     | 20%  |

Table 5: Expected installed RE capacity in ECOWAS region in 2020 and 2030 (in MW and related %, excluding medium and large-scale hydro)<sup>13</sup>

|                          | 2020      | 2030        |
|--------------------------|-----------|-------------|
| <b>Wind</b>              | 318 (18%) | 993 (13%)   |
| <b>Solar PV</b>          | 686 (28%) | 1,156 (15%) |
| <b>Solar CSP</b>         | N/A       | 1000 (13%)  |
| <b>Small scale hydro</b> | 787 (33%) | 2,449 (32%) |
| <b>Biomass</b>           | 634 (26%) | 2,008 (27%) |
| <b>Total</b>             | 2,425     | 7,606 (30%) |

Table 6: Current and planned installed solar capacity in 2020 and 2030 in WAEMU countries (source: 2017 data for total installed solar capacity from IRENA (2018) Renewable Energy Statistics; data for planned solar installed capacity from National Renewable Energy and Energy Efficiency Action Plans)

<sup>7</sup> Plan d'Action National des Énergies Renouvelables du Benin, July 2015

<sup>8</sup> Plan d'Action National des Énergies Renouvelables du Burkina Faso, Juillet 2015

<sup>9</sup> Plan de Ação Nacional no Sector das Energias Renováveis (PANER) para a Guiné-Bissau, October de 2017

<sup>10</sup> Plan d'Action National des Énergies Renouvelables du Mali, Novembre 2015

<sup>11</sup> Plan d'Action National des Énergies Renouvelables du Niger, Mars 2015

<sup>12</sup> Plan d'Action National des Énergies Renouvelables du Togo, Octobre 2015

<sup>13</sup> Grid-connected RE scenario of ECOWAS 2020/2030.

| Country         | Total installed solar capacity (MW) | Planned solar installed capacity 2020 (MW) | Planned solar installed capacity 2030 (MW) |
|-----------------|-------------------------------------|--------------------------------------------|--------------------------------------------|
| Benin           | 8                                   | 50                                         | 227                                        |
| Burkina Faso    | 44                                  | 107                                        | 205                                        |
| Guinea-Bissau   | 0                                   | 15                                         | 15                                         |
| Mali            | 6                                   | 308                                        | 528                                        |
| Niger           | 9                                   | 75                                         | 150                                        |
| Togo            | 3                                   | 45                                         | 67                                         |
| <b>Total MW</b> | <b>70</b>                           | <b>600</b>                                 | <b>1,192</b>                               |

**The total installed solar capacity in the six target countries was 70MW by the end of 2017 and the sum of the national targets by 2030 is 1,192MW.**

#### **Other existing sustainable energy initiatives operating in Africa**

There are a wide range of sustainable energy funds covering African countries. They usually differ in terms of type of support provided, geographical areas supported, type of funding provided and funding level. Types of technical assistance provided vary greatly and can include: dialogue and networking (e.g. SE4All, AEMP); studies such as feasibility, due diligence, mapping, investment report, etc. (e.g. DBSA); policy support and reform (e.g. PPIAF, Energy Africa Campaign); institutional capacity building and training (e.g. ACCF, Rural Energy Fund, Power Africa); technical assistance for project preparation (e.g. DevCo, GEF, Powering Agriculture); or technical assistance for project implementation (e.g. EU-Africa Infrastructure Trust Fund, NEPAD IPPF). An increasing number of funds such as the Rural Energy Fund (REF), Africa Climate Change Fund (ACCF), Africa Energy Market Place (AEMP), SE4All, Energy Africa Campaign and Public-Private Infrastructure Advisory Facility (PPIAF) help governments facilitate private sector investment and achieve targets in their Nationally Determined Contributions (NDCs) by providing technical assistance in the form of policy dialogue, capacity building, networking, policy and reform support as well as knowledge sharing.

The type of funding provided is also diverse and can include grants, repayable grants, equity (direct, intermediated, quasi), senior debt, subordinated debt, mezzanine debt, guarantees, loans (debt, concessional, direct, long-term, short-term, bridging), interest rates subsidies, risk-sharing, risk capital and risk-mitigation products, or low-interest credit. For instance, the Electrification Finance Initiative (ElectriFI) funded by the European Union (EU) offers flexible support options, including technical assistance, junior debt, senior debt, and equity. In terms of size of investment, there is no clear gap in funding across the different ranges (<1 million /1 million -10 million/>10 million) while in terms of type of funding provided, within the different ranges, findings indicate that funding below 1 million USD tends to be in the form of grants, whereas larger investments above 10 million USD tend to be a mixture of equity and debt. For the middle range, a combination of grants, equity and debt is provided. Some funds like ABREC only provides TA for private sector and governments facilitate entry of private investors for renewable energy projects in Africa. The Africa Renewable Energy Fund (AREF), a Sub-Saharan Africa (excl. South Africa) is a private equity fund solely focused on small and medium sized (5-50 MW) independent power projects utilizing solar, wind, biomass, hydro, as well as geothermal and stranded gas technologies. In practice, AREF is more active in small hydro and also in geothermal and biomass. The Sustainable Energy Fund for Africa (SEFA) is another existing initiative. It is a USD 90m multi-donor trust fund administered by the African Development Bank (AfDB) to support small- and medium-scale Renewable Energy (RE) in Africa. SEFA operates under three financing windows: project preparation (grants), equity investments and enabling environment support (grants). SEFA does not invest in projects and only focuses on project development phase. It has also distinguished itself from other funds with regards to its focus on more innovative technologies in small/medium scale projects.

In terms of technology, the largest beneficiary sector is the off-grid solar sector, then come hydro, biomass and wind energy, while more innovative technologies are under covered. Several funds have chosen to focus on off-grid solutions such as the United-States African Development Foundation (USADF), the ResponsAbility Innovative Investment Fund, the Africa Enterprise Challenge Fund (AECF), the Rural Energy Fund (REF), the Energy Africa Campaign and this confirms the growth potential of decentralized energy solutions in Africa. The DfID-led Energy Africa campaign channels funding to accelerate off-grid solar power generation for households. Only a few programmes such as OFID Energy



Poverty Program will explicitly emphasize their willingness to support projects that demonstrate technology or business model innovation. The larger the funding available, the more likely the projects supported will be grid-connected (Power Africa, EU-Africa Infrastructure Trust Fund (EU-AITF), PIDG and subsidiaries, IFC Infraventures). Accordingly, the smaller the funding available, the more likely the projects supported will be off-grid.

Although there are several other similar funds, there are sufficient important differences that the field is not too crowded and the proposed GCF programme does not face strong competition. As explained above, several funds focus on solar off-grid solutions while other funds will tend to look to the more challenging route to market (such as AREF or SEFA). In terms of type of funding provided there are many funds providing TA through grants, and the proposed programme will complement the existing programmes by also providing concessional loans and standby loans to private sector actors. No other similar support has been provided to BOAD. The Bank has received financial supports from the Adaptation Fund in the agriculture sector and from the Global Environment Facility (GEF) but the projects supported were at a smaller scale and at the country level.

## C.2. Project / Programme Objective against Baseline

The current status of the energy system hampers the social, economic and industrial development of the West African region. The countries from West Africa are challenged with threatening factors of energy security such as poor system reliability, limited infrastructure, fuel import dependence, and heavy reliance on fossil fuels, hydropower, and traditional biomass resources. **Fossil fuels account for 76% of the energy mix in the targeted countries, while hydropower and solar energy respectively account for 14% and 5%.** These factors have made the region have the lowest modern energy consumption rates in the world with average electricity consumption of 116<sup>14</sup>kWh/capita compared to the continental and global averages of 529 and 2570 kWh/capita, respectively. In the context of climate change, diversification of energy sources is necessary to transit to the clean energies.

In the RE sector, the institutional, regulatory, legal, and tariff structures and frameworks are weak or insufficiently. So far there are only a few incentives for private capital to invest in the renewable energy (RE) sector in West Africa. Investments in RE power projects have had a predominant share of Official Development Assistance (ODA) funding. The policy environment has not helped improve the situation significantly, but there has been a growing realisation over the last few years that drastic action is required to ensure adequate generating capacity in the region, and to develop renewable energy sources to make them an important share of the region's energy and electricity mix.

Although hydropower has been used throughout the region for many decades, deployment of non-hydro renewables—including wind, solar PV and to a lesser extent CSP, and biomass—is accelerating. Within WAEMU countries, electricity has traditionally been provided through conventional grid systems. As of the end of 2017, there was an estimated of 354 MW of grid-connected renewable installed capacity<sup>15</sup> in the target countries comprising hydropower and solar power. Although the target region is well endowed with RE resources, their share in the countries' energy mix is very little when large hydro power generation is excluded.

Table 7: Breakdown of MW capacity installed by technology and by country (in MW)

| Country                       | Thermal <sup>16</sup> | Hydro      | Solar     | Total generation capacity |
|-------------------------------|-----------------------|------------|-----------|---------------------------|
| Benin                         | 288                   | 1          | 8         | 297                       |
| Burkina Faso                  | 258                   | 32         | 44        | 334                       |
| Guinea-Bissau                 | 12.5 <sup>17</sup>    | 0          | -         | 12.5                      |
| Mali                          | 124                   | 184        | 6         | 314                       |
| Niger                         | 174                   | -          | 9         | 183                       |
| Togo                          | 263                   | 67         | 3         | 333                       |
| <b>Total target countries</b> | <b>1,119.5</b>        | <b>284</b> | <b>70</b> | <b>1,473.5</b>            |

<sup>14</sup> Source: <https://www.se4all-africa.org/seforall-in-africa/country-data/mali/>

<sup>15</sup> IRENA (2018) Renewable Energy Statistics

<sup>16</sup> 2017 data from ECREEE (2018) Regional Off-Grid Solar Market Assessment & Private Sector Support Facility Design, Country Reports, August.

<sup>17</sup> Poor network maintenance and planning result in an irregular electricity supply. Total grid capacity of around 5MW (ECREEE, 2018)

Including 354 MW of RE

### Installed solar capacity

Technologies to convert solar energy into electricity generally fall into one of two categories: photovoltaic (PV) modules that convert light directly into electricity and concentrating solar thermal power (CSP) systems that convert sunlight into heat energy that is later used to drive an engine. Although solar power can be generated at any scale, PV technology is modular and can be scaled for anything from household use to a large network of PV farms, whereas CSP is typically considered viable only as a utility-scale plant. The resource potential for solar PV is generally good and relatively homogenous throughout all of West Africa, except in Mali and northern Niger, where the resource potential is particularly strong and deemed “highly suitable”<sup>18</sup>.

There is a huge technically and economic feasible potential for RE development in the target countries. Market trends indicate a price decrease for RE technologies on the one hand and a price increase for fossil fuel costs on the other hand. Among the RE solutions, PV system is noted to be the best option in many cases. In addition, after originally considering CSP as suitable only in the Sahel region<sup>19</sup>, ECREEE has recently helped implement a utility-scale CSP project in Ghana<sup>20</sup>, in line with the 1,000 MW installed CSP capacity target for the ECOWAS region as a whole by 2030. CSP technology would offer excellent results in zones endowed with high solar potential, as in Mali and northern Niger. Furthermore, CSP technologies can be operated through the day, unlike PV without battery storage, which makes it attractive for large-scale solar production. In favour of its geographical location, solar power – either PV or CSP – appears to be good for increasing access to electricity in West Africa and to improve people’s living conditions.

Recently, several Member States have demonstrated a growing interest in grid-connected and large-scale solar projects. As of 2018, the region’s use of solar PV technology is split between distributed, on-grid and off-grid functions, including for rural electrification and to power community centers, health clinics, and individual homes, as well as street lights. Distributed solar PV generation has often been linked to development-orientated projects implemented by governments or donor partners to supply schools, health centers, or rural communities. In the last couple of years, large-scale solar PV projects have blossomed in Senegal (through the Scaling Solar Initiative), Burkina Faso and Mali<sup>21</sup>. **However, these developments have been slow as the total installed solar capacity in the six target countries was only around 70MW by the end of 2017 while the national targets in these countries by 2030 is 1,192 MW for solar energy.**

Table 8: Total solar installed capacity by country (in MW)<sup>22</sup>

| Country                                                  | Total solar installed capacity |
|----------------------------------------------------------|--------------------------------|
| Benin                                                    | 8                              |
| Burkina Faso                                             | 44                             |
| Guinea-Bissau                                            | 0                              |
| Mali                                                     | 6                              |
| Niger                                                    | 9                              |
| Togo                                                     | 3                              |
| <b>Total solar installed capacity in the target LDCs</b> | <b>70</b>                      |

### Market barriers

<sup>18</sup> IRENA (2016) Investment opportunities in West Africa: Suitability Maps for Grid-connected and Off-grid Solar and Wind projects

<sup>19</sup> ECOWAS (2014) Renewable Energy and Energy Efficiency Status Report, p. 37

<sup>20</sup> <https://www.solarpaces.org/ghana-tap-abundant-solar-thermal-energy-potential/>

<sup>21</sup> <https://www.pv-tech.org/editors-blog/large-scale-solar-blossoms-in-africa>

<sup>22</sup> IRENA (2018) Renewable Energy Statistics

A number of barriers continue to hold back solar investment at scale, despite the dynamism of the sector, the proven performance of solar technologies worldwide and the increased number of national policies supporting solar market development. The barriers to solar private-led development can be found throughout the region and seem to impact similar areas. These barriers impede and delay the development of solar projects and therefore increase the difficulty to achieve national targets in terms of solar generation and grid-connected capacity.

The proposed GCF programme is designed to overcome the challenges facing the target countries by providing financial support to private sector actors as well technical support to private sector actors and national institutions.

The table below describes the main market barriers and how the proposed programme will address them.

*Table 9: Market barrier analysis*

| Type of barrier           | Description of the barrier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Objectives of the proposed programme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Financial barriers</b> | <p>The lack of climate finance for West Africa's private sector is the main barrier and it is due to</p> <ul style="list-style-type: none"> <li>• unattractive risk-return balance because of high perceived risk of solar lending or insufficient risk-taking capacity of local financial institutions,</li> <li>• maturity and liquidity risk (local commercial banks are unable to extend loans with tenors of no more than 5-7 years)</li> <li>• inappropriate financial instruments (interest rate and tenor) to incentivize investments,</li> <li>• lack of financing structures and systems,</li> <li>• limited local capital which contributes to bankers' conservatism.</li> </ul> <p>Together, these translate into inadequate sources of necessary concessional long-term debt finance.</p> | <p>The specific objectives of the Programme will address financial barriers by scaling up long-term and affordable climate finance and by providing local financial institutions with a liquidity guarantee for extension of loan tenor that will encourage them to invest in the solar sectors.</p>                                                                                                                                                                                                                                                                                                                                  |
| <b>Technical barrier</b>  | <p>Technical barriers are related to the firm size, the ownership structure, the insufficient experience of private sector actors in originating commercially viable RE projects; the lack of technical knowhow among private sector actors to assess the financial and technical feasibility of solar investment; lack of awareness among private sector actors about the benefits of solar projects on energy costs saving opportunities and on the positive impacts that sustainable energy production and consumption have on energy security, environmental benefits, health and livelihoods in general.</p>                                                                                                                                                                                      | <p>The specific objectives of the Programme will address the technical barriers</p> <ul style="list-style-type: none"> <li>• by supporting private sector actors in the development phase of their project. Effective project preparation is key to attract and mobilize private sector capital. Given the risks and uncertainties of project preparation, the private sector has been hesitant to get involved during the early stages of a project.</li> <li>• by disseminating technical know-how and good practices and awareness raising activities among private sector actors about the benefits of solar projects.</li> </ul> |
| <b>Legal barriers</b>     | <p>The creation of an enabling environment is of crucial importance to ensure the attractiveness of the solar energy sector to prospective private sector investors. Most UEMOA countries have implemented some form of regulatory framework</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <p>The specific objectives of the Programme will address policy and regulatory barriers by participating in institutional debates and carrying out policy dialogue activities that will be informed by the activities of the proposed programme to support the creation of simplified</p>                                                                                                                                                                                                                                                                                                                                             |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                           |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>and financing mechanisms, though this does not constitute a guarantee of result.</p> <p>Legal barriers are related to the lack of legal structures and systems to push the market and open up the sector; the lack of legal and binding targets which is often the hindrance for progress in the RE sector; the lack of planning, policy and regulation capacity in energy planning and modelling integration of solar into energy planning; and lack of incentives mechanism.</p> | <p>regulatory frameworks for Feed-in-Tariff schemes for on-grid solar solutions, support on standard PPAs development, support on IPP tenders and procurement process, on tariff structure, support on the national regulatory and legal frameworks to drive investment in low emission technologies.</p> |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### C.3. Project / Programme Description

The goal of the proposed programme is to create a market for investments in solar technologies in the six LDCs of West Africa, namely Benin, Burkina Faso, Guinea-Bissau, Mali, Niger and Togo. The programme aims to incentivize and leverage private sector actors based in these countries to scale up private investment in solar technologies. The programme is fully aligned with regional and national efforts to drive low-carbon development in West Africa.

While the region has a huge solar potential capacity, market development has been slow because of the unattractive business environment. Even where the policy and regulatory framework are in principle supportive of the goals of increasing RE investment, the magnitude of the barriers to doing business effectively and in a timely manner has meant that progress has been slower than expected. The proposed programme is aimed at removing or reducing the financial, technical and legal barriers faced by the private sector of LDC countries in West Africa looking to invest in the solar sector. The programme is expected to increase by 400% the existing solar capacity in the target countries and this is expected to benefit to around 2.9m people<sup>23</sup>.

By removing the multiple barriers that are preventing the development of the solar market in West Africa, the programme will facilitate project funding in the target countries and will develop the capacity of local investors to deploy solar technologies widely. Without addressing these barriers, the market for solar investments is likely to remain limited in these countries and private sector actors will not be willing to take the risk to invest.

#### Structure of the proposed programme

The Programme will set up a new direct lending facility for the purpose of supporting private sector actors to make viable solar investments. The programme will enable BOAD to offer tailor-made debt financing with credit enhancement in the form of tenor extension to private sector investors alongside technical assistance for local project developers and public institutions. The total programme cost is EUR 122 million, of which EUR 61 million is GCF financing and EUR 61 million are co-financing from BOAD. The GCF finance includes EUR 4 million in grants, EUR 57 million in senior loans and standby loans to BOAD. The BOAD finance includes EUR 1 million in grants and EUR 60 million in senior loans.

BOAD will offer an integrated approach that is based on a combination of financial and technical support to facilitate access to finance to private sector actors and assist them in structuring their solar investments. It is expected that the Facility will support at least 6 projects in the six target LDCs. The programme will have two main objectives and interrelated components:

1. To scale up commercial and sustainable financing for solar investments through senior debt and standby loans
2. To develop technical capacity of private and public sector actors and raise awareness about the benefits of solar technologies

#### Component 1: To scale up commercial and sustainable financing for solar investments - Direct lending (loan component)

##### Sub-component 1.1: Senior loans facility

Financial barriers are a significant barrier to the mobilization and scaling up of climate finance i.e. inadequate pricing, inflexible grace periods and loan tenors that are not adapted to the characteristics of renewable energy investments.

<sup>23</sup> Assuming an electricity consumption per capita of 116 kWh/ person-year (SE4ALL)

The lack of available and accessible commercial finance by local banks hinders the ability of businesses to invest in solar projects. The Programme will address financial barriers to private sector investment specific to each country's market context and contribute to increasing market confidence, reducing risk perception and attracting private investors into the target market segment.

The programme will address financial barriers by providing affordable and long-term debt financing for medium to large-scale solar investments. BOAD will provide long term affordable debt to qualifying private sector companies for solar projects that are commercially viable but that cannot currently attract affordable financing at scale. The GCF funding will be on concessional terms and will be blended with BOAD own funding to cover barriers and risks, and thereby catalyse private sector investment in the target region at scale. The programme will provide appropriate financial resources to selected private sector companies in order to reduce the barriers and risks and thus make these investments attractive to commercial project developers. The outcome will be an increased investment portfolio of medium to large-scale solar projects leading to an increased capacity of 215MW as indicated in Table 10.

The financial arrangements between BOAD and the project developers will be structured in a way that it will not crowd out other private sector investors. In order to catalyse additional private finance, the blending of GCF financing will be tailored to make a project viable. BOAD will ensure that concessionality levels are as low as possible so as to produce the desired market effect while minimising undesirable market distortion with highly concessional loans. GCF's concessionality will be passed down to projects.

The GCF will provide senior loans to BOAD for on-lending to solar projects. BOAD will provide its own commercial financing priced at market terms based on risk factors on a case-by-case basis using clear and established evaluation criteria. The projects funded under this programme will have to comply with a set of technical and financial criteria which are described hereafter. In terms of technical criteria, eligible projects should be located in the six LDCs from West African region and target solar grid connected power generation projects with a capacity of approximately 10 MW to 70 MW. Eligible technologies include: solar photovoltaic (PV) and Concentrated Solar Power (CSP).

In terms of financial criteria:

- Sub-Projects will be required to apply cost-benefit analysis techniques that adequately evaluate all (positive and negative) externalities to ensure holistic cost-effectiveness and the highest possible climate and social return;
- Sub-Projects shall be financially structured such that the present value of cash inflows is greater than the present value of cash outflows – both discounted at the weighted average cost of capital;
- Sub-Projects shall generate an adequate risk adjusted rate of return by identifying and assessing relevant project risks to attract commercial investment;
- BOAD will appropriately size the blending of GCF funds in the financing of Sub-Projects to make them bankable with limited concessionality to minimise the risk of market distortion with highly concessional loans;
- Sub-Projects shall provide clarity on the ultimate source of revenue that would cover operating costs, in order to mitigate risks and ensure financial viability. All Sub-Projects must be backed by power purchase agreements (PPA) covering the tenor of the loans to be provided by BOAD;
- Sub-Project's developer will be required to contribute to the cost of the Sub-Project as equity participation;
- Sub-Project's developers will have to estimate the expected cost of the investment per lifetime impact potential in tonnes of CO<sub>2</sub>eq reduced.

BOAD has identified a number of projects in its pipeline which are listed below. The total expected MW to be installed is 215 MW.

*Table 10: Indicative pipeline of projects to be funded under the programme*

|   | Country      | Project                                                   |
|---|--------------|-----------------------------------------------------------|
| 1 | Burkina Faso | Construction of a solar PV power plant of 10MW in Banfora |



|    |               |                                                                                                          |
|----|---------------|----------------------------------------------------------------------------------------------------------|
| 2  | Burkina Faso  | Construction of a solar PV power plant of 25MW in the North Western part of the country in Ouahigouya    |
| 3  | Burkina Faso  | Construction of a solar PV power plant of 50 MW in the South East of the country at Tenkedogo            |
| 4  | Niger         | Construction of a solar PV power plant of 20 MW                                                          |
| 5  | Mali          | Construction project for a Solar PV power plant of 50MW in Koulikoro                                     |
| 6  | Mali          | Construction of a solar PV power plant of 20 MW                                                          |
| 7  | Togo          | Construction of a solar PV power plant of 20 MW in Blitta                                                |
| 8  | Togo          | Construction of a solar PV power plant of 30 MW in Dapaong                                               |
| 9  | Guinea Bissau | Construction of a solar PV power plant of 10MW in Bissau                                                 |
| 10 | Benin         | Improvement of the energy in Cotonou City of Cotonou with the construction of 10 MW solar PV power plant |

Note that the total capacity to be installed based on this list of projects is 245MW. However, the conservative estimate of 215MW has been considered for the calculation of the GHG emission reductions.

### **BOAD Solar Project Example**

BOAD has recently participated in the financing of the largest private PV power plant in West Africa located in Kita, Mali. In January 2018, BOAD, the Emerging Africa Infrastructure Fund (EAIF), National Agricultural Development Bank of Mali (BNDA), Green Africa Power (GAP) and GuarantCo have secured a €77 million credit facility to finance the construction and operation of a 50MW PV power plant<sup>24</sup>. The financing will be used to develop the project, which will be built under a Build-Own-Operate and Transfer (BOOT) contract by local company Akuo Kita Solar SA, a subsidiary of French IPP Akuo Energy<sup>25</sup>. Funding for the project was co-arranged by both BOAD and EAIF, with the 50MW installation standing as the first PV plant owned and operated by an independent producer in Mali.

Please see five concepts of Sub-projects in the pipeline in the supporting documents for the Funding Proposal.

### **Sub-component 1.2: Standby loan facility (credit enhancement in the form of tenor extension)**

One of the biggest challenges for commercial banks in the region is the maturity and liquidity risk. The length of tenor is a key limitation encountered by project developers seeking local financing in the RE sector. The local commercial banks are unable to extend loans with tenors of no more than 5-7 years due to the short-term maturity profile of their deposit and the absence of deep capital markets to source long-term financing. To overcome this barrier, the GCF can incentivise the participation of local commercial banks in the Programme by providing up to EUR 12 million of Standby Loans to BOAD which will be on-lent to sub-projects for extension of the commercial banks loan tenor to reasonably match the project's maturity.

For example, assuming the project developer is seeking a 14-year loan, but local banks can only provide 7-year money. The commercial bank loan can be structured as a 7-year loan partly amortized over the tenor with a bullet payment at maturity. If prior to the final maturity date of the local commercial bank loans, such lenders are unable to sufficiently extend the tenor of the outstanding bullet payment by another 5-7 years considering the cash flow of the sub-project; GCF shall guarantee the repayment of the final bullet instalment due from the sub-project to the relevant commercial

<sup>24</sup> <https://www.boad.org/en/boad-participates-in-the-financing-of-the-largest-private-photovoltaic-power-plant-in-west-africa-located-in-kita-mali/>

<sup>25</sup> <https://www.pv-tech.org/news/akuo-solar-secures-financing-for-50mw-pv-project-in-mali>

banks by providing a standby loan facility ("Standby Loans") to BOAD which shall be on-lent to the sub-borrower (sub-project) to repay the scheduled final bullet instalment only.

The standby loans will automatically provide a credit enhancement to the underlying sub-project by extending the tenor of the outstanding commercial bank loan at maturity to match the tenor of BOAD's loan to the sub-project. Note that the standby loan facility will apply only when commercial banks cannot extend the tenor at maturity.

**Component 2: To develop technical capacity of private and public sector actors and raise awareness - Technical assistance (Grant Component)**

Component 2 will provide capacity building to private sector companies while supporting the enabling environment through marketing and awareness raising activities and TA to national institutions in order to stimulate the demand for solar projects.

**Sub-component 2.1: Technical assistance to BOAD and local project developers for investment preparation**

Local project developers do not always have sufficient skills to prepare feasibility studies and offer attractive and bankable projects. Development cost is a key barrier to project developers. The programme will address the technical barriers faced by local project developers by building the capacity of mid-size project developers to structure their investments (project preparation and management).

This will include capacity building and training for project developers to build their capacities during the development phase of their projects. For this sub-component, BOAD will hire qualified long-term international and local consultants for the implementation of the TA activities, based on its procurement policies as approved by the GCF. Training will target local developers active in target countries and include support:

- to originate commercially viable solar related projects;
- to develop technical and financial feasibility studies to develop bankable solar projects;
- to develop a financial & commercial framework and risk mitigation strategy for the solar energy projects;
- to develop environmental and Social Impact Assessments (ESIA) and environmental and social action plan (ESAP); and
- to prepare to reach financial closure for the Sub-Projects

It is expected that approx. 300 people from the private sector will be trained. The transfer of knowledge will be ensured through the provision of proper operational manuals and guidelines and practical tools.

TA to local project developers will be complemented by capacity building to BOAD staff during the development and implementation of the project portfolio. BOAD experience in solar project investment and in mitigation and adaptation projects in general is relatively new. There is a significant need to strengthen BOAD staff expertise in several areas. TA will be provided to remove the technical barriers faced by BOAD staff. TA will include:

- Support for the development of eligible and bankable solar projects: This will include support to build BOAD pipeline of projects, to assess the eligibility sponsors' projects, to screen project financial and technical feasibility studies, to review and improve projects' business models carried by project developers, etc.
- support in the implementation of the solar projects: This will include support to track progress of projects implementation and in the monitoring of project performance indicators (e.g. MW installed capacity, tCO<sub>2</sub>eq avoided, etc.); in the management of environmental and social risks and in the implementation of the gender action plan; support for marketing and communication activities (defining a marketing plan and providing communication tools to promote the programme with the aim to stimulate the demand for solar projects)
- support for the integration of climate change considerations into BOAD project cycle: This will include support to enhance BOAD capacities on how to integrate mitigation and adaptation dimensions into the project activities. Training participants will learn how climate change is interlinked with development cooperation, where to find relevant climate information and how to use it, how to plan and support processes of mainstreaming mitigation and adaptation in BOAD project cycle and how climate change measures should be integrated in development planning.

The recruitment of long term local and international consultants will allow for a 'hands-on' approach to accompanying BOAD staff through project identification, development, implementation, management and monitoring. This 'hands-on' approach will help to ensure the long-term sustainability of BOAD interventions in terms of impacts achieved.

Technical trainings will be provided to around 50 operational permanent staff from BOAD (including financial analysts, environmentalists, engineers, economists, etc.). The aim will be to strengthen their operational skills in terms of formulation and development of mitigation and adaptation projects and on how to integrate the climate dimensions in BOAD operations (in the private and public sectors). The knowledge acquired through the trainings will be captured and institutionalised in proper operational manuals and practical tools such as financial model guidelines or GHG emission calculation tool.

### **Sub-component 2.2: Technical assistance (TA) to national institutions to address legal and regulatory barriers in the solar market**

The small installed capacity of solar PV in the Region reveals important regulatory, structural, and technical constraints such as inadequate Independent Power Producer (IPP) regulatory framework, limited planning and dispatch capacity, and overall grid integration challenges. The programme will also support the enabling environment for solar investment through BOAD's policy dialogue activities with governments in the region to improve the regulatory environment for solar investments. The outcome will be an increased pipeline of fundable solar projects.

BOAD will carry out policy dialogue activities that will be informed by the activities of the proposed programme. BOAD will hire qualified long-term international and local consultants for the implementation of the policy dialogue activities, based on its procurement policies as approved by the GCF. The rationale of this sub-component is to remove the market barriers to the development of solar projects by supporting the implementation of regulatory and policy regimes that provides clear and predictable rules for solar project development in the region. The aim of this sub-component is to prepare the way for the entry of solar private investment and to create clarity and a stable reliable environment for solar investment. TA under this sub-component will be focused on aspects of the enabling environment that have an immediate impact on stimulating solar private sector investment. Support can include

- Support to the creation of a simplified regulatory framework for feed-in-tariffs for on-grid solar solutions;
- Support on the development of standard power purchase agreements (PPAs);
- Support on Independent Power Producer (IPP) tenders and procurement process;
- Support on tariff structure; and
- Support on a framework of financial and technical incentives.

### **Sub-component 2.3: Marketing and awareness raising activities on the benefits of solar system technologies, and broad dissemination of experiences and good practices:**

BOAD will work to ensure that relevant information relating to the deployment of solar technologies is publicised to demonstrate the viability of these technologies. This will be done mainly through awareness raising campaign, BOAD website and business networks.

BOAD will develop, manage and maintain organizational brand and execute a communications and outreach effort that promotes the objectives of the programme; lead direct outreach to project developers, commercial banks, DFI's and other relevant stakeholders; build the reputation and brand of the programme with donors, foundations, corporations, the media and public officials and agencies; establish strategic partnerships with lenders, contractors, developers, government agencies, utilities, business and industry associations, and community groups.

### **Programme benefits**

In creating favourable market environments for investment in the solar sector, the programme is expected to increase by 400% the existing total solar installed capacity which will lead to sizable greenhouse gas (GHG) emissions reductions (4.8 million tCO<sub>2e</sub> over the 25-year average lifetime of the investments) and to green job creation and private sector business growth. Better access to green finance will enable private sector actors to have access to new markets related to green growth. Potential co-benefits include increased access to more sustainable energy, more affordable energy and reduced costs. The programme will also directly contribute to achievement of countries Nationally Determined Contributions (NDCs) and the Sustainable Development Goals (SDGs).



#### C.4. Background Information on Project / Programme Sponsor (Executing Entity)

The West African Development Bank (BOAD) is the common development finance institution of the member countries of the West African Economic and Monetary Union (WAEMU), including Benin, Burkina-Faso, Côte d'Ivoire, Guinea Bissau, Mali, Niger, Senegal and Togo. It was established by an Agreement signed on 14 November 1973, and became operational in 1976. By Treaty of the West African Economic and Monetary Union (WAEMU) signed on 10 January 1994 and entered into force on 1st August 1994, BOAD is a specialized and autonomous institution. It contributes "in full independence to the attainment of the objectives of the WAEMU without prejudice to the objectives assigned to it under the WAMU Treaty". BOAD is an international public institution whose purpose, as provided under Article 2 of its Articles of Association, is to promote the balanced development of its member countries and foster economic integration within West Africa by financing priority development projects.

BOAD delivers on its mandate by contributing towards the mobilization of domestic resources in its member state countries, outsourcing foreign capital through loans as well as providing funding through equity investments, loans, guarantees, and interest rebates. BOAD uses the financial resources that it mobilizes to invest in public and private sector projects and programmes aimed at building basic and modern infrastructure, improving rural livelihoods, generating energy, and climate change adaptation and mitigation.

##### **BOAD and the private sector**

Convinced that the private sector is a real development vector and a key source of economic growth, wealth and job creation in the WAEMU countries, BOAD continuously adjusts, intensifies and diversifies its action towards the promotion and financing of private productive investments. BOAD has a number of strategies for the promotion and financing of the private sector. Since 2013, BOAD has organized capacity building activities and outreaches with IFI such as the World Bank, AfDB and the UNFCCC secretariat to bring more awareness to the West African NFIs and commercial bankers in the financing of energy transition projects. In June 2017 BOAD launched the creation of the West African network for climate finance. BOAD is also promoting the use of green bonds and green credit line mobilization throughout the region. Out of FCFA 4,202.1 billion of total net commitments as at 30 September 2016, BOAD devoted to the private sector about 24.3%, representing FCFA 1,022.4 billion for 319 projects.

BOAD's private sector financing operations and the promotion of financial inclusion are carried out through the "Private Sector" window. The Private Sector Window prioritizes the fight against poverty and support for inclusive growth in the member states. The portfolio of the Bank's shareholding capital of companies and financial institutions includes 82 transactions. BOAD has set a number of initiatives as explained below:

**The Regional Initiative for Sustainable Energy (IREN):** In order to better overcome recurrent energy crises in many of the WAEMU member countries, the Conference of Heads of State and Government decided in January 2008 to set up a committee tasked with proposing a strategy called "Regional Initiative for Sustainable Energy (IREN)". The initiative aims to provide, by 2030, 100% access to electricity in the WAEMU region at lower cost and as part of a large integrated and harmonized power pool market in West Africa. This regional market will be based on a dynamic public-private partnership. It also targets an increase by 82% of renewable and sustainable energies in the WAEMU's power generation base. The financial and institutional mechanism for the implementation of the initiative includes a concessional Energy Development Fund (FDE), and an investment fund operating as a private infrastructure development facility.

**The Energy Development Fund (Fonds de Développement Energie - FDE):** Established on 25 September 2009, the Fund aims to contribute to improved access to energy in the WAEMU member countries and promote the development of sustainable energy under concessional conditions in order to finance projects eligible for IREN initiative. It is managed by BOAD. The Fund started its operations in the first quarter of 2010. As of the July 2018, 18 projects and 1 programme were financed through loans accounting for a total amount of XOF 239,205 billion (EUR 364 million). The status of implementation of FDE financed operations is summed up in the table below.

Table 11: List of projects financed by the FDE<sup>26</sup>

|    | Country       | Name of project                                                                                                       | Date of approval | Technology                  | Status                                                                                                  | Amount committed (million Euro) |
|----|---------------|-----------------------------------------------------------------------------------------------------------------------|------------------|-----------------------------|---------------------------------------------------------------------------------------------------------|---------------------------------|
| 1  | Benin         | Construction project of 161 kV Bembèrèkè-Kandi-Malanville line                                                        | January, 2011    | Network construction        | Works ongoing with expected due date October/December 2018                                              | 30.5                            |
| 2  | Benin         | Strengthening of SBEE distribution networks                                                                           | N/A              | Network improvement         | N/A                                                                                                     | N/A                             |
| 3  | Benin         | Construction of 120 MW thermal plant in Mari Gléta                                                                    | N/A              | Diesel and HFO              | N/A                                                                                                     | N/A                             |
| 4  | Burkina Faso  | Project to strengthen the production capacity of the Komsilga power station                                           | September, 2010  | Diesel and HFO              | Works completed. Power station in operation since June 2014                                             | 30.5                            |
| 5  | Burkina Faso  | Strengthening/reinforcement of Burkina Faso National Interconnected Network (PR-RNI) Project                          | May, 2017        | General network improvement | Works have not started. BOAD had meeting in April 2018 to support start of project implementation phase | 14.5                            |
| 6  | Ivory Coast   | Project for implementation of the emergency phase of reinforcement program to strengthen the electricity sector       | January 2011     | Network improvement         | Works ongoing. Completion rate 95%                                                                      | 38.1                            |
| 7  | Ivory Coast   | Project for the reinforcement and rehabilitation of electric power transmission and distribution works of CI-ENERGIES | June, 2013       | Network improvement         | Works completed. Operational since December 2015                                                        | 19.8                            |
| 8  | Ivory Coast   | Project of extension in combined cycle of production capacity of AZITO thermal plant                                  | N/A              | Diesel and HFO              | N/A                                                                                                     | N/A                             |
| 9  | Ivory Coast   | Project to extend the capacity of the CIPREL SA thermal plant                                                         | N/A              | Diesel and HFO              | N/A                                                                                                     | N/A                             |
| 10 | Guinea-Bissau | Project to strengthen electricity production through the construction of a 15MW thermal power plant                   | June, 2011       | Diesel and HFO              | Works are delayed ESIA/ESAP complete                                                                    | 20.8                            |
| 11 | Mali          | Construction project of the Center National Code of Conduct (CNC) of Bamako and doubling of production                | January, 2011    | Hydro                       | Works have started                                                                                      | 30.5                            |

<sup>26</sup> Etat de mise en œuvre des projets financés par le Fonds de Développement Energie, 2018

|              |         |                                                                                                                                                             |             |                                          |                                  |              |
|--------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------------------------------|----------------------------------|--------------|
|              |         | capacity of Sotouba Hydroelectric plant                                                                                                                     |             |                                          |                                  |              |
| 12           | Mali    | Rehabilitation of hydroelectric power plants of Sélingué and Sotouba                                                                                        | N/A         | Hydro                                    | N/A                              | N/A          |
| 13           | Niger   | Strengthening of production capacity of Goudel plant                                                                                                        | April, 2012 | Diesel and HFO                           | Works are being finalised        | 30.5         |
| 14           | Niger   | Construction of 100 MW thermal plant in Gorou Banda/Niamey: first 30 MW                                                                                     | N/A         | Diesel and HFO                           | Works are being finalised        | 19.8         |
| 15           | Senegal | Construction project to build the 225 kV loop of Senelec Phase I                                                                                            | June, 2011  | Network construction                     | Works completed in November 2014 | 15.2         |
| 16           | Senegal | Construction project to build the 225 kV loop of Senelec Phase II                                                                                           | March, 2015 | Network construction                     |                                  | 38.1         |
| 17           | Togo    | Project to strengthen and construction of electric power supplies system                                                                                    | June, 2011  | Network improvement                      | Works ongoing                    | 8.8          |
| 18           | Togo    | Decentralized of rural electrification of 62 localities by SPV in the 5 regions in substitution for the reinforcement and construction of the power supply. | Sept 2016   | Solar PV                                 | Loan agreement not signed        | 21.6         |
| 19           | UEMOA   | Regional Program for the Development of Renewable Energies and Energy Efficiency (PRODERE) - Part 2 of Phase 1                                              | March 2016  | Renewable energy – solar PV among others | Ongoing                          | 22.8         |
| <b>Total</b> |         |                                                                                                                                                             |             |                                          |                                  | <b>341.5</b> |

**The Infrastructure Development Fund:** It is dedicated to private investments. It is a profit-making fund meant for financing access to electricity in the WAEMU region according to market terms and conditions. In addition, it aims to support the restructuring of electricity companies, the emergency programme and the medium and long-term projects. It will be a lever for public-private partnerships in the energy sector in the WAEMU region. The due diligence towards its implementation is underway at the level of BOAD. The Bank participated in the implementation of major projects including:

- Mali-Côte d'Ivoire electric power interconnection of by Energie du Mali (EDM);
- Electrification of the townships of Bakel, Selibaly and Goureye in Senegal by Société de Gestion de l'Energie de Manantali (SOGEM);
- Expansion of CIPREL SA's power plant capacities in Côte d'Ivoire (CIPREL IV A);

Expansion in combined cycle of the generation capacities of AZITO power plant by SOCIETE AZITO ENERGIE SA in Côte d'Ivoire.

#### BOAD and climate change

Like most of other African countries, the WAEMU countries are increasingly facing the adverse effects of climate change (floods, prolonged droughts, strong winds, coastal erosion, etc.). These phenomena have, over the last years, exacerbated vulnerability of both populations and natural ecosystems useful to life. BOAD shares concerns raised by climate change. Thus, it finances climate-related projects, a true lever for economic and social development of WAEMU member countries. Thus, it mapped out an environmental and climate strategy for the 2015-2019 period, whose aim is

“making environment a new growth area for the Bank”. To achieve this goal, BOAD’s action is structured around three specific directions or strategic areas, namely:

- strengthening the capacities of operational units in the area of environmental assessments;
- developing effective tools and mechanisms for financing and resource mobilization in order to support environmental management and combat climate change;
- strengthening the Bank’s leadership in the sub-region on environmental and climate issues.

In conformity with its statutory missions and as part of the implementation of directions included in the 2015-2019 strategic plan, BOAD has committed itself to providing support to member countries in order to make them resilient to climate shocks and help them achieve sustainable and inclusive growth. In this regard, the Bank promotes climate change mitigation and adaptation projects in various sectors: agriculture, fisheries, handicraft, energy, natural resource rehabilitation and preservation, agroforestry. At last, it promotes resource mobilization by forging ad hoc partnerships with financial institutions such as KfW, AFD, EIB, etc., as well as with the UNFCCC financing mechanisms. BOAD seeks to raise, at international level, funds dedicated to environment and climate, such as the Climate Change Adaptation Fund, the Global Environmental Facility, and the GCF.

Since 2009, the Bank sits as an observer to the UNFCCC and actively participates in discussions relating to the establishment of an international architecture for climate finance. In 2011, the Bank was accredited as the first regional implementing entity of the Adaptation Fund. In addition, since January 2013, BOAD has hosted the first Regional Collaboration Centre (RCC) on Clean Development Mechanism (CDM) and climate change with the UNFCCC. The aim of the RCC is to provide direct support to governments, NGOs and the private sector to identify and develop CDM projects and mitigation activities such as NAMA while supporting BOAD climate finance resource mobilization. This centre has already provided direct support to projects under execution in areas like energy, improved cook stoves, efficient power lighting, solar water heating, etc.

Since January 2015, BOAD has set up at internal level a new department in charge of environment and climate finance to closely look at climate finance aspect in projects that involve environment. In addition to the environmental component, this new department aims to prepare and finance adaptation and mitigation projects, as well as carbon finance.

## BOAD Financial Position

BOAD is a traditional regional multilateral bank that serves the West African Economic and Monetary Union (WAEMU). It provides loans to sovereigns of the WAEMU primarily, as well as to its broader public sector and private sector.

Some key facts are given below:

|                      |                        |
|----------------------|------------------------|
| Number of operations | 1138                   |
| Total commitments    | 5 293,5 billion FCFA   |
| Total assets         | 2 667 870 Million FCFA |
| Total loans          | 1 806 283 Million FCFA |
| Total equity         | 677 905 Million FCFA   |

Since 2015, BOAD is rated « investment grade » by Moody’s (Baa1) and Fitch (BBB), reflecting the solid financial position of BOAD which has one of the highest financial institution credit ratings in the region. Moody’s Investors Service has affirmed the BOAD Baa1 long-term issuer and senior unsecured ratings and maintained the stable outlook.

## C.5. Market Overview (if applicable)

### Electricity access in the target countries

The energy systems within the targeted countries are facing interrelated challenges of energy poverty, energy security and climate change. The region has one of the lowest access rates to modern energy services in the world. Although much progress has been achieved towards increasing electricity access in the west African countries, particularly in Senegal and Côte d’Ivoire, electricity access remains insufficient in most countries. Furthermore, electricity networks are outdated and prone to significant transmission losses and interruptions in supply. The resultant high tariffs and loss of productivity are costly to the households of the target countries and to the regional economy as a whole.

Table 12: Access to electricity rates in target countries (in % of total, rural or urban population)

| Country       | Access to electricity<br>(% of population) | Access to electricity,<br>rural (% of rural<br>population) | Access to electricity,<br>urban (% of urban<br>population) |
|---------------|--------------------------------------------|------------------------------------------------------------|------------------------------------------------------------|
| Benin         | 41.4                                       | 18.0                                                       | 70.8                                                       |
| Burkina Faso  | 19.2                                       | 0.8                                                        | 60.7                                                       |
| Guinea-Bissau | 14.7                                       | -                                                          | 29.8                                                       |
| Mali          | 35.1                                       | 1.8                                                        | 83.6                                                       |
| Niger         | 16.2                                       | 4.7                                                        | 65.4                                                       |
| Togo          | 46.9                                       | 19.4                                                       | 87.4                                                       |

Source: World Bank, 2016

### Electricity tariffs in the target countries

Electricity tariffs show important differences among the target countries. In Guinea-Bissau, where major infrastructure and capacity difficulties occur, the average tariff is around \$0.40/kWh, while in Mali and Niger, prices are significantly lower with an average retail price is at \$0.16/kWh. Regarding the regional average, it stands at around \$0.23/kWh which is substantially higher than in most African countries.

Table 13: Electricity tariffs in WAEMU countries

| Country                       | Electricity tariffs in 2017                                                                                                                                             |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Benin                         | \$0.22/kWh for all end users<br>Subsidy for low-income households                                                                                                       |
| Burkina Faso                  | Average retail price \$0.24/kWh                                                                                                                                         |
| Guinea-Bissau                 | \$0.40/kWh for on-grid consumers                                                                                                                                        |
| Mali                          | Average retail price \$0.16/kWh for on-grid consumers                                                                                                                   |
| Niger                         | \$0.16/kWh on average for on-grid consumers<br>Consumers below 3kWh per day \$0.11/kWh<br>\$0.11/kWh for industrial consumers<br>\$0.07/kWh for agricultural facilities |
| Togo                          | \$0.22/kWh for all end users<br>\$0.31/kWh for residential<br>\$0.20/kWh for commercial/industrial                                                                      |
| Regional average retail price | \$0.2333/kWh<br>Rounded: \$0.23/kWh                                                                                                                                     |

Source: ECREEE (2018) Regional Off-Grid Solar Market Assessment & Private Sector Support Facility Design, Country Reports

### Electricity demand in the target countries

Scaling up renewable energy generation capacity is of crucial importance to enable the target countries to sustain economic growth and development, and to prepare for the upcoming population boom. Indeed, electricity demand is expected to increase fivefold by 2030 to 250 TWh in West Africa<sup>27</sup>, and the energy sector will not be able to meet the electricity demand in its current state. The figure below shows the projected electricity demand by country in West Africa up to 2030.

<sup>27</sup> IRENA (2015) Africa Power Sector Synthesis

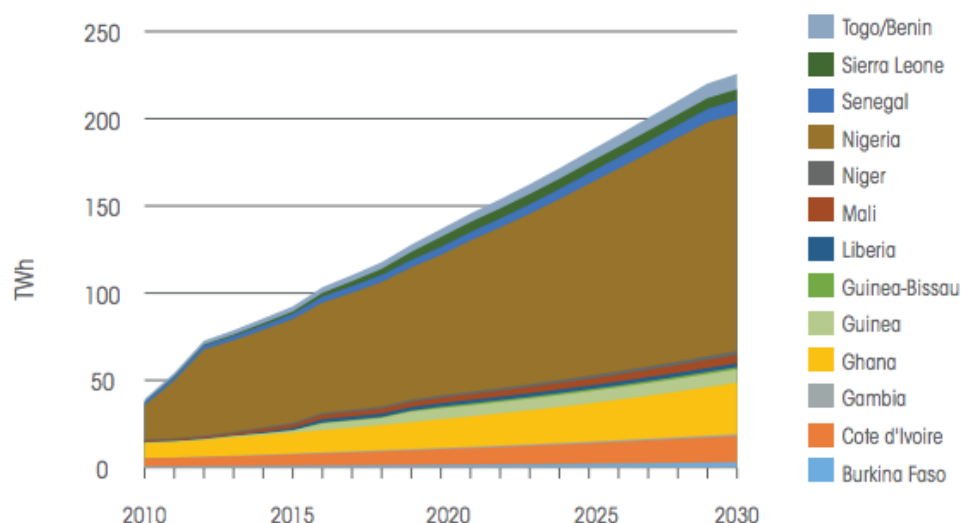


Figure 1: Projected Electricity Demand by country – West Africa

Source: IRENA (2015) Africa Power Sector: Planning and Prospects for Renewable Energy

### Energy mix in the target countries

Despite the considerable potential of RE and the dynamism of WAEMU economies, RE resources remain largely untapped. Scaling up RE investment is of crucial importance considering the direct correlation between sustained economic growth, social progress and energy access. Among other challenges, which are discussed by country below, national energy sectors in WAEMU states are affected by the monopoly of one corporation over the electricity supply, issues related to the insufficient energy supply compared to the demand, and sometimes severe infrastructure deficiencies. Low electricity access rates have negatively impacted the modernization of the economy and social development. Improving electricity access rates and scaling up non-hydro renewable energy will diversify the countries' energy mix while providing clean access to electricity to the population.

Table 14: Energy mix by target country (in %) (See Table 7 for the breakdown in MW)

| Country       | Thermal <sup>28</sup> | Hydro | Solar |
|---------------|-----------------------|-------|-------|
| Benin         | 97                    | 0     | 3     |
| Burkina Faso  | 77                    | 10    | 13    |
| Guinea-Bissau | 100                   | 0     | -     |
| Mali          | 36                    | 53    | 11    |
| Niger         | 95                    | -     | 5     |
| Togo          | 79                    | 20    | 1     |

Source: IRENA (2018) Renewable Energy Statistics (unless otherwise specified)

<sup>28</sup> 2017 data from ECREEE (2018) Regional Off-Grid Solar Market Assessment & Private Sector Support Facility Design, Country Reports, August.



Many clean energy projects in West Africa are technically and commercially viable but have yet to become bankable investment opportunities. High initial development costs, lack of start-up capital, limited know-how of project financing, and inadequate enabling environments all present barriers to such progress. As a result, RE projects in West Africa tend to be financed by a combination of private and public financing, with the public sector contributing the lion's share. Traditionally, the public-private financing models being used do not necessarily guarantee timely or efficient investments. The procedures associated with these financing models are bureaucratic, complex and prolonged. The commitment from investors is not consistent and is largely influenced by global dynamics and perceptions beyond the control of project developers.

The potential private sector finance for RE projects in West Africa is significant. However, the challenge is to provide affordable and attractive financing frameworks that sustainably exploit commercial lending and investment opportunities in the private sector. RE technologies often have high upfront costs and low rates of return. The consequent high-risk profiles of investments and little access to affordable financing in West Africa make RE unattractive and challenging for investors.

## Renewable energy investment scenario

In the six target countries, financial instruments are the predominant means of supporting the renewable energy sector. Tax incentives take a number of forms, including investment or production tax credits, as well as reduction or elimination of taxes such as import duties, sales, and value-added tax (VAT). In Burkina Faso and Mali, import duties on RE components have been reduced or removed and value-added tax reductions for RE projects have been established. In addition, Niger and Togo all offer some form of tax incentives for RE. In Mali additional financial support has come from public financing mechanisms such as public investment, project grants or low interest loans.

Both public and private financing have played an important role in the sector's development to date. While data on private finance flows in the region is not widely available, an analysis of regional projects such as the Cabeolica Wind Farm Project shows that private financing has played a key role in project development.

As mentioned above, millions of dollars of support have been lined up through specialised regional and Africa-wide funds and programmes such as the ECOWAS-led Renewable Energy Investment Initiative (EREI) and Renewable Energy Facility (EREF), the Sustainable Energy Fund for Africa (SEFA), the African Renewable Energy Fund (AREF), and the Power Africa Initiative. International financing through mechanisms established under the United Nations Framework Convention on Climate Change (UNFCCC) process—such as the Global Environment Facility (GEF), Climate Investment Funds (CIF), Clean Development Mechanism (CDM), and Nationally Appropriate Mitigation Actions (NAMAs)—have all supported RE development in the region and offer an opportunity to continue scaling up future investments. While public and private funding from national, regional, and international institutions has been instrumental in the funding of RE development in West Africa, it has yet to significantly penetrate the private sector market.

An overview of the energy sector in each target country is provided below, along with key indicators demonstrating the need to scale up RE investment in target countries.

*Table 15: Overview of the energy sector in each target country*

| Country | Overview of the energy sector                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Benin   | <p>Benin's electricity sector faces significant challenges in terms of regulatory arrangements, financial sustainability and technical capacity. Benin's electricity needs are not met by its 288 MW of non-renewable energy capacity and 9 MW of renewable energy capacity. The missing electricity supply is covered by the cross-country grid shared with Ghana, Nigeria and Ivory Coast. One of the country's major energy challenges is its heavy dependence on electricity imports from its neighbours, and its insufficient non-renewable generation capacity.</p> <p>Indeed, Benin is locked in its fossil fuel dependency unless an enabling environment is implemented to allow for the private sector to invest and scale up the country's generation capacity. When in 2014, electricity production from oil sources represented 99.5% of the total, it still accounted for over 94% a year later. Electricity production from renewable energies represented a mere 1.5% in 2015, though showing a 1-point increase from the past year, and energy imports accounted for 46.6% of the country's total energy use in 2014. Nonetheless, while in 2011, the national electrification rate was low at around</p> |



|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | <p>28%, significant progress has been made as the electricity access rate now is 41.4% of the total population (2016). However, there is an important disparity between rural and urban electricity access rates, with 18% and 70.8% respectively. These results show that efforts have been made to scale up electricity access, but little has changed in terms of the diversification of Benin's energy mix. A number of outstanding issues have hampered clean energy development in Benin:</p> <ul style="list-style-type: none"> <li>• Despite objectives stated in the Electricity Code (2007) to open the energy market to private actors, the CEB (Communauté Électrique du Bénin) remains the sole actor responsible for buying and supplying the electricity produced. Recently, the CEB has implemented a feed-in tariff for renewable energy sources fixed at 73XOF/kWh, applied to all RE technologies.</li> <li>• Even though national authorities agree on the potential of renewable energies in Benin, no national targets for RE project development have been defined.</li> <li>• The CEB financial balance rests on currency exchange from USD\$ for cross-country connections with Nigeria, Ghana and Ivory Coast</li> <li>• Overall ambitions have not been translated into actions in terms of creating an enabling environment for private-led RE project development, though a handful of small-scale solar projects are currently under consideration</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Burkina Faso</b>  | <p>Burkina Faso has one of the lowest rankings in terms of electricity access, with only 19.2% of total population with access to electricity in 2016, a figure mainly accounted by urban areas, with 60.7% of urban population having access to electricity. The rural electricity access rate is among with lowest in Africa with less than 1% (0.8%) as of 2016. The country's energy supply is heavily dependent on imported hydrocarbons from neighbouring countries, such as Ghana, Togo and Ivory Coast for use in electricity production, transport and other industries. Dependence on imports is exacerbated by high production costs, fluctuating oil prices and a steadily increasing demand for electricity.</p> <p>Furthermore, in recent years, Burkina Faso's power utility, SONABEL has faced system transmission and distribution losses of over 20%, much higher than the internationally accepted norm of 10%. In addition, the high tariffs barely cover the operating costs of the expensive SONABEL power. The resultant disruptions in supply and the high cost of energy for consumers constitute significant barriers to private-led energy development and negatively impact productive capacity. These hidden costs represent a significant burden for the expansion of the private sector and national growth. While recent initiatives have placed an emphasis on improving energy access through off-grid solar solutions, much remains to be done to achieve SE4All objectives by 2030<sup>29</sup> with regard to energy access and diversification of the energy mix.</p>                                                                                                                                                                                                                                                                                                                       |
| <b>Guinea-Bissau</b> | <p>The RE market has emerged rather recently in Guinea-Bissau due its difficult economic and political environment. Should the political scene remain stable, the RE market offers good potential even though the vast majority of the population lives on less than 1USD\$/day and therefore cannot afford electricity. Currently, a few solar panels are installed in urban areas (for public lighting), but there is almost no electrification in rural areas, for which data is not available.</p> <p>The national electricity service struggles immensely to meet the electricity demand, with national capacity standing at 12.5 MW while demand for the capital alone is around 30MWh. Guinea-Bissau is one of few countries in the world which has seen its electricity access rate decline in recent years, from 17.2% in 2014 to around 14.7% in 2016. The urban electricity access rate is estimated to be around 29.8% as of 2016. The distribution network suffers from significant losses ranging from 30 to 40%. They ought to be extensively renovated before focusing on attracting private investment because the deficiencies may result in the deterioration of newly installed equipment. This constitutes an important barrier to private investment in the energy sector.</p> <p>Another barrier to private investment is the systemic theft of electricity and fraud. National authorities are currently developing a regulatory framework for the energy sector and have confirmed the considerable potential for the development of RE projects in the NREAP. The greatest barrier to the development of the RE sector remains the lack of finances and facilities available (credit access for RE firms, poverty of the population, insufficient resources of local banks, VAT scrapped for solar panels but fully applied to all other solar components: batteries, controllers, inverters etc.).</p> |
| <b>Mali</b>          | <p>Mali has a strong renewable energy potential, especially for solar power thanks to its above average irradiation levels. The country has seen its national average rate of access to electricity increase steadily, from 22.3% in 2010 to 35.1% in 2016. However, rural and urban electricity access rates show a significant disparity, with 1.8% of the total rural population with access to the grid against 83.6% of the total urban</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

<sup>29</sup> 65% national electricity access, 32% national access to clean cooking.

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              | <p>population. In Mali, the electricity consumption per capita is about 90 kWh, against an average of 116 kWh in ECOWAS countries and 500 kWh on average in Sub-Saharan Africa. The average consumption is also characterised by a strong disparity between rural and urban areas.</p> <p>Key constraints and challenges in the energy sector faced by the private sector include:</p> <ul style="list-style-type: none"> <li>• The national electricity access rate is still very low, especially in rural areas</li> <li>• Mali exempts solar panels, solar lamps, and other renewables from import levies and duties, however, the current regulatory framework does not adequately promote private investment in renewable energy, though the local private sector has been increasingly involved in rural electrification.</li> <li>• The impacts of climate variability are making the country's electricity supply – which depends on hydroelectric power for about 59% of its on-grid supply – increasingly vulnerable.</li> <li>• Regulatory (including tariffs) and legal frameworks do not favour RE investment by the private sector on a large scale.</li> <li>• The financial status of the electricity utility EDM SA is extremely weak.</li> </ul> <p>Private sector contributions to RE are currently not accounted for in national statistics. However, it is estimated that there are more than 20 local, private operators active in the RE sector. These include decentralised energy service companies operating a few pilot RE rural electrification schemes, and retailers and distributors involved in the sale of electrical (mainly SHS), lighting and energy efficiency equipment.</p>                                                                                                                                                                                                                                                                                                                                                  |
| <b>Niger</b> | <p>Electricity, in terms of both quality and access, is a key challenge for Niger. The existing power infrastructure is underdeveloped, and the country continues to rely heavily on imported electricity from neighbouring Nigeria. There are occasional power-supply interruptions from Nigeria due to failing electricity infrastructure, highlighting Niger's dependence and vulnerability. In 2016, the electricity access rate stands at 16.2% of the total population, with important disparities between rural and urban areas (4.7% and 65.4% of the population respectively). The country's energy mix is characterised by a carbon-intensive energy production, despite having significant RE potential, particularly for solar in the north.</p> <p>The Renewables Readiness Assessment, carried out by IRENA, has highlighted several technical opportunities and barriers facing Niger's private sector and the renewable energy generation and transmission infrastructure. It also brought to light some of the institutional, financial and capacity issues associated with PV and wind-based power generation to feed the grid. These are listed below.</p> <ul style="list-style-type: none"> <li>• Lack of information on solar and wind energy resources is a major obstacle to investment due to the difficulties in developing robust project proposals.</li> <li>• At present, there is little in the way of private sector participation and incentives to draw investors to the development and distribution of solar products in Niger.</li> <li>• The market is limited by Niger's small size and constrained by taxes of over 52% (import tariffs and VAT), a lack of regulation on poor quality products smuggled from neighbouring countries, and the absence of a meaningful government renewables stimulus.</li> <li>• Lack of institutional coordination and collaboration between financial institutions and other stakeholders</li> <li>• Technical, legal and administrative barriers reducing investor confidence.</li> </ul> |
| <b>Togo</b>  | <p>As of 2016, 46.9% of Togo's total population had access to electricity, with 19.4% of the total population in rural areas and over 87.4% in urban centres.</p> <p>The energy sector in Togo can be characterized by the following:</p> <ul style="list-style-type: none"> <li>• A dominance of traditional sources of energy (biomass consisting of wood energy and agricultural residues). Strong energy dependence on biomass energy in a context of scarcity of wood resources;</li> <li>• Widespread poverty and low purchasing power of populations for access to electricity and modern fuels;</li> <li>• The predominance of hydroelectricity and biomass based renewable energies;</li> <li>• Strong dependence on imports (electricity and hydrocarbons - 100% of miscellaneous petroleum products needs and 79% of electricity needs are covered by imports). and limited control over the economic and energy policies of the countries of supply (Ghana, Côte d'Ivoire and Nigeria);</li> <li>• Virtually no upgrading of new and renewable sources of energy (solar, wind, conventional biomass energy applications, etc.);</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <ul style="list-style-type: none"> <li>• Very low overall energy efficiency due to the predominance of biomass, with poor energy yields for traditional carbonisation (charcoal production through traditional wheels: 12 to 20% weight) and final consumption (yields energy use of wood and charcoal equipment between 7 and 15%).</li> </ul> <p>The development of private-led RE in Togo faces the following capacity and financial barriers:</p> <ul style="list-style-type: none"> <li>• Lack of specialised institutional and operational structures for the governance of rural electrification and the promotion of more efficient domestic fuels;</li> <li>• Lack of financial capacity to meet investment needs and high dependence on international financing;</li> <li>• Poor exploitation of domestic and international private sector financing potentials (including banking and microfinance institutions);</li> <li>• Low national financing capacity (public and private) and high dependence on public and private external financing (financial obstacle);</li> <li>• Enormous technological delays in renewable energy, energy efficiency and the exploitation of new and renewable sources of energy and a high dependence on the transfer of technology and know-how (technological obstacle);</li> <li>• Poverty of populations, especially rural populations and its impact on purchasing power (poverty and affordability of modern energy services).</li> </ul> |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### C.6. Regulation, Taxation and Insurance (if applicable)

BOAD is an international public institution whose purpose is to "promote the balanced development of its Member States and foster economic integration within West Africa" by financing priority development projects. Created in 1973 by the member states of the West African Economic and Monetary Union (WAEMU), BOAD has a legal status of a development finance institution. With this status, BOAD is exempt from tax, therefore interest to the GCF will be free of any taxes including withholding. This is outlined in the text depicting the legal status of the Bank<sup>30</sup>:

- Section 2.2: privileges and immunities, Article 4, H) the assets and operations of the bank benefit from the tax exemptions specified in Article 44.
- Article 44 stipulates that "The Bank, its revenues, its property and other assets, as well as transactions and operations that it carries out under its present status, are exempt from all direct or indirect taxes". Additionally, "it will not be levied on the bonds issued by the bank or the interests which come from them regardless the holder of these securities, nor tax by states or collectivities of the union"

With regards to taxes applicable to solar technologies, in the WAEMU member states, fiscal incentives have been unanimously chosen as a tool to create incentives to invest in the RE sector. These measures take the form of tax credits on investment or production, taxes on sales and Value Added Tax (VAT). In Burkina Faso, Benin, Ivory Coast, Niger, Togo and Mali, customs and import tax have been reduced or even scrapped for solar PV products, solar panels and/or solar system components.<sup>31</sup> Burkina Faso has reduced VAT on RE projects and has entirely suppressed it for solar energy projects. Niger has introduced a tax exemption on domestic solar systems and solar lamps.

The project operators will have to provide a letter of authorization for the construction of the project in the site, a preliminary permit from the electrification agency, the Environmental and Social Permit for category B project, the agreement between the regulatory authority and the operator, the authorization from the Minister in charge of energy, the specifications for the operation in the site, the tariff structuring and the PPA.

### C.7. Institutional / Implementation Arrangements

BOAD as part of its AE role will among others carry out the following:

- BOAD will be responsible for the overall oversight of the framework implementation and will report to GCF as per the terms to be agreed under the Accreditation Master Agreement (AMA);
- BOAD will report to GCF as per the terms to be agreed under the AMA and the Funded Activity Agreement (FAA).
- BOAD will prepare environmental and social impact analysis reports in line with the bank's environmental and social safeguards (including ensuring that the sub-project is within allowed environmental impact classification, i.e. only up to category B)

<sup>30</sup> Status de la Banque Ouest Africaine de Développement (B.O.A.D), 2016, pp.2, pp. 13

<sup>31</sup> Ibid.

- BOAD will ensure that a robust climate rationale is in place for all projects.
- BOAD will ensure that an MRV framework will be established to ensure activities of the facility and other relevant stakeholders are in compliance with all applicable policies, procedures and other requirements of the GCF to enhance transparency. An independent party will be procured by BOAD for Monitoring, Reporting and Verification (MRV); BOAD will be ultimately responsible for reporting to the GCF.

**Component 1: To scale up commercial and sustainable financing for solar investments**

The GCF will provide concessional debt financing of EUR 57 million to BOAD for on-lending to medium/large scale on-grid solar projects in the target countries. The GCF loan facility can be drawn down for two (2) purposes, respectively presented under sub-components 1.1 and 1.2. BOAD shall blend the proceeds of the GCF loan with its own funding to on-lend to the projects in a single blended tranche.

**Sub-component 1.1: Senior loans facility**

Of the total EUR 57 million GCF loan, a minimum of EUR 45 million shall be drawn down by BOAD to finance the construction of solar energy projects for on-lending to solar energy projects (sub-projects) in Francophone West Africa ("**Senior loan Facility**").

**Sub-component 1.2: Standby loan facility (credit enhancement in the form of tenor extension)**

Of the total EUR 57 million GCF loan, a maximum of EUR 12 million can be drawn down by BOAD to be used for credit enhancement in the form of tenor extension to solar projects financed under the programme ("**Standby Loan Facility**"). Such tenor extension instrument is intended to crowd-in participation from interested and credible local commercial banks. If less than the maximum cap has been committed for standby loans by the Accredited Entity at financial close, the remaining amount shall be drawdown for the purpose of sub-component 1.1.

It is expected that local commercial banks are unable to extend loans with tenors of more than 5-7 years. To overcome this barrier, the GCF will incentivise the participation of local commercial banks in the Programme by providing Standby Loans to the sub-project for extension of the commercial banks loan tenor (as described in Section C3. of the FP).

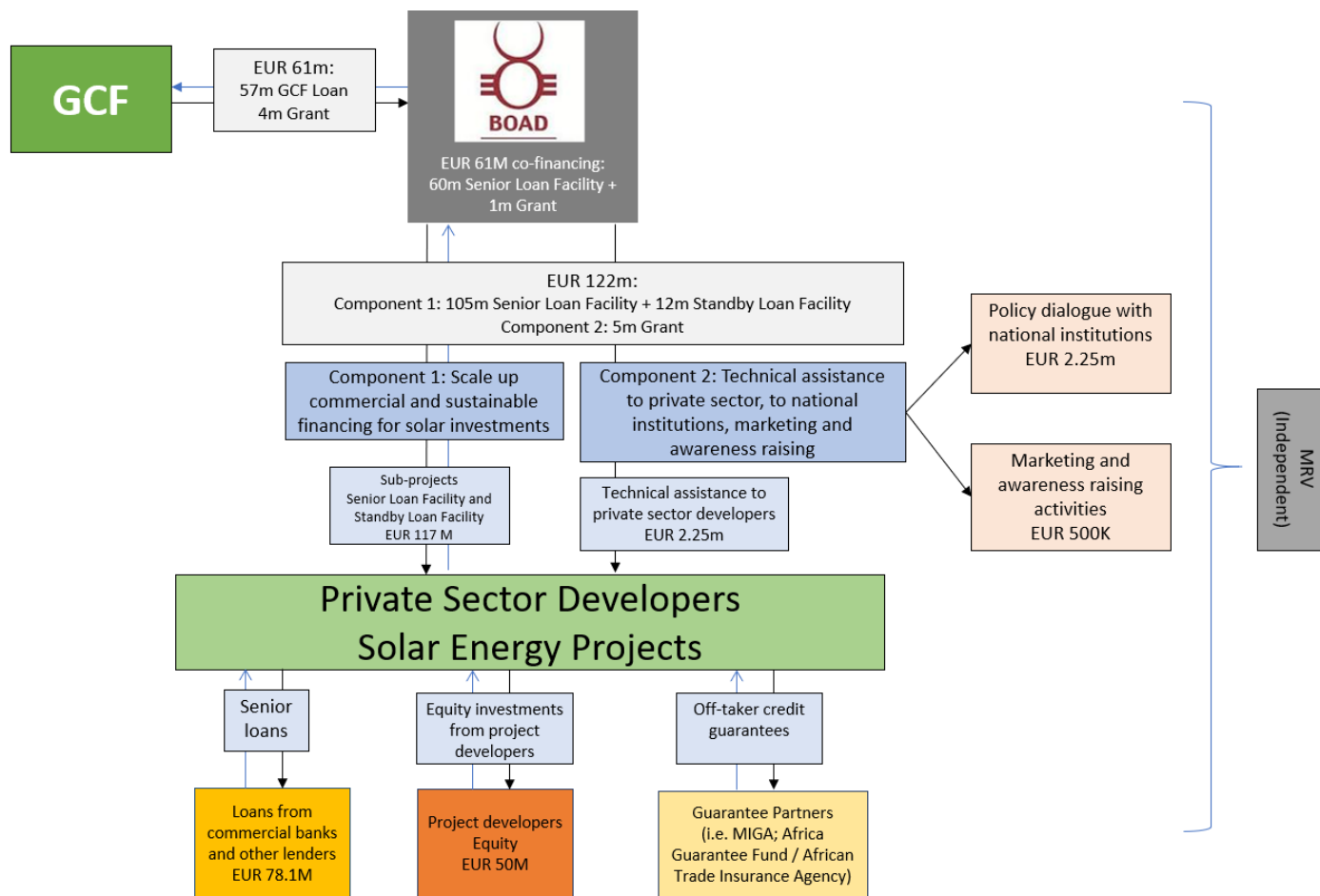
**Component 2: Technical assistance (grant component)**

BOAD as part of its EE role will be responsible for the implementation of the grant component. BOAD will hire qualified local and international consultants for the implementation of the technical assistance (component 2), based on its procurement policies as approved by the GCF. The consultants will work under the supervision of BOAD and will provide technical assistance:

- to local private sector actors to build their capacities during the development phase of their projects (Sub-component 2.1);
- to BOAD staff to build their capacities during the development and implementation of the project portfolio (Sub-component 2.1);
- to national institutions to support them in the implementation of regulatory and policy regimes that provides clear and predictable rules for solar project development (Sub-component 2.2).

**Financial Management (Flow of Funds)**

***Figure 2: Flow of Funds Diagram***



### C.8. Timetable of Project/Programme Implementation

Please provide a project/programme implementation timetable in [section I \(Annexes\)](#). The table below is for illustrative purposes. If the table format below is used, please refer to the activities as numbered in Section H. In the case of outputs, please mark when all the required activities will be completed.

| TASK                                                                                                                                                                                                                                                                                                                                                                        | 2019<br>q2, q3, q4 | 2020<br>q1, q2, q3, q4 | 2021<br>q1, q2, q3, q4 | 2022<br>q1, q2, q3, q4 | 2023<br>q1, q2, q3, q4 | 2024<br>q1, q2, q3, q4 | 2025<br>q1, q2, q3, q4 | 2026<br>q1, q2, q3, q4 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
| <b>Component 1</b>                                                                                                                                                                                                                                                                                                                                                          |                    |                        |                        |                        |                        |                        |                        |                        |
| <b>Sub-Component 1.1 Senior Loans Facility</b> <ul style="list-style-type: none"> <li>• Origination of eligible solar projects</li> <li>• Financial appraisal and technical feasibility, due diligence</li> <li>• Approval by BOAD Executive Committee and financial close</li> <li>• Disbursement of Senior loans</li> <li>• Construction of solar power plants</li> </ul> | X<br>(START Q2)    | X                      | X                      | X                      | X                      | X                      | X                      | X<br>(END Q2)          |
| <b>Sub-component 1.2 Standby Loans Facility</b> <ul style="list-style-type: none"> <li>• Reach out to commercial banks to participate in originated solar projects</li> <li>• Structuring of standby loans at financial close</li> </ul>                                                                                                                                    | X<br>(START Q2)    | X                      | X                      | X                      | X                      | X<br>(END Q2)          |                        |                        |
| <b>Component 2</b>                                                                                                                                                                                                                                                                                                                                                          |                    |                        |                        |                        |                        |                        |                        |                        |
| <b>Sub-component 2.1: Technical assistance to BOAD and local project developers</b> <ul style="list-style-type: none"> <li>• Recruitment of consultants to carry out the TA activities under this sub-component</li> <li>• Trainings of project developers</li> <li>• Trainings of BOAD staff</li> </ul>                                                                    | X<br>(START Q2)    | X                      | X                      | X                      | X                      | X (END Q2)             |                        |                        |
| <b>Sub-component 2.2: Technical assistance to national institutions</b> <ul style="list-style-type: none"> <li>• Recruitment of consultants to carry out the TA activities under this sub-component</li> <li>• Trainings of national institutions</li> </ul>                                                                                                                | X<br>(START Q2)    | X                      | X                      | X                      | X                      | X (END Q2)             |                        |                        |



|                                                                                                                                                                                                                                                                                                                                                     |                         |          |          |          |          |                   |  |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|----------|----------|----------|----------|-------------------|--|--|
| <p><i>Sub-component 2.3: Marketing and awareness raising activities</i></p> <ul style="list-style-type: none"> <li>• <i>Organisation of events to raise awareness on the benefits of solar system technologies</i></li> <li>• <i>Communications and outreach effort that promotes the objectives and good practices of the programme</i></li> </ul> | <p>x<br/>(START Q2)</p> | <p>x</p> | <p>x</p> | <p>x</p> | <p>x</p> | <p>x (END Q2)</p> |  |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|----------|----------|----------|----------|-------------------|--|--|

The programme implementation period is 7 years including the construction of solar projects (from Q2 2019 to Q2 2026).

## D.1. Value Added for GCF Involvement

The GCF involvement in the programme is critical for the following reasons:

### **Triggering private investment in the solar sector at scale through the appropriate risk-return profile**

As explained in previous sections, the solar potential is significant in West Africa but the installed capacity remains low because of an unattractive business environment. The national governments have made substantial commitments to address climate change by investing massively in RE. The policy and regulatory frameworks are in principle supportive of the goals of increasing RE investment, however the magnitude of the barriers to doing business effectively and in a timely manner has meant that progress has been slower than expected.

In West Africa, financing for solar projects has traditionally been dominated by the Development Finance Institutions (DFIs) with little space of participation from commercial and local financial players. Technological progress and associated cost reductions have created opportunities for the private sector. However, these technology solutions are still new, currently limited in scale and operating in an unstable regulatory environment. Furthermore, solar projects usually have a long-term investment horizon of over 10 years and as such require long-term financing. Commercial banks are not able to provide long-term credit for solar projects partly because they depend largely on short-term deposits for loan transactions. This mismatch of using short-term deposits for long-term loans and the financing constraints it presents to the project developers of solar projects will persist without availability of long-term credit facility. Further, the cost of financing from local commercial banks is high. This crowds out private investment from the solar sector. Still, maintaining the status quo cannot bring a transformative change that is required for dramatically increasing access to energy in the region with stable and sustainable energy source like solar energy.

The partnership created with the support from the GCF is required to provide private sector debt financing with support and comfort to enter into financing this new asset class. The GCF's involvement in supporting the proposed Programme will directly allow BOAD to offer more attractive lending to project developers and to crowd private banks in the solar sector, therefore, mobilizing and increasing private sector leverage. Involvement of the GCF will enable a financing scheme that will offer long-term and reasonably priced loans to solar investments, which will at the same time open up the market for participation of local private financiers. The provision of adapted financial tools such as concessional loans and standby loans will give private sector actors the confidence to invest in solar technologies. Standby loan facility is proposed to enable the participation of commercial banks who are not capable of making long-term investment into solar projects. It will give confidence to commercial banks, incentivizing their participation to the projects. Without addressing the financial barriers, the market for solar investments is likely to remain limited in the West African region and private sector actors will not be willing to take the risk to invest in the target sector.

The proposed financial structure will allow BOAD to "crowd in" private sector investors and support solar interventions on a significant scale. By blending its own financing with the GCF concessional funding and the mobilized private investor funding, the risk exposure will be lower for private sector developers and the increased participation of private sector will over time lead to sustainable private sector support without GCF participation.

### **Project developers require training and capability building measures to prepare bankable solar related businesses**

Market barriers continue to hold back solar related investment at scale, including insufficient access to finance, low internal capacity and a limited pipeline of bankable projects. Local private sector companies have insufficient experience in originating commercially viable solar related projects and lack technical knowhow to assess the financial and technical feasibility of solar investment proposals. The programme will address these technical barriers to scaling up the market for solar investments by building the capacity of project developers to structure their investments (project preparation and management). The GCF funds, through the financing of TA, will provide support to mid-size local project developers for developing and appraising projects. Trainings will include support to private sector actors to develop technical and financial feasibility studies and to prepare bankable projects (more information on the type of trainings provided can be found in section C.3).

The relevance of this type of support has been confirmed by a study from IRENA (“Unlocking Renewable Energy Investment”, 2016) which highlights the lack of funds to support project preparation and development phase. The proposed TA support is therefore filling a crucial gap in the availability of funding to support early stage development projects.

#### **No alternative source of public sector funding at scale is available**

Given the large investment needs in the sector, a significant volume of investments into the solar market in West Africa is required. Currently, only the GCF is able to provide such finance within the given timeframe. BOAD’s strategic objective is to increase its portfolio of investments in the solar directly to private sector actors, but so far this has been low because of the lack of alternative source of public sector funding available. The involvement of the GCF is required as it will allow a combination of concessional lending and grant for technical assistance required to achieve the objectives of the programme and the consequent paradigm shift in the target countries.

The programme will further contribute to a paradigm shift to low-emission development pathways through the mobilisation at scale of private climate finance and through the creation of a replicable model for eventual scale-up of the Programme’s success into the region. Transformational impacts will be achieved through the growth in commercially viable solar investments and the scaling up of financial products and services offered to the private sector in the region.

Without GCF’s contribution, it would take considerably longer for existing market barriers to be removed. In the meantime, access to finance for climate investments would remain limited to a certain number of borrowers and remain too costly for many others. GCF’s contribution will unlock the considerable solar potential in LDC countries from Francophone West Africa. The total solar installed capacity in the NDC countries of West Africa will be increased twofold thanks to the GCF contribution. This will therefore help to operationalize the commitments made by these countries in the Paris Agreement.

### **D.2. Exit Strategy**

Projects financed under the programme will benefit from long-term power purchase agreements (PPA) that guarantee income over most of the asset life and longer than the tenor of the GCF and BOAD loans. Financial viability of projects is expected to be backed by the PPA that is longer than the tenor, creating a tail of at least 5 years. Solar technology is well proven and highly durable. Overall therefore, sustainability of the projects throughout their operating lives will be assured.

At the market level, the deal flow under this programme will support mobilise private sector actors with experience at scale in the region. Furthermore, supporting the implementation of regulatory and policy regimes that provides clear and predictable rules for solar project development in the region will ensure that the programme will have a lasting transformative (“paradigm shifting”) impact on the medium-large scale solar sector in the region.

For individual sub-projects, the loans (both standard loans and standby loans) will be paid back according to the repayment schedule. At the programme level, once the facility is provided, GCF does not need to be highly involved in program operations. BOAD will report on the facility disbursement to GCF periodically on the usage of the facility. The GCF will gradually exit as BOAD will be repaying the GCF loan facility. Exit risk is minimal.

In this section, the accredited entity is expected to provide a brief description of the expected performance of the proposed project/programme against each of the Fund's six investment criteria. Activity-specific sub-criteria and indicative assessment factors, which can be found in the Fund's [Investment Framework](#), should be addressed where relevant and applicable. This section should tie into any request for concessionality made in [section B.2](#).

## E.1. Impact Potential

Potential of the project/programme to contribute to the achievement of the Fund's objectives and result areas

### E.1.1. Mitigation / adaptation impact potential

Energy access are key to the development of the NDC countries from West Africa. Given that the energy mix in the target region already heavily depends on fossil-fuel based electricity, further adding fossil-fuel based emissions is not an option from a climate perspective. Low-carbon energy production can be achieved by scaling up the use of solar sources given the significant potential in the region.

The climate impact potential of the Programme will be the reduction of CO<sub>2</sub> emissions through displacement of fossil-fueled electricity use at the user level by implementing and installing of clean and solar energy sources.

It is expected that the implementation of the programme will deliver approx. 215 MW of new generating capacity in the solar sector, generating approx. 339 GWh of clean electricity annually and thereby directly avoiding emissions of 193,398 tCO<sub>2</sub>e per year (equivalent to approx. 4.8mtCO<sub>2</sub>e over 25 years).

Based on per capita electricity consumption for the region (116 kWh/ person-year<sup>32</sup>), the programme will deliver clean energy services for approximately 2.9 million beneficiaries.

### E.1.2. Key impact potential indicator

Provide specific numerical values for the indicators below.

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                |                                                                     |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------------------------------------------------------------------|
| GCF<br>core<br>indicators       | Expected tonnes of carbon dioxide equivalent (t CO <sub>2</sub> eq) to be reduced or avoided (Mitigation only)                                                                                                                                                                                                                                                                                                                                                                        | Annual         | 193,398 tCO <sub>2</sub> e (once all projects will be in operation) |
|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Lifetime       | 4,834,955 tCO <sub>2</sub> e                                        |
|                                 | <ul style="list-style-type: none"> <li>Expected total number of direct and indirect beneficiaries, disaggregated by gender (reduced vulnerability or increased resilience);</li> <li>Number of beneficiaries relative to total population, disaggregated by gender (adaptation only)</li> </ul>                                                                                                                                                                                       | Total          | 2.9 million                                                         |
|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Percentage (%) | NA                                                                  |
| Other<br>relevant<br>indicators | <p>Examples include:</p> <ul style="list-style-type: none"> <li>Expected increase in the number of households with access to low-emission energy: 700 000 households with improved access to low-emission energy sources</li> <li>Expected increase in the number of medium and large low-emission power suppliers, and installed effective capacity: the programme expects to support at least 6 projects and to add 215MW of new solar capacity installed into the grid.</li> </ul> |                |                                                                     |

<sup>32</sup> <https://www.se4all-africa.org/seforall-in-africa/country-data/mali/>

BOAD developed a methodology to conservatively assess CO<sub>2</sub> reductions driven by the Programme. The methodology calculates the CO<sub>2</sub> savings based on the total estimated installed capacity of PV power plants expected to be deployed by the programme. Based on the pipeline of projects to be funded under the programme (See Table 10), the total solar capacity to be installed in the six target LDCs is 215MW.

The calculation uses the grid emission factor (EF) from the CDM methodology “Standardized baseline - Grid emission factor for West African Power Pool” (ASB0034: Grid emission factor for West African Power Pool (version 01.0) for the five countries included in the programme connected to the Power Pool: Benin, Burkina Faso, Mali, Niger and Togo. This standardized baseline provides the values for grid emission factors (i.e. the carbon dioxide (CO<sub>2</sub>) emission factors) for the West African Power Pool (WAPP) – interconnected electricity system of nine countries: Benin, Burkina Faso, Côte d’Ivoire, Ghana, Mali, Niger, Nigeria, Senegal, and Togo. The grid emission factor used is 0.561 tCO<sub>2</sub>/MWh. More information on the assumptions used can be found on the Standardized baseline - Grid emission factor for West African Power (available at: [https://cdm.unfccc.int/methodologies/standard\\_base/2015/sb102.html](https://cdm.unfccc.int/methodologies/standard_base/2015/sb102.html)).

For Guinea-Bissau, the emission factor has been calculated using a bottom up approach to find emissions per unit of MWh produced based on certain assumptions. The grid emission factor used is 0.76 tCO<sub>2</sub>/MWh.

*Table 16: Grid emission factor (EF) for West African Power Pool as defined in the CDM methodology ASB0034*

**Table 1. Grid emission factors for the project electricity system of the WAPP**

| Parameter        | Unit                  | Description                                                                         | Applicable project types                                                         | Applicable values      |                         |                        |
|------------------|-----------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|------------------------|-------------------------|------------------------|
|                  |                       |                                                                                     |                                                                                  | First crediting period | Second crediting period | Third crediting period |
| $EF_{grid,OM,y}$ | tCO <sub>2</sub> /MWh | Operating margin CO <sub>2</sub> emission factor for the project electricity system | All project activities                                                           | 0.559                  |                         |                        |
| $EF_{grid,BM,y}$ | tCO <sub>2</sub> /MWh | Build margin CO <sub>2</sub> emission factor for the project electricity system     | All project activities                                                           | 0.565                  |                         |                        |
| $EF_{grid,CM,y}$ | tCO <sub>2</sub> /MWh | Combined margin CO <sub>2</sub> emission factor for the project electricity system  | Wind and solar power generation project activities                               | 0.561                  |                         |                        |
| $EF_{grid,CM,y}$ | tCO <sub>2</sub> /MWh | Combined margin CO <sub>2</sub> emission factor for the project electricity system  | All project activities except wind and solar power generation project activities | 0.562                  | 0.563                   | 0.563                  |

For the capacity factor, a conservative approach has been used for the 2 technologies (solar PV and CSP) of 18%. The operational lifetime of solar technologies is 25 years.

The annual emission avoidances of the Programme are calculated as follow:

| Technology   | Countries                                          | Installed capacity | Capacity factor | Annual elec. Gen (MWh) | Grid EF (tCO2/MWh) | Annual emissions avoided | Lifetime (yr) | Lifetime emissions avoided |
|--------------|----------------------------------------------------|--------------------|-----------------|------------------------|--------------------|--------------------------|---------------|----------------------------|
| Solar        | Benin, Burkina Faso, Mali, Niger, Nigeria and Togo | 205                | 18%             | 323,244                | 0.561              | 181,340                  | 25            | 4,533,497                  |
| Solar        | Guinea-Bissau                                      | 10                 | 18%             | 15,768                 | 0.7647             | 12,058                   | 25            | 301,458                    |
| <b>TOTAL</b> |                                                    |                    |                 | <b>339,012</b>         |                    | <b>193,398</b>           |               | <b>4,834,955</b>           |

## E.2. Paradigm Shift Potential

Degree to which the proposed activity can catalyze impact beyond a one-off project/programme investment

### E.2.1. Potential for scaling up and replication (Provide a numerical multiple and supporting rationale)

In recent years the six target countries have grown economically. Evidence shows, however, that economic growth alone does not automatically increase access to sustainable, clean and reliable energy. Availability of inexpensive clean energy solutions in urban and rural areas, is a prerequisite for inclusive development. Individual solutions are not adequate to tackle the dual issues of energy poverty. In order to achieve a paradigm shift in these economies, systemic change is necessary in the availability of, and access to, affordable RE finance and technical support. The proposed programme will initiate this systemic change and contribute to a paradigm shift towards a low-emission development pathway through the mobilisation of the private climate finance at scale for the solar market. Accelerating private sector involvement in the solar sector will enable market transformation that is required for the paradigm shift.

Medium and large solar project financing in the six LDC countries is still at an early development stage. Private sector companies have limited opportunity to invest into solar projects in the region, and the current demand for debt of businesses is not met by local financial institutions due to the market barriers listed in section C.2. Given the ambitious solar generation targets in the region as specified in the NDC and other national strategies, systematic market failure must be avoided and the GCF involvement can make a critical contribution.

Under the programme, BOAD will address the market barriers by playing a catalytic role using a blended finance approach to provide long-term financing at affordable rates to increase the bankability of projects and attract private sector investments. The involvement of the GCF is required as it will allow a combination of senior debt, standby loans (tenor extension instrument) and TA to up to 215MW of solar projects to support their financial closures, a comprehensive and innovative financing package that can be replicated in other countries of the region.

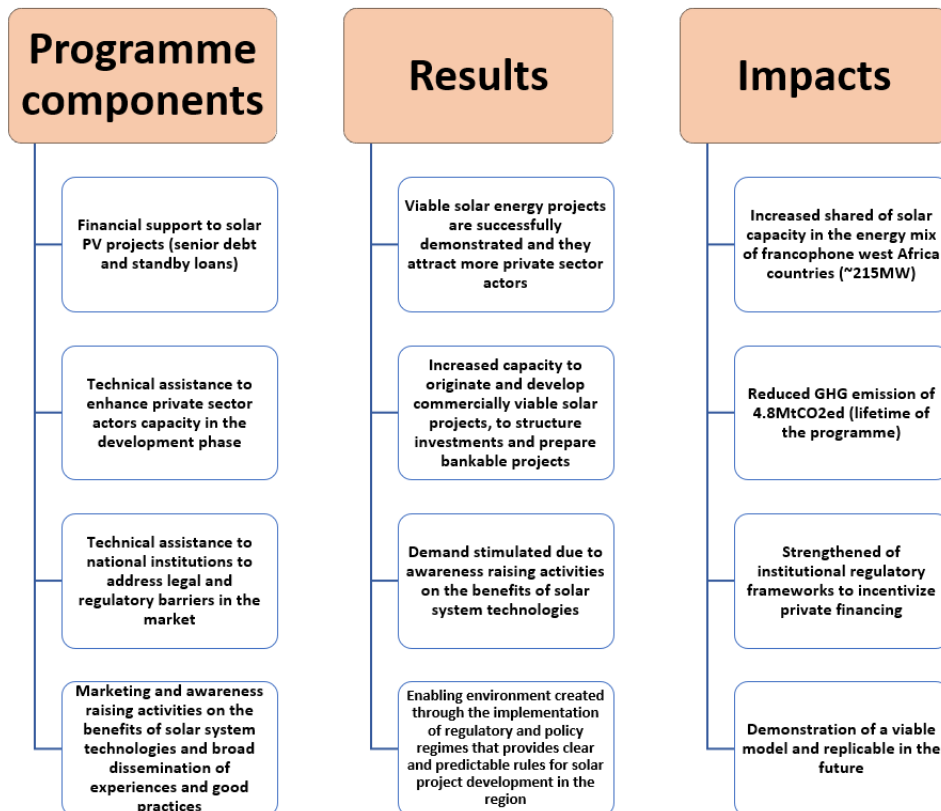
The programme will contribute to creating a replicable model for eventual scale-up of the Programme's success into the region as it is expected to showcase a viable model for all upcoming solar projects across the continent. Following the successful track-record in lending to these projects gained through the programme, it is expected that there will be a greater demand for financing solar projects and that local financial institutions will start using their own resources for climate lending.

Furthermore, supporting the implementation of regulatory and policy regimes that provides clear and predictable rules for solar project development in the region will ensure that the programme will have a lasting transformative ("paradigm shifting") impact on the medium-large scale solar sector in the region.

Transformational impacts will be achieved through the growth in commercially viable solar investments and the scaling up of financial products and services offered to the private sector in the region. This will essentially accelerate the GHGs emission reduction thereby contribute to national low-carbon development pathways.



**Figure 3: Theory of Change**



Assumptions: Technical and financial barriers exist for private sector participation to on-grid solar projects  
 Solar projects are not financially viable in the absence of a well-developed financial package  
 Short tenor and high price are the barriers of solar financing from commercial banks

## E.2.2. Potential for knowledge and learning

The Programme will disseminate technical know-how and good practices and consist of a significant amount of capacity development for private sector actors and national institutions. Transfer of expertise and skills is an important objective of this programme under Component 2. In this context, the following learning and know-how transfers and benefits will be achieved: private sector companies will benefit from technical support on how to structure their business plans to make them bankable. Trainings will include support to originate commercially viable solar related projects, to develop technical and financial feasibility studies and to prepare bankable projects.

BOAD will also benefit from the technical assistance to improve its capacity to develop, manage and implement the programme portfolio. TA will include:

- Support for the development of eligible and bankable solar projects: This will include support to build BOAD pipeline of projects, to assess the eligibility sponsors' projects, to screen project financial and technical feasibility studies, to review and improve projects' business models carried by project developers, etc.
- support in the implementation of the solar projects: This will include support to track progress of projects implementation and in the monitoring of project performance indicators (e.g. MW installed capacity, teqCO2 avoided, etc.); in the management of environmental and social risks and in the implementation of the gender action plan; support for marketing and communication activities (defining a marketing plan and providing communication tools to promote the programme with the aim to stimulate the demand for solar projects)
- support for the integration of climate change considerations into BOAD project cycle: This will consist in training BOAD on how to integrate mitigation and adaptation dimensions into the project activities.

The interventions will be bolstered by marketing and awareness raising activities among private sector actors on the benefits of solar system technologies, on the positive impacts that sustainable energy production have on energy security, environmental benefits, health and livelihoods. BOAD will develop, manage, and maintain organizational brand and execute a communication and outreach effort that promotes the objectives of the programme, disseminate good practices and experiences.

## E.2.3. Contribution to the creation of an enabling environment

The programme will contribute to creating an enabling environment for solar investments through targeted capacity building, awareness raising activities and connecting a diverse range of private sector stakeholders. Activities will include organizing various marketing and networking opportunities as well as consultation with local stakeholders in the solar technology supply chain (from manufacturers, suppliers and installers, to buyers, to finance providers, and to other stakeholders like the NGOs). By raising awareness among a diverse range of private sector stakeholders about the benefits of solar technologies at market levels and building the capacity of the private sector actors, the programme will stimulate demand for solar investments and directly contribute to the creation of the solar market in the target countries.

## E.2.4. Contribution to regulatory framework and policies

Although the target LDCs are well endowed with renewable energy resources, their share in the countries' energy mix is almost zero when large hydro power generation is excluded. The policy environment has not helped improve the situation significantly, but there has been a growing realisation over the last few years that drastic action is required to ensure adequate generating capacity in the region, and to develop both conventional and renewable energy sources.

A stable policy and enabling regulatory framework is a prerequisite to attract and capitalize on the interest from the private sector. Investors want transparency, longevity, certainty and consistency. Such frameworks would have to be developed. The ECOWAS Renewable Energy Policy (EREPP, pp.26) emphasizes that weak policy frameworks are a significant barrier to the development and proliferation of solar energy in the region. In the six target countries, the institutional, regulatory, legal, and tariff structures and frameworks are weak or insufficiently implemented. So far there are only a few incentives for private capital to invest in the RE sector in West Africa. Investments in RE projects have had a predominant share of Official Development Assistance (ODA) funding.

The six target countries of this programme are characterized with similar market and policy barriers. Constraints vary from lack of clear institutional roles between different agencies, weak planning capacity, the absence of a clear framework for PPPs and IPPs, the mismatch between tariffs in urban and rural areas (with a necessary harmonization of tariffs), and capacity to channel funds for systematic large-scale projects, lack of legal structures and systems to push the market and open up the sector, the lack of legal and binding targets which is often the hindrance for progress in the RE sector, and lack of incentive mechanisms.

To address the regulatory and legal barriers in the market, BOAD will carry out policy dialogue activities that will be informed by the activities of the proposed investments. The rationale of sub-component 2.2 is to remove the market barriers to the development of solar projects by supporting the implementation of regulatory and policy regimes that provides clear and predictable rules for solar project development in the region.

The aim of sub-component 2.2 is to prepare the way for the entry of solar private investment and to create clarity and a stable reliable environment for solar investment. TA under this sub-component will be focused on aspects of the enabling environment that have an immediate impact on stimulating solar private sector investment. Support can include the creation or revision of a simplified regulatory frameworks for Feed-in-tariffs for on-grid solar solutions, support on standard PPAs, support on IPP tenders and procurement process, on tariff structure, on a framework of financial and technical incentives and capacity building.

## E.3. Sustainable Development Potential

### Wider benefits and priorities

#### E.3.1. Environmental, social and economic co-benefits, including gender-sensitive development impact

The programme is aligned with, and will broadly contribute to achieving, a number of the Sustainable Development Goals (SDGs) in each target country including:

- Goal 1. No Poverty – end poverty, in all forms, everywhere
- Goal 5. Gender Equality – achieve gender equality and empower all women and girls
- Goal 7. Affordable and clean energy – ensure access to affordable, reliable, sustainable and modern energy for all
- Goal 8. Industry Innovation and Infrastructure - build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation;
- Goal 10: Reduced Inequalities - reduce income inequality within and among countries
- Goal 11: Sustainable Cities and Communities - make cities and human settlements inclusive, safe, resilient and sustainable
- Goal 12: Responsible Consumption and Production - ensure sustainable consumption and production patterns

- Goal 13: Climate Action - Take urgent action to combat climate change and its impacts by regulating emissions and promoting developments in renewable energy
- Goal 17: Partnerships for the Goals - Strengthen the means of implementation and revitalize the global partnership for sustainable development

The programme is expected to deliver wider environmental and socio-economic benefits on top of adding solar generation capacity to the grid and diversifying the energy mix in the target countries. Such co-benefits will contribute to achieving the SDGs as outlined below.

#### **Economic and social co-benefits**

The six target countries have abundant renewable resources that present commercial opportunities to generate clean electricity at relatively low cost. The programme will contribute to increasing the share renewables have in the respective national energy mix. This will reduce the requirement to import fossil fuel carriers and therefore reduce existing trade deficits that are caused by the import of refined oil products and secure energy supply through diversified energy sources. The increased, stable, and affordable energy access will bring about a positive impact to the sustainable growth of the region.

In addition, the programme will create new job opportunities during the construction and operations of solar power plants. Among other development impacts, this programme will contribute to the economy by creating about 500 green jobs during the construction phase ranging from highly skilled (field technicians, operators, corporate staff, etc.) to unskilled labour work (support and service), and additional ~200 jobs for operations and maintenance. This will be a starting point of building the ecosystem of technicians and skilled experts in solar technologies in the LDC countries of West Africa.

#### **Environmental co-benefits**

The programme will have the following key environmental co-benefits. The reduced reliance on fossil-fuel based generation will lead to the cleaner air quality by reducing the emission of pollutants including GHGs. Finally, the stable supply of electricity will reduce the dependency on charcoal and firewood for cooking, contributing to the reduction of deforestation and associated carbon emissions. This will have additional positive health impact to the population.

#### **Gender co-benefits**

Improved and more reliable access to electricity will have strong co-benefits for gender-sensitive development. The programme will take into account the different ways that women and men access energy, and the role that energy can play in increasing opportunities for women from the initial design of the programme and through the provision of technical support on the promotion of gender equality throughout the programme lifetime. The programme will add significant value in the target sector where gender equality is poorly addressed. By mainstreaming gender throughout the programme lifetime, the programme will be in line with the Gender Policy and the Gender Action Plan of GCF. Activities targeting women's participation and training as part of component 2 will make sure that the program is gender-sensitive and the benefits are shared among men and women. More information can be found in the Gender Assessment and Action Plan.

## E.4. Needs of the Recipient

Vulnerability and financing needs of the beneficiary country and population

### E.4.1. Vulnerability of country and beneficiary groups (Adaptation only)

About half of the population in the West Africa Region do not have access to electricity. All target countries are classified as Least Developed countries<sup>33</sup> and according to the ND-GAIN Country Index, they are all among the most vulnerable countries to climate change. The target countries face interrelated challenges of energy access, energy security and high cost of electricity. This is attributed to the high dependence on fossil fuels, the insufficient availability of public funds to invest in energy infrastructure to meet the growing electricity demand, the inability to attract private capital in power generation at scale, and the inefficiencies along the value chain. This situation hampers the social, economic, and industrial development of the region.

The target LDCs' energy mix mostly comprises diesel, heavy fuel oil and hydropower. While they are endowed with high solar irradiation especially in the Sahel region, the installed solar PV capacity in the target countries was around 70 MW at the end of 2017, representing only 5% of the total installed electricity generation capacity. Given the continuous decline in the cost of solar electricity generation, the target LDCs would benefit from investing massively in solar technologies to reduce its dependence on expensive fossil fuels for power generation, while supporting a shift towards cheaper, cleaner and reliable sources of electricity generation. With large financing needs over the sector's entire value chain, the solar sector increasingly relies on the private sector to deploy grid-connected solar power generation projects.

The proposed programme will address the needs of the beneficiary countries by improving the access to clean and reliable energy for approximately 2.9 million people. In creating favourable market environments for investment in the solar sector, the programme is expected to increase by 400% the existing total solar installed capacity which will lead to sizable GHG emissions reductions (approx. 4.8 million tCO<sub>2</sub>e over the 25-year average lifetime of the investments) and to green job creation and private sector business growth. Better access to green finance will enable private sector actors to have access to new markets related to green growth. Potential co-benefits include increased access to more sustainable energy, more affordable energy and reduced costs. The programme will also directly contribute to achievement of countries NDCs and the SDGs.

*Table 17: Ranking of the beneficiary countries in terms of poverty and vulnerability to climate change*

| Country              | LDC | ND-GAIN<br>INDEX<br>Ranking <sup>34</sup> | Score | Vulnerability |       | Readiness |       |
|----------------------|-----|-------------------------------------------|-------|---------------|-------|-----------|-------|
|                      |     |                                           |       | Ranking       | Score | Ranking   | Score |
| <b>Benin</b>         | Yes | 155                                       | 35.5  | 163           | 0.577 | 147       | 0.288 |
| <b>Burkina Faso</b>  | Yes | 164                                       | 34.6  | 166           | 0.582 | 157       | 0.274 |
| <b>Guinea-Bissau</b> | Yes | 171                                       | 32.2  | 176           | 0.625 | 158       | 0.268 |
| <b>Mali</b>          | Yes | 166                                       | 33.6  | 173           | 0.614 | 149       | 0.286 |
| <b>Niger</b>         | Yes | 175                                       | 30.5  | 181           | 0.680 | 146       | 0.290 |
| <b>Togo</b>          | Yes | 143                                       | 38.2  | 143           | 0.540 | 137       | 0.303 |

<sup>33</sup> United Nations, List of Least Developed Countries (as of March 2018):

[https://www.un.org/development/desa/dpad/wp-content/uploads/sites/45/publication/ldc\\_list.pdf](https://www.un.org/development/desa/dpad/wp-content/uploads/sites/45/publication/ldc_list.pdf)

<sup>34</sup> <https://gain.nd.edu/our-work/country-index/rankings/>, scores for 2016

**E.4.2. Financial, economic, social and institutional needs**

**Economic and social development level of the country and affected population**

In recent years the six target countries have grown economically. Evidence shows, however, that economic growth alone does not automatically increase access to sustainable, clean and reliable energy. Availability of inexpensive clean energy solutions in urban and rural areas, is a prerequisite for inclusive development. Individual solutions are not adequate to tackle the dual issues of energy poverty. In order to achieve a paradigm shift in these economies, systemic change is necessary in the availability of, and access to, affordable RE finance and technical support. The proposed programme will initiate this systemic change and contribute to a paradigm shift towards a low-emission development pathway through the mobilisation of the private climate finance at scale for the solar market. Accelerating private sector involvement in the solar sector will enable market transformation that is required for the paradigm shift.

*Table 18: Overview of the economic and social development in the target countries*

| Country              | Overview of the economic and social development in the target countries                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Benin</b>         | <p>In 2017 Benin had a population of 11,175,692 million, with a GDP per capita (current US\$) of 829. The country performed well in terms of economic growth rates with 5,6% while the inflation stood at around 0.1%. Important economic and structural reforms financed by the IMF and World Bank have helped the country sustain this growth rate over the past decade. However, poverty remains widespread throughout the country and the economy is vulnerable, due to a lack of diversification and exogenous shocks.</p> <p>Politically, Benin continues to benefit from a stable and democratic regime, though terrorist acts occur in the north of the country. President Patrice Talon has pledged to tackle terrorism and to promote national unity.</p> <p>Over the past decade, Benin has considerably reinforced its macroeconomic stability, which resulted in a fast-increasing growth rate, though it is modest. GDP growth stood at 3,7% between 2007 and 2011, it increased to 5,4% in 2012 to then reach 5,6% in 2013. It is expected to remain steady.</p> <p>However, economic growth has not contributed to reduce poverty rates significantly, resulting from both the uneven distribution of resources and the rapid growth rate of the population. The share of the population living under the national poverty line has increased, from 37,5% in 2006 to 40.1% in 2015. To remedy this, the government has developed and implemented its third Stratégie de Croissance pour la Réduction de la Pauvreté (2011-2015), which aims to make Benin an emerging country by 2025. In order to achieve its objectives, the government ought to sustain economic growth on the medium-term while meeting the MDGs targets. The GoB counts on its agricultural potential as well as its central position for regional trade, but developing affordable energy is the key to achieving these targets.</p> |
| <b>Burkina Faso</b>  | <p>In 2017 Burkina Faso had a population of 19.2 million, a GDP of USD 12.87 billion, an estimated economic growth rate of 6.7% and 40.1% of the population lived below the national poverty line in 2014. In 2016, national access to electricity was estimated at 19.2%, with an urban electricity access rate of 60.7% of total urban population and a rural electricity access rate of 0.8% of total rural population. After a period of social and political unrest, the economy was expected to bounce back strongly in 2018, driven by extractive industries and public investment. The Africa Development Bank (2017) concluded that the country needs to underpin this revival by improving its ability to absorb investment costs. Furthermore, economic reforms under the 2011-20 plan for the industry, commerce and small producers sector have created more entrepreneurs, but new industrial sector firms are in extractive industries rather than agro-food. Significant physical, technical and socioeconomic constraints have been limiting the performance of the agricultural sector.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Guinea Bissau</b> | <p>Guinea Bissau is one of the most fragile countries in Sub-Saharan Africa, with a population of around 1.8 million (2017). It has a long history of political and institutional instability and is highly coup-prone, which results in significant difficulties creating a favourable environment for investment and economic development.</p> <p>The country's economy continues to grow despite the political turmoil, with a real GDP growth rate above 5.9% in 2017. However, because of the high political and institutional instability the economic outlook remains uncertain and reducing poverty rates as well as promoting sustainable development will be challenging without political incentives.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Niger</b>         | <p>In 2017 Niger had a population of 21.47 million, a GDP of USD 8.1 billion and an estimated economic growth rate of 4.9%. As of 2014, 44.5% of the total population lived under the national poverty line. In</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |



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|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             | <p>2016, national access to electricity was estimated at only 16.2%, with an urban electricity access rate of 65.4% of total urban population and a rural electricity access rate of 4.7% of total rural population.</p> <p>The government of Niger has long recognized that lack of energy access seriously handicaps development, and efforts to remedy this problem need to be scaled up. Indeed, in many of these successful cases elsewhere, energy intervention in the social and economic sectors namely health, water, agriculture and education were made as part of a comprehensive development strategy.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Mali</b> | <p>In 2017 Mali had a population of 18.5 million, a GDP of USD 15.28 billion, and an estimated economic growth rate of 5.3%. In 2016, national access to electricity was estimated at 35.1%, with an urban electricity access rate of 83.6% and a rural electricity access rate of 1.8%.</p> <p>The economy of Mali is based to a large extent upon agriculture, with a mostly rural population engaged in subsistence agriculture. A security crisis prior to 2015 slowed economic growth however, medium-term macroeconomic prospects are good, with overall growth forecast as 5.0% in 2017, driven partly by more public investment and foreign aid and by the agricultural and service sectors. There are considerable constraints holding back economic development in Mali, particularly factors facing the private sector. Mali's financial system is under-developed and does not allow the financing and thus the emergence and development of dynamic employment-generating private sector. Steps have been made towards creating these enabling environments in 2008 where a Financial Sector Development Strategy was developed. One aim of the FSDS was to broadening access to financial institutions and products, in particular in rural areas. The Poverty Reduction and Growth Strategic Framework (GPRSP) cover all relevant country development policies and strategies 2012-2017. The GPRSP will be updated in the coming years. Economy-wide constraints help explain the slow progress in financial deepening and financial sector outreach. These include weaknesses in the legal framework and judicial system that result in a slowdown in loan recovery, shortcomings in the financial reporting of private sector actors, and lack of collateral for bank loans due to problems with land tenure and land registry.</p> <p>Also, poor physical access and infrastructure, including a limited electricity supply, poor road conditions, and distance from central bank offices, which deter rural branch expansion, along with difficulty in finding skilled human resources, increase the costs of extending credit or offering financial services. Owing to these unfavorable conditions, Mali's economy remains under-diversified and the country is still over-dependent on cotton and agriculture in general as well as extractive/mining industry. It is therefore very exposed to exogenous factors such as the price of raw materials, the euro and dollar exchange rate, the price of oil and climate change.</p>                      |
| <b>Togo</b> | <p>In 2017 Togo had a population of 7.7 million, a GDP of USD 4.8 billion, and an estimated economic growth rate of 5.6%. In 2015, 55.1% of population lived below the national poverty line. In 2016, national access to electricity was estimated at 46.9%, with an urban electricity access rate of 87.4% and rural electricity access rate of 19.4%. By 2030, Togo will have about 9.5 million inhabitants, 5.9 million of whom live in urban areas, compared with 3.5 million people living in rural areas. This demographic growth and above all the change in the distribution of the population between rural and urban areas will have a very strong impact on the volume of energy needs and on the structuring of energy consumption in 2030. The rate of access to electricity in Togo is in progression (from 17% in 2000 to 46.9% in 2016) but with huge disparities between urban areas. Renewable energies (solar, wind, etc.) are present in the electricity generation capacity of the country. The productive economy is dominated by agriculture and in 2015, still 90.1% of the total population lived under \$5.50 a day, making it one of the world's poorest countries. Togo's Strategy for Boosting Growth and Promoting Employment (SCAPE—Stratégie de Croissance Accélérée et de Promotion de l'Emploi) offers a medium-term development framework for implementing the Government's General Policy Statement, the Millennium Development Goals (MDGs), and the Government's vision for making Togo an emerging economy in 15 to 20 years.</p> <p>The Togolese financial system plays a minor role in mobilizing resources for the economy, particularly private sector actors, with total financial system assets amounting to about 30 percent of GDP. The cost of energy and communications since the 1990s, as well as market weakness and limited access to funding, are serious obstacles to the growth of towns and cities and to the country's structural transformation and poverty reduction. The Government intends to establish an Autonomous National Investment Fund (FINA—Fonds d'Investissement National Autonome) for financing private sector actors, particularly where young people are involved. Other measures taken to assist the private sector and allow it to act as an engine of growth pertain to gradually establishing banking and financial guarantee mechanisms in support of private sector actors in their search for financing at preferential rates, and capacity-building for microfinance institutions.</p> |

## Absence of alternative sources of financing

The six target countries are characterized by low market share of RE technologies (excluding large hydro). Despite the substantial economic, social and environmental co-benefits of RE investment, these have not been prioritized by private sector actors due to a number of barriers described in section C.2, in particular the limited local debt funding with the required tenors and interest rates.

Given the large investment needs in the sector, a significant volume of investments into the solar market in West Africa is required. BOAD's strategic objective is to increase its portfolio of investments in the solar directly to private sector actors, but so far this has been low because of the lack of alternative source of public sector funding available. Currently, only the GCF is able to provide such finance within the given timeframe. Without GCF's contribution, it would take considerably longer for existing market barriers to be removed. In the meantime, access to finance for climate investments would remain limited to a certain number of borrowers and remain too costly for many others. GCF's contribution will unlock the considerable solar potential in LDC countries from Francophone West Africa by providing a combination of concessional lending and grant for technical assistance. The total solar installed capacity in the NDC countries from West Africa will be increased twofold thanks to the GCF contribution.

## Need for strengthening institutions and implementation capacity.

In the renewable energy sector, the institutional, regulatory, legal, and tariff structures and frameworks are weak or insufficiently implemented. So far there are only a few incentives for private capital to invest in the RE sector in West Africa. Investments in RE projects have had a predominant share of ODA funding. The policy environment has not helped improve the situation significantly, but there has been a growing realisation over the last few years that drastic action is required to ensure adequate generating capacity in the region, and to develop renewable energy sources to make them an important share of the region's energy and electricity mix.

The six target countries of this programme are characterized with similar market and policy barriers. To address the regulatory and legal barriers in the market, BOAD will carry out policy dialogue activities that will be informed by the activities of the proposed investments. The rationale of component 2.2 is to remove the market barriers to the development of solar projects by supporting the implementation of regulatory and policy regimes that provides clear and predictable rules for solar project development in the region. The aim of component 2.2 is to prepare the way for the entry of solar private investment and to create clarity and a stable reliable environment in for solar investment. TA under this sub-component will be focused on aspects of the enabling environment that have an immediate impact on stimulating solar private sector investment. Support can include the creation or revision of a simplified regulatory frameworks for Feed-in-tariffs for on-grid solar solutions, support on standard PPAs, support on IPP tenders and procurement process, on tariff structure, on a framework of financial and technical incentives and capacity building.

## E.5. Country Ownership

Beneficiary country (ies) ownership of, and capacity to implement, a funded project or programme

### E.5.1. Existence of a national climate strategy and coherence with existing plans and policies, including NAMAs, NAPAs and NAPs

#### Alignment with country's national climate strategy

Each country in the target region has adopted national climate strategies which highlight the crucial role of RE deployment in mitigating climate change. The national strategies also recognize the private sector's important role in deploying RE investments to support the NDCs, and its potential to contribute to the paradigm shift to a low-carbon, climate resilient economy.

As indicated in the table below, each country has the promotion of RE sources high on their political agenda. The proposed programme aims to provide EUR 117 million of senior debt and standby loans (tenor extension instrument) to selected solar projects in LDCs from West Africa region which are expected to result in the reduction or avoidance of approximately 4.8 million tCO<sub>2e</sub> during the lifetime of the programme. The proposed programme therefore is in line

with national climate policies and action plans of the target countries and it will help fill the current investment shortfall to achieve West Africa's NDCs.

*Table 19: Common priorities as stated in the national climate strategies of the target countries*

| Country      | National targets                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Basis for Execution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Benin        | <p>Mitigation (all targeted sectors in BAU scenario):<br/>The measures envisaged in the sectors of Energy and Agriculture are likely to contribute to reduce the cumulative GHG emissions (without LULUCF) by approximately 49.49 Mt CO<sub>2</sub> eq in the BAU scenario, that is a reduction of 16.17 % over the period 2021-2030, including 12.55% of conditional contribution and 3.62% of unconditional contribution.</p> <p>Measures relating to the energy sector:<br/>Developing electric power generation using natural gas and RE:</p> <ul style="list-style-type: none"> <li>- Construction of thermal biofuel power plants (500 MW by 2030)</li> <li>- Construction in Port of Cotonou of a Liquified Natural Gas regasifying floating terminal</li> <li>- Operating with natural gas the installed thermal generation capacity</li> <li>- Construction of hydroelectric power plants of Adjarala (147MW) and Dogo (128MW)</li> <li>- Establishing solar PV farms for 95MW</li> <li>- Structuring a biomass fuel investment for 15MW</li> </ul> <p>Increasing households' access to electric lighting in place of kerosene lighting:</p> <ul style="list-style-type: none"> <li>- Electrification of localities by connecting them to the network (600 localities between 2021 and 2030)</li> <li>- Promoting the access of 424,000 kerosene-lighting using households to electric lighting in the localities which will be connected to the grid</li> <li>- Pursuing actions for EE in all sectors</li> </ul> <p>Promoting low wood-energy consuming technologies<br/>Promoting partial substitution of firewood-energy consumption with butane gas<br/>Addressing current shortcomings as regards energy databases<br/>Emissions reduction for the energy sector:<br/>The implementation of these measures will contribute to reduce the cumulative GHG emissions in this sector in the BAU scenario by 23,35 Mt CO<sub>2</sub> eq over 2021 to 2030 period, that is 11,51 % including 9,53% of conditional contribution and 1,98 % of unconditional contribution. The avoided cumulative emissions are distributed as follows:(i) power generation 7,80 % including 7,06 % of conditional contribution;(ii) promotion of electric lighting in households 3,62 % including 2,42 % of conditional contribution;(iii) saving in wood-energy by promoting cleaner cookstoves 0.084% including 0.056 % of conditional contribution;(iv) partial substitution of charcoal with butane gas 0,001%.</p> | <p>INDCs<br/>CCNUCC (Convention-Cadre des Nations Unies sur les Changements Climatiques)<br/>FNEC (National Fund for the Environment and Climate)<br/>MEGCCRPRNF (Ministère de l'Environnement Chargé de la Gestion des Changements Climatiques du Reboisement et de la Protection des Ressources Naturelles et Forestières)<br/>CNCC (Comité National sur les Changements Climatiques)<br/>CMEICB (Commission de Modélisation Économique des Impacts du Climat et de l'Intégration des Changements Climatiques dans le Budget Général de l'État</p> |
| Burkina Faso | <p>Mitigation: Basis year 2014<br/>Reduction of emissions at the 2030 horizon:</p> <ul style="list-style-type: none"> <li>- 21,574.63 GgCO<sub>2</sub> eq.</li> <li>- 18.2% reduction</li> <li>- Investment cost: 1,840,953,571 US\$</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>INDC (2015)<br/>National Climate Change Learning Strategy (SNACC) (2017)<br/>Programme National de Développement Economique et social (PNDES) (2016)</p>                                                                                                                                                                                                                                                                                                                                                                                          |

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                   |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | <p>In line with SE4ALL objectives, Burkina Faso aims to achieve:</p> <ul style="list-style-type: none"> <li>• Universal access to modern energy services.</li> <li>• Double the rate of improvement of energy efficiency.</li> <li>• Double the share of renewable energy in the world energy mix.</li> </ul> <p>Burkina Faso in its INDC analyses scenarios for reductions according to contributions. In an Unconditional scenario for 2030 aims to reduce 6.6% of emissions. In a Conditional Scenario, the goal is 11.6%.</p> <p>In 2007, the agricultural sector contributed 88% of the national GHG emissions.</p> <p>In the conditional scenario, Burkina Faso expects to reduce GHG emissions in electricity production by 5% in 2020, 4% in 2025 and 4% in 2030.</p>                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                   |
| Guinea-Bissau | <p>Guinea-Bissau is an absolute sink of GHG, given the high potential for sequestration of its forest sector. Main mitigation measure to be adopted is related to reforestation. The second sector that contributes more to GHG emissions is the energy sector. In this sector an increase in electric power capacity of at least 90MW by 2020 using petroleum products (diesel and HFO) is planned.</p> <p>Guinea Bissau's key aspects/key mitigation measures to meet its energy Intended Nationally Determined Contributions (INDCs)</p> <p>Conduct studies on the energy potential and set the energy development incorporating the largest possible potential of RE in the energy mix</p> <p>Attain 80 per cent renewable energy in the national energy mix by 2030;</p> <p>Improve energy efficiency by reducing energy losses up to 10 per cent in the 2030-time span;</p> <p>Reach 80 per cent of universal access to electricity by 2030</p> <p>Investment required: at least 200 million USD by 2020 and 500 million USD between 2020 and 2030</p> | <p>INDCs</p> <p>National Poverty Reduction Strategy (PRSP II)</p> <p>National Strategic Plan – TERRA RANKA 2015-2025</p> <p>National Action Plan for Climate Change Adaptation (NAPA, 2006)</p>                                                                                                                                                                   |
| Mali          | <p>Significant RE potential, but GHG emissions in progression: 83.63% between 2000 and 2012, accounted for by the residential use of wood (82.20%). Between 2007 and 2014, GHG emissions from production and consumption of energy progressed from 3,434 kTeq CO<sub>2</sub> to 5,268 kTeq CO<sub>2</sub>, or 6.44% annually.</p> <p>Mitigation measures:</p> <ul style="list-style-type: none"> <li>• Reduction of GHG of 29% for agriculture, 31% for energy, and 21% for forestry and land use. Overall, the level of reduction expected for 2030 is of 27%.</li> <li>• Installation of at least 100 MW renewable energy sources, with a goal of having 10% of renewables in the energy matrix for 2020.</li> </ul> <p>For the energy sector:</p> <ul style="list-style-type: none"> <li>- Large-scale valorisation of RE for an amount of 258 million USD</li> <li>- Manantali II project between 2016 and 2021 will cost 150 million USD</li> </ul>                                                                                                     | <p>INDC (2016)</p> <p>NAPA (2007), National Policy for Climate Change (2011), Strategic Framework for a Resilient and Green Mali.</p> <p>Mali has set in place several sectoral policies including, National Policy for the Protection of the Environment, Energy National Policy, Water National Policy.</p> <p>Global Summit for Climate in New York (2014)</p> |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                 |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | <ul style="list-style-type: none"> <li>- Villages' electrification project using RE will cost 7.2 million USD between 2015 and 2020</li> <li>- Kénié hydroelectric power plant implemented between 2015 and 2020 will cost 165 million USD</li> </ul> <p>GHG emissions reductions: 4,750 kTeq CO<sub>2</sub> by 2020, 6,336 kTeq CO<sub>2</sub> by 2030, or 31.6% by 2030.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                 |
| Niger | <p>Reduction goals:</p> <ul style="list-style-type: none"> <li>• Unconditional reduction of 2.5% (BAU 2020) and 3.5% (2030) of GHG.</li> <li>• Conditional reduction 25% (BAU 2020), and 34.6% (2030) of GHG.</li> </ul> <p>Energy: 2000 base year: Energy sector accounts for 8.5% of GHG emissions<br/>Mitigation potential of the sector: 700 GgCO<sub>2</sub>e or 0.7% of national emissions.<br/>Note: a reduction in this emission is noted in the electricity for all scenario beginning in 2025, explained by the growth of production from gas and by the entry of nuclear power production into the system.</p> <ul style="list-style-type: none"> <li>• Electricity: Improvement of the rate of access to electricity (overall, exceed 10% in 2010, 60% in 2030, of which 47% to 100% is in the urban zone and 0.4% to 30% in 2030 is in the rural zone.</li> <li>• Cooking energy: reduction in the demand for wood energy per inhabitant by the mass spread of improved cook stoves, with a rate of penetration of 100% in urban areas and 30% in rural areas; promotion as domestic gas of biogas and biofuels at both the industrial and family level. Spread of multifunctional platforms.</li> </ul> <p>Renewable energy:</p> <ul style="list-style-type: none"> <li>• Unconditional mitigation options: rural electrification, replacement of wood energy, management of the sector concerned with the demand for and transformation and dissemination of renewable energies, promoting solar PV for pumping and electrification</li> <li>• Exceed a capacity of 4 MW in 2010, 250 MW in 2030, 130 MW of which comes from the Kandadji hydroelectric plant and 20°MW comes from wind energy (currently 0.035 MW).</li> <li>• Double the rate of energy mix to reach 30% energy mix in the primary and final energy balance.</li> </ul> | <p>INDC (2015)<br/>Strategic Framework for Sustainable Land Management (SF-SLM)<br/>Economic and Social Development Plan (2016-2020)<br/>Sustainable Development and Inclusive Growth Strategy Niger 2035 (SDDCI)<br/>National Policy on Climate Change (PNCC)<br/>National Strategy and Plan of Action for Climate Change and Variability (SNPA-CVC).</p>                      |
| Togo  | <p>With a 2010 baseline, the Unconditional reduction target for 2030 is 11.14%, and the conditional target is of 31.14%.<br/>Funding needs: US\$ 3.54 billion (Adaptation = 1.54; Mitigation = 1.10; Tech transfers = 0.5; Capacity building = 0.4)<br/>Targets: The conditional target for additional GHG emissions reduction, according to the most ambitious scenario, is estimated at 20% compared to the dynamic BAU. The conditional target for the total reduction would therefore be 31.14% in 2030, compared to the projections if no measures were to be applied.<br/>GHG mitigation measures and options:<br/>In the energy sector:</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <p>INDC (2015)<br/>Togo Vision 2030<br/>Strategy for accelerated growth and employment promotion (SCAPE) 2013-2017<br/>National Environmental Action Plan (NEAP)<br/>National Sustainable Development Strategy (NSDS)<br/>National Strategy for Reducing the Risk of Catastrophes in Togo<br/>National Programme for Reducing GHG from Deforestation and forest Degradation</p> |



|  |                                                                                                                                                                                                                                                                                                                                                                   |  |
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|  | <ul style="list-style-type: none"> <li>- Solar electricity: emphasis will be placed on the introduction of solar equipment in households and on capacity-building for the various actors concerned.</li> </ul> <p>Thanks to measures already in progress, Togo will unconditionally reduce its emissions by 11.14% compared to the baseline scenario in 2030.</p> |  |
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## Benin

In order to meet its targets and achieve a climate-resilient economy, Benin has introduced a Commission Nationale du Développement Durable (CNDD). The country has pledged to combine climate-resilient, sustainable development and poverty reduction via the implementation of mitigation and adaptation measures. As 68% of Benin's GHG emissions are accounted by the agriculture sector and the sequestration capacity keeps on decreasing, mitigation measures aim at reducing cumulated GHG emissions by 21,4% based on the BAU scenario by 2030 (conditional contribution). The unconditional contribution involves reducing GHG emissions by 3,5% by 2030 based on the BAU scenario. Another measure envisions to increase the sequestration potential by 5,7% based on a similar scenario, through the reduction of the deforestation rate by 41,7%. This would enable Benin to reduce GHG emissions by 110 Mt E-CO<sub>2</sub> over the 2021-2030 period.

Benin has developed a set of mitigation measures relating to the energy sector. Among others, Benin has pledged to increase the share of renewable energies in the country's energy mix, using natural gas, biofuel, hydropower and solar power; and to increase the national electricity access rate by connecting 600 localities to the grid. These measures demonstrate the country's commitment to invest in clean energy to contribute to the reduction of GHG emissions and follow a low-carbon development pathway.

## Burkina Faso

Burkina Faso's National Assembly on the Environment and Sustainable Development, held in November 2011, strongly recommended the development of a National Sustainable Development Policy (NSDP) accompanied by a law. Prepared in 2013, the NSDP was an effective framework for the Strategy for Accelerated Growth and Sustainable Development (SAGSD). This economic framework document, together with "Outlook Burkina 2025" and policy framework instruments, contributes to placing the concept of sustainability at the heart of public action and the activities of other non-state actors (technological and financial partners, civil society organisations, non-governmental organisations and the private sector) (Government of Burkina Faso, 2015).

On the mitigation side, three scenarios have been considered for evaluating the emissions trends and the possible reductions on the basis of a reference situation and the potential for financing:

- A "trend" scenario (Business as Usual - BAU), which corresponds to continuation of the past under the assumption that economic development continues without interruption.
- An "unconditional" scenario, taking into account all the public policies adopted after 2007, technological developments and recent studies, with financing that has been acquired or is being acquired.
- A "conditional" scenario that takes into account all the mitigation projects that have been developed and/or are being developed, but without any acquired financing.
- On the mitigation side, the measures foreseen in the country's INDC are to double the share of renewable energies in the energy mix, and to provide universal access to electricity by 2030, in line with SE4ALL objectives.

## Guinee-Bissau

Guinea-Bissau has integrated mitigation and adaptation measures as integral parts of its National Agricultural Investment Plan – Programme 4.7 – Adaptation of agriculture to climate change. A wide array of policies was developed and adopted, among which a NREAP (National Plan for Renewable Energy - 2014) and the Sustainable Financing Strategy of Adaptation to Climate Change in the short, medium and long-term (2013). The country also aims at mainstreaming climate change into strategies and policies in order to achieve the MDGs (Millennium



Development Goals). Guinea-Bissau is an absolute sink of GHG, because of the high sequestration potential of its forests, according to the GHG Inventory (2006) and the CARBOVEG-GB (2010). The successful implementation of the following measures will largely depend on IFIs and development partners' capacities to sufficiently fund them.

As deforestation accounts for the majority of GHG emissions, the main mitigation measure to be adapted is related to reforestation. This contribution includes: i) Establish and schedule a new forestry policy to enhance a socio-economic balance that meets the needs of communities, ii) conduct studies on the energy potential of the country and set energy development to make the economy increasingly resilient to climate change and iii) develop and establish a legal framework through a national strategy for long-term, low-carbon development.

Mitigation measures with regards to the energy sector are aligned with SE4ALL objectives and Guinea-Bissau's NREAP. These are to attain an 80% share of renewable energies in the national energy mix, to improve energy efficiency by reducing energy losses by 10%, and to reach universal access to electricity by 2030.

### **Mali**

In terms of mitigation of climate change, Mali has developed an unconditional scenario (where Mali is the only country financing climate projects). The forestry sector plays a major role due to the level of sequestration. The GHG reduction ambition levels are 29% for agriculture, 31% for energy and 21% for forests, taking into account the baseline of 2015. The overall cost for mitigation goals is \$34.68 billion (Republique du Mali, 2015). While the focus of mitigation measures is primarily on the agriculture and forestry sectors, Mali has developed a set of measures with regards to the energy sector. These measures place an important emphasis on the diversification of the country's energy generation mix and the increase of the share of renewable energies.

### **Niger**

Since Niger is a non-Annex I Party to the UNFCCC, it has no quantitative obligations with respect to emission reductions. However, Niger is contributing to the reduction of worldwide climate change impacts through a double results approach. In 2000, emissions represented 2.8 tons per inhabitant and 0.07% of the worldwide emissions of CO<sub>2</sub>. Niger's ambition is to limit its emissions to 2.1% in 2030.

The objectives of Niger's INDC are to assure food security, combat poverty and contribute to the reduction of world greenhouse gas emissions. For Niger's INDC, the adaptation options considered top priority are those that will permit the higher co-benefits with respect to climate change mitigation, particularly good adaptation practices and techniques which, at the level of the country's six regions, will permit carbon sequestration and a reduction of GHG emissions at the same time. These adaptation options have already been clearly defined in the existing strategic frameworks, such as the Economic and Social Development Plan (PDES 2012-2015 and 2016-2020), which flows from the Sustainable Development and Inclusive Growth Strategy - Niger 2035 (SDDCI), the 3N Initiative (Nigerans Feed Nigerans), the National Policy on Climate Change (PNCC), the Strategic Framework for Sustainable Land Management (SF-SLM), and the National Strategy and Plan of Action for Climate Change and Variability.

The National Communications indicate that the AFOLU (agriculture/animal husbandry and land use) and energy sectors represent on average 89% and 9% respectively of Niger's total GHG emissions. Given the potential of the country's resources, the national concerns are focussed on issues related to adaptation, particularly in the AFOLU sector, and on issues related to mitigation, principally in the AFOLU sector and the energy sector (transportation and residential and industrial energy). The strategy is based on the vision of climatically intelligent agriculture and on access to modern energy services for everyone in 2030. However, while the AFOLU and energy sectors are the priority action areas of the INDC on climate change in Niger, the implementation of the INDC actually represents cross-sectoral support for all sectors of the economy (Government of Niger, 2015).

Regarding mitigation measures for the energy sector, Niger has emphasised the need to use renewable energies applied to the agricultural sector, for pumping and rural electrification. In addition, Niger has a target to double the overall share of renewable energy in the energy mix to reach 30% in the primary and final energy balance.

## Togo

The national strategic orientations underlying sectoral programmes and policies can be found in the document Togo Vision 2030, currently under production, and in the Strategy for Accelerated Growth and Employment Promotion (SCAPE) 2013-2017. Togo's SCAPE defines a development framework for the medium term in order to fulfil its General Policy Declaration (DPG). It is based on the Millennium Development Goals (MDGs) covering the 2006-2015 period. It aims to transform Togo into an emerging country within the next 15 to 20 years.

For several years, Togo has been engaged in a proactive strategy for sustainable development and against global warming. Its efforts focus mainly on bad production practices in the economic sectors; lack of population control; and the high poverty rate. The poverty rate is exacerbated by the negative impacts of climate change, further reinforcing the vulnerability of the production sectors and the pressure on natural resources. This political will can be seen, amongst others, in the National Environmental Action Plan (NEAP), the National Environmental Management Programme (NEMP), the National Sustainable Development Strategy (December 2011), the National Environmental Management Capacity-building Strategy (October 2008), the National Programme for Reducing Greenhouse Gas Emissions from Deforestation and Forest Degradation (REDD+) 2010-2050, the National Strategy for Reducing the Risk of Catastrophes in Togo (December 2009), the National Medium Term Priority Framework (NMTPF) for Togo (2010-2015), and the National Action Plan for Marine and Coastal Environmental Resources Management. Further, Togo's membership of the Climate & Clean Air Coalition (CCAC) means it could raise funds to finance its short-term GHG and climate pollutant mitigation actions (Republic of Togo, 2015).

### E.5.2. Capacity of accredited entities and executing entities to deliver

*Please describe experience and track record of the accredited entity and executing entities with respect to the activities that they are expected to undertake in the proposed project/programme.*

This programme is in line with BOAD's priority to intensify its activities in favour of the private sector by providing adequate support to private sector development in the LDCs of West Africa region. The sections below highlight the experience and track record of BOAD - as the accredited entity – with respect to financing private sector actors.

In order to maximise the programme impact, technical assistance to BOAD staff will be provided to improve its capacity to develop, manage and implement the programme portfolio and to integrate climate change dimensions into project activities. Further details on the type of trainings provided are described in section C.3.

#### **BOAD ability to promoting and financing private sector actors**

BOAD uses the financial resources that it mobilizes to invest in public and private sector projects and programmes aimed at building basic and modern infrastructure, improving rural livelihoods, generating energy, and climate change adaptation and mitigation.

BOAD's contribution to promoting and financing private sector actors has materialized into promoting financing vehicles and funding through commercial banks and financial institutions as well as direct loans to projects and financing of studies. To provide for the needs of private sector actors that are customarily facing difficulty in mobilizing adequate equity and guarantees to make their projects bankable, BOAD has, in conjunction with several partners like Agence Française de Développement (AFD), German Development Agency (DEG), European Investment Bank (EIB) and African Development Bank (AfDB), contributed to setting up specific vehicles including GARI Fund (providing guarantee for private investments in West Africa) and CAURIS Investissement, a venture capital company.

BOAD continuously adjusts, intensifies and diversifies its action towards the promotion and financing of private productive investments. BOAD has a number of strategies for the promotion and financing of the private sector. Since 2013, BOAD has organized capacity building activities and outreaches with IFI such as the World Bank, AfDB and the UNFCCC secretariat to bring more awareness to the West African NFIs and commercial bankers in the financing of energy transition projects. In June 2017 BOAD launched the creation of the West African network for climate finance.

BOAD is also promoting the use of green bonds and green credit line mobilization throughout the region. Out of FCFA 4,202.1 billion of total net commitments as at 30 September 2016, BOAD devoted to the private sector about 24.3%, representing FCFA1,022.4 billion for 319 projects.

BOAD's private sector financing operations and the promotion of financial inclusion are carried out through the "Private Sector" window. The Private Sector Window prioritizes the fight against poverty and support for inclusive growth in the member states. The portfolio of the Bank's shareholding capital of companies and financial institutions includes 82 transactions. BOAD has set a number of initiatives as explained below:

- **The Regional Initiative for Sustainable Energy (IREN):** In order to better overcome recurrent energy crises in many of the WAEMU member countries, the Conference of Heads of State and Government decided in January 2008 to set up a committee tasked with proposing a strategy called "Regional Initiative for Sustainable Energy (IREN)". The initiative aims to provide, by 2030, 100% access to electricity in the WAEMU region at lower cost and as part of a large integrated and harmonized power pool market in West Africa. This regional market will be based on a dynamic public-private partnership. It also targets an increase by 82% of renewable and sustainable energies in the WAEMU's power generation base. The financial and institutional mechanism for the implementation of the initiative includes a concessional Energy Development Fund (FDE), and an investment fund operating as a private infrastructure development facility.
- **The Energy Development Fund (FDE):** Established on 25 September 2009, the Fund aims to contribute to improved access to energy in the WAEMU member countries and promote the development of sustainable energy under concessional conditions in order to finance projects eligible for IREN initiative. It is managed by BOAD. The Fund started its operations in the first quarter of 2010. As of the July 2018, 18 projects and 1 programme (PRODERE) have been financed in the form of loans involving a total amount of EUR 341.5 million (more details in section C.4).
- **The Infrastructure Development Fund:** It is dedicated to private investments. It is a profit-making fund meant for financing access to electricity in the WAEMU region according to market terms and conditions. In addition, it aims to support the restructuring of electricity companies, the emergency programme and the medium and long-term projects. It will be a lever for public-private partnerships in the energy sector in the WAEMU region. The due diligence towards its implementation is underway at the level of BOAD. The Bank participated in the implementation of major projects including:
  - Mali-Côte d'Ivoire electric power interconnection of by Energie du Mali (EDM);
  - Electrification of the townships of Bakel, Selibaly and Goureye in Senegal by Société de Gestion de l'Energie de Manantali (SOGEM);
  - Expansion of CIPREL SA's power plant capacities in Côte d'Ivoire (CIPREL IV A);
  - Expansion in combined cycle of the generation capacities of AZITO power plant by SOCIETE AZITO ENERGIE SA in Côte d'Ivoire.

### E.5.3. Engagement with NDAs, civil society organizations and other relevant stakeholders

Consultations are on-going with many stakeholders in the region and particularly in the six targeted countries. On-going consultations continue to take place with the NDAs of the six countries, as well as potential project developers.

Stakeholder consultations have been conducted so far with a number of organizations. BOAD plans to organize stakeholder workshops in the target countries to introduce the programme to local project developers as well as local commercial financial institutions. BOADs regional presence will ensure a strong on-the-ground presence with networks that enable the programme to develop solutions adapted to local contexts and needs.

Table 20: BOAD engagement with key stakeholders

| Main actors | BOAD engagements with each actor |
|-------------|----------------------------------|
|-------------|----------------------------------|

|                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Private sector                                                                                                  | BOAD has reached out and engaged with several local project developers in the 6 target LDCs as well as local financial institutions who have expressed strong interest in participating in the programme.<br><br>The African Guarantee Fund (AGF) and the ECOWAS Bank for Investment and Development (EBID) have confirmed their interest in participating in the programme. The Letters of Interests of AGF and EBID have been included in the annexes of the FP. Furthermore, Orabank has indicated its interest in co-financing solar projects through the proposed Facility. |
| NDAs                                                                                                            | BOAD has already engaged with the NDAs of the six target countries. The NDAs have communicated that they are willing to support the programme. All letters of non-objection of the six target countries have been obtained.                                                                                                                                                                                                                                                                                                                                                      |
| National authorities                                                                                            | BOAD works very closely with governments and authorities to uphold the principle of country ownership via direct partnerships; via country representatives on BOAD's Board of Directors (all member countries are represented), and via the provision of country and sector strategies that are developed with the relevant country's involvement and in consultation with key stakeholders.                                                                                                                                                                                     |
| West African Economic and Monetary Union (WAEMU), Economic Community of West African States (ECOWAS) and ECREEE | BOAD works closely with WAEMU and ECOWAS on a number of regional initiatives. BOAD has consulted with ECOWAS and ECREEE who have both provided key inputs to the design of the programme.                                                                                                                                                                                                                                                                                                                                                                                        |
| Members of West Africa Network for Climate Finance (WANCF)                                                      | BOAD has been an active participant in climate finance dialogue with key stakeholders in the region since the bank inception. The WANCF was established in June 2017 as a leader in climate finance initiatives in the region and to ensure stakeholders have a platform to provide inputs to initiatives such as the proposed programme and to help identify climate finance opportunities. BOAD began consultations with WANCF in June 2017 and consultations are ongoing.                                                                                                     |

## E.6. Efficiency and Effectiveness

Economic and, if appropriate, financial soundness of the project/programme

### E.6.1. Cost-effectiveness and efficiency

As described in section B there is a strong need to mobilise long term debt capital to private sector to scale up cost-competitive solar technologies. Public sector funding through development institutions will not suffice to serve the capital needed to reach a transformative scale towards low-emission pathways. As already explained, the main financial barriers that continue to hold back solar private sector capital are the high-risk perception of private sector investors and the maturity and liquidity risk of local commercial banks which are unable to extend loans with tenors of no more than 5-7 years.

The programme has been designed to overcome these barriers by incentivising the participation of private sector actors and commercial banks:

- private sector actor mobilisation will be achieved through the provision of concessional loans to BOAD (component 1.1). The optimal level of concessionality will be targeted in order to avoid providing excess subsidy to the private sector or crowding out private sector investors. It is expected that the concessionality received from the GCF will enable BOAD to offer a final blended cost of capital which will be below commercial rates. GCF's contribution will allow to trigger access to private investment by providing an adequate risk-return and stable revenues to project developers. The concessional funds from the GCF will also allow for bringing the electricity tariff to attractive level that are competitive with electricity produced from fossil fuels energy.

- Private banks crowd in will be achieved through the standby loan facility (component 1.2). One of the objectives of the programme is to crowd-in the private sector investment from commercial banks into the solar sector by providing a tenor extension instrument that can remove the barrier for commercial banks' participation. The length of tenor is a key limitation encountered by project developers seeking local financing. The local commercial banks are unable to extend loans with tenors of no more than 5-7 years due to the short-term maturity profile of their deposit and the absence of deep capital markets to source long-term financing. To overcome this barrier, the GCF will incentivise the participation of local commercial banks in the Programme using a liquidity guarantee for extension of the commercial banks loan tenor.

The efficiency of the programme is measured based on two indicators:

- **The amount of private sector leveraged:** The GCF's support is critical to secure private sector investors. Projects developers are expected to contribute their own capital to the projects financed by BOAD. Local private and public banks will also contribute to the total project cost in the form of loans. The programme is expected to leverage EUR 128.1m of private investment.
- **The amount of GHG emissions being mitigated in absolute numbers:** BOAD will track the amount of tCO<sub>2</sub>e emissions being mitigated by the solar investments financed. Considering an expected lifetime of emission reductions of 25 years (most likely the impacts will continue beyond that period) the abatement cost of the programme for the GCF is of EUR 12 per tCO<sub>2</sub>e. The projects supported are expected to contribute to savings of approx. 4.8m tCO<sub>2</sub> over their life.

## E.6.2. Co-financing, leveraging and mobilized long-term investments (mitigation only)

The programme will be designed to maximize total investment, using limited public funds to leverage greater private investment. The GCF contributions will account for 50% of the total financing of EUR 122m required to deliver the Programme. BOAD will provide co-financing of EUR 61m from its own resources. The co-financing ratio is therefore 1:1

| BOAD Co-financing (m EUR) | GCF (m EUR) | Co-financing ratio |
|---------------------------|-------------|--------------------|
| 61                        | 61          | 1:1                |

In addition, this funding is expected to leverage a substantial amount of financing from project developers and private sector banks. The leverage ratio is calculated based on funding brought by the equity investments from project developers (EUR 50m) and the debt finance from local banks (EUR 78.1m).

| GCF (m EUR) | BOAD (m EUR) | Equity (m EUR) | Local banks (m EUR) | Total Leverage ratio |
|-------------|--------------|----------------|---------------------|----------------------|
| 61          | 61           | 50             | 78.1                | 1:3                  |

## E.6.3. Financial viability

BOAD depends on the participation of the GCF without which the programme would not come into realisation. The financial return without and with GCF contribution is as follow:

- Without GCF: IRR of 11% with an NPV of EUR -5.4m. At this return levels, the private sector would not be willing to invest in this sector.
- With GCF: IRR of 18% with an NPV of EUR 14.3m. This level of return is expected to secure private investment at scale.



BOAD is the borrower of GCF funds. The GCF will gradually exit as BOAD will be repaying the GCF loan facility. At maturity (within 20 years), GCF principal commitments will be fully repaid. Exit risk is minimal.

## E.6.4. Application of best practices

Equipment and technology will be selected from the best options available on the market. During the appraisal process of the projects to be funded under the programme, BOAD will support the application of best available technologies by requiring projects to demonstrate that they use solar technologies that are 1) effective in addressing climate-related goals, 2) market-tested, technically performing and commercially viable; and 3) contribute to market transformation in terms of demonstrating the viability and commercial application of new technologies.

BOAD will ensure that all solar systems will be sized appropriately for anticipated loads, localized irradiation, and annual seasonality. Best practices in terms of system design, installation, and maintenance will be ensured through rigorous quality testing and trainings for employees.

Under component 2.1, TA can support project developers in selecting the technology fitting the needs identified, to ensure the viability of the technology and the soundness of the investment, and in identifying the most adapted certifications. If needed, the TA can be asked to assess the quality of equipment available in the market to help project developers to select a robust technology and avoid underperformance of the equipment. TA may also help project developers in designing a monitoring system which will be used to assess the equipment's performance and to identify any potential technical dysfunctions that may put impacts (carbon savings) and efficiency (€/tCO<sub>2</sub>eq) at risk. Project developers are also encouraged to engage maintenance contracts and may be advised by TA along this procedure. Maintenance contracts guarantee the effective repairation of the equipment and therefore are a complementary tool to ensure the overall investment effectiveness. TA team will secure continuous follow-up of the selection of technologies and equipment to avoid underperforming technologies.

## E.6.5. Key efficiency and effectiveness indicators

GCF core indicators

Estimated cost per t CO<sub>2</sub> eq, defined as total investment cost / expected lifetime emission reductions (mitigation only)

(a) Total project financing

EUR 122 000 000

(b) Requested GCF amount

EUR 57 000 000

(c) Expected lifetime emission reductions overtime

4 834 955 tCO<sub>2</sub>eq

(d) Estimated cost per tCO<sub>2</sub>eq (d = a / c)

EUR 25 / tCO<sub>2</sub>eq

(e) Estimated GCF cost per tCO<sub>2</sub>eq removed (e = b / c)

EUR 12/ tCO<sub>2</sub>eq

Describe the detailed methodology used for calculating the indicators (d) and (e) above.

Please describe how the indicator values compare to the appropriate benchmarks established in a comparable context.

| Technolgy | Countries                                          | Installed capacity | Capacity factor | Annual elec. Gen (MWh) | Grid EF (tCO2/MWh) | Annual emissions avoided | Lifetime (yr) | Lifetime emissions avoided |
|-----------|----------------------------------------------------|--------------------|-----------------|------------------------|--------------------|--------------------------|---------------|----------------------------|
| Solar     | Benin, Burkina Faso, Mali, Niger, Nigeria and Togo | 205                | 18%             | 323,244                | 0.561              | 181,340                  | 25            | 4,533,497                  |
| Solar     | Guinea-Bissau                                      | 10                 | 18%             | 15,768                 | 0.7647             | 12,058                   | 25            | 301,458                    |
| TOTAL     |                                                    |                    |                 | 339,012                |                    | 193,398                  |               | 4,834,955                  |



## Description of the assumptions

- BOAD is seeking EUR 61 m of concessional loan financing from the GCF and will raise a further EUR 61 m from its own resources, hence the total commitment is EUR 122m. In addition, this funding is expected to leverage a substantial amount of financing from project developers and private/public sector banks, namely EUR 128.1m.
- Using a figure of EUR 1,140,000 (IRENA, Solar PV in Africa: Costs and Markets, 2016) – equivalent to USD 1,300,000 CAPEX per MW using an exchange rate of 1 EUR= 1,149 USD, the total capex is EUR 245.1m given the total solar capacity to be installed of 215MW (See Table 10 Pipeline of projects to funded under the programme).
- For the capacity factor, a conservative approach has been used for the two technologies (solar PV and CSP) of 18%.
- The operational lifetime of solar technologies is 25 years.
- The grid emission factor used for Benin, Burkina Faso, Mali, Niger and Togo is 0.561 tCO<sub>2</sub>/MWh. This emission factor has been calculated using the CDM “Tool to calculate the emission factor for an electricity system (version 05.0)”. It applies for Benin, Burkina Faso, Côte d’Ivoire, Ghana, Mali, Niger, Nigeria, Senegal, Togo. More information on the assumptions used can be found on the Standardized baseline - Grid emission factor for West African Power (available at: [https://cdm.unfccc.int/methodologies/standard\\_base/2015/sb102.html](https://cdm.unfccc.int/methodologies/standard_base/2015/sb102.html)).
- For Guinea-Bissau, the emission factor has been calculated using a bottom up approach to find emissions per unit of MWh produced based on certain assumptions. The grid emission factor used is 0.76 tCO<sub>2</sub>/MWh.

The climate impact potential of the Programme will be the reduction of CO<sub>2</sub> emissions through displacement of fossil-fueled electricity use at the user level by implementing and installing of clean and solar energy sources. The programme’s cost of mitigation (cost per t CO<sub>2</sub>eq) is described in table below:

Table 21: Programme’s cost of mitigation

|                                                                |             |
|----------------------------------------------------------------|-------------|
| 10 Year Estimated GHG Reduction, tCO <sub>2</sub> e            | 1 933 982   |
| 15 Year Estimated GHG Reduction, tCO <sub>2</sub> e            | 2 900 973   |
| 25 Year Estimated GHG Reduction, tCO <sub>2</sub> e            | 4 834 955   |
| Total Project cost, EUR                                        | 122 000 000 |
| GCF investment, EUR                                            | 57 000 000  |
| 25 Year average GCF cost per tonne, USD/tCO <sub>2</sub> e     | 12          |
| 25 Year average project cost per tonne, USD/tCO <sub>2</sub> e | 25          |

|                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                      | <p>Expected volume of finance to be leveraged by the proposed project/programme and as a result of the Fund's financing, disaggregated by public and private sources (mitigation only)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                                                                                                      | <p><i>Describe the detailed methodology used for calculating the indicators above.</i></p> <p><i>Please describe how the indicator values compare to the appropriate benchmarks established in a comparable context.</i></p> <p>The programme will be designed to maximize total investment, using limited public funds to leverage greater private investment. The GCF contributions will account for 50% of the total financing of EUR 122m required to deliver the Programme. BOAD will provide co-financing of EUR 61m from its own resources.</p> <p>This total commitment is expected to leverage a substantial amount of financing from project developers and private sector banks.</p> |
| <p>Other relevant indicators (e.g. estimated cost per co-benefit generated as a result of the project/programme)</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

\* The information can be drawn from the project/programme appraisal document.

## F.1. Economic and Financial Analysis

An economic and financial model was produced to analyze the impact of the GCF's funding in terms of financial return to private sector actors.

Table 22: Assumptions of the financial model

|                                          |   |           |
|------------------------------------------|---|-----------|
| Wholesale electricity tariff (EUR/MWh)   | € | 87.72     |
| Inflation rate on electricity tariff     |   | 1%        |
| Annual O&M as % of capital costs         |   | 1%        |
| Construction and drawdown period (years) |   | 2         |
| Installed cost per MW (EUR)              |   | 1,140,000 |
| Total installed capacity (MW)            |   | 215       |
| Capacity factor                          |   | 18%       |
| Annual power generation (MWh)            |   | 339 012   |
| Average corporate income tax rate        |   | 25%       |

### Justification for concessionality:

Concessional and grant funding from the GCF will bring a range of benefits to private sector actors. Concessionality will consist of reduced pricing compared to BOAD pricing and market finance. The provision of such reduced pricing is a key element attracting private sector actors. The concessional funds from the GCF will allow:

- reducing the blended cost from borrowing from BOAD and from commercial banks
- increasing the tenor and the grace period of lending from BOAD and from commercial banks
- enhance the bankability of solar projects at electricity tariff that are competitive with electricity produced from fossil fuels energy.

With the GCF funding included, the financial model presents a convincing financial case for the programme. Without GCF, the electricity tariff will need to increase to achieve bankability. The blended financing structure will assist the projects in achieving financial viability with a lower average electricity tariff and at an acceptable NPV and IRR level attractive to the private sector.

All these factors will make solar projects financially attractive as indicated in table below.

Table 23: Summary of the key performance indicators

| KPI                                             | Without GCF  | With GCF     |
|-------------------------------------------------|--------------|--------------|
| 20-year Internal Rate of Return - leveraged (%) | 11%          | 18%          |
| Discount rate (%)                               | 12%          | 12%          |
| Net Present Value (EUR)                         | -€ 5,400,000 | € 14,300,000 |
| Payback period (Years)                          | 15           | 7            |
| Minimum Debt Service Coverage Ratio             | 0.86         | 1.26         |

Overall, through this programme, GCF and BOAD financing will demonstrate and scale up commercially viable solar projects in the region while catalysing a paradigm shift within the electricity sector in the LDCs of West Africa.

## F.2. Technical Evaluation

Under this programme, eligible projects should target solar grid connected power generation projects with a capacity of approximately 10 MW to 70 MW. Eligible technologies will include: solar photovoltaic (PV) and Concentrated Solar Power (CSP).

Technologies to convert solar energy into electricity generally fall into one of two categories: photovoltaic (PV) modules that convert light directly into electricity and concentrating solar thermal power (CSP) systems that extract heat energy transferred by solar radiation. Although solar power can be generated at any scale, PV technology is modular and can be scaled for anything from household use to a large network of PV farms. Due to the scarcity of transmission and distribution infrastructure in the region, ECREEE originally estimated that CSP is currently technically feasible only within a certain geographic band through the Sahel, though it has helped implement a solar thermal energy project in Ghana last year. Solar thermal technologies, unlike solar PV, can operate through the day, making it attractive for large-scale energy production. Heat can be stored during the daytime and excess daytime heat can potentially be stored and converted to electricity when required. Such thermal plants would improve storage capacities to optimise the economics and the dispatch-ability of solar electricity.

### Solar PV

The solar PV sector is so dynamic that today's cost of generation may become tomorrow's benchmark. Prices have dramatically reduced over the last decade, combined with the availability of economic financing and policy pushes. IRENA estimates that by 2030, total installed costs could fall between 50% and 60%. Battery lifetimes and performance will also keep improving. Total installed costs for solar PV technology show regional cost differences because of different market maturity levels, differences in local labour, manufacturing costs and different support policy structures. Between 2010 and 2017, the global capacity weighted average total installed cost of newly-commissioned utility-scale PV projects decreased by 68%, with a 10% decrease in 2017 from 2016. Projects in newer or undeveloped markets are being developed at costs that are increasingly at par, and sometimes even cheaper than the averages in most cost mature markets.

The global weighted average capacity factor of utility-scale PV systems increased by 28% between 2010 and 2017, from an average of 13.7% to 17.6%. This has been driven by three major factors, the trend towards greater deployment in regions with higher irradiation levels, the increased use of tracking and improvements in the performance of systems as losses have been reduced.

### Concentrating Solar Thermal Power (CSP)

Total installed costs for CSP plants that include thermal energy storage tend to be higher than those without, but storage also allows for higher capacity factors. For example, for parabolic trough systems (the technology with the highest share of installed projects so far), total installed plant costs can range between USD 2 550 and USD 11 265/kW for systems with no storage. Adding four to six hours of storage, however, can see this range increase to between USD 6 050 and USD 13 150/kW.

Capacity factors have increased over time as a shift towards newer technologies, with larger thermal storage capacities has coincided with a trend towards the growth of markets in higher irradiation locations. Similarly, higher levels of irradiance were likely the main factor behind lower levelized costs. During 2016 the capacity weighted average LCOE of CSP plants was estimated to be USD 0.27/kWh (a fifth lower than in 2009) although IRENA data suggests that the LCOE, although about 18% during 2017 to USD 0.22/kWh.

## F.3. Environmental, Social Assessment, including Gender Considerations

In order to ensure a better consideration of environmental and social issues in the funded-operations, BOAD adopted in 2003, a document of policies and procedures for environmental and social management intervention in the financing of projects, an environmental and social procedures manual and a manual of environmental classification of projects. This allowed BOAD to participate effectively alongside its borrowers to the protection of populations, their ecosystems and their living environment.

Such policies and procedures adopted in 2003 by BOAD recommend revising periodically the content of the documents based upon international evolution. Also, in order to comply with current international standards for environmental and social management, BOAD undertook an update of its safeguards, like institutions such as the World Bank, the African Development Bank, etc. with which BOAD cooperates closely in its operations.

BOAD requires that projects that are submitted for funding are subject to environmental and social impact assessment (ESIA), which helps to ensure that the projects are environmentally sound and socially viable to facilitate decision making.

In addition to the regulation of the countries of intervention, BOAD will refer to the rules, good practices and guidelines produced by international organizations such as: (i) United Nations Conventions on Environment and Development; (ii) United Nations Convention on the Elimination of All Forms of Discrimination against Women; (iii) United Nations Universal Declaration on Human Rights; (iv) International conventions of ILO on human rights and labor law; (v) Safeguard policies, guidelines and other relevant documents of the World Bank on environmental and social issues; (vi) the safeguard policies of the IFC on environmental and social issues; (vii) Good international sectoral practice, published by major trade associations on environmental and social issues; etc.

Under this programme, environmental and social best practices will be applied by requiring all projects funded to follow BOAD environmental and social safeguard policies, standards and guidelines which follow best international practices such as the International Finance Corporation (IFC) environmental and social performance standards.

Risks will be mitigated through technical assistance provided to project developers to undertake technical feasibility studies, Environmental Impact Assessments (EIA) and environmental and social action plan (ESAP). These measures will identify potential environmental and social impacts and provide detailed mitigation measures. The project-level ESAP will be an integral part of the required agreement with the project developers. BOAD will only finance the projects classified under the category B or below and apply its highest standards in the screening and managing of E&S related issues.

All subprojects funded will come under BOAD's ESS policy and the agreed cooperation framework between GCF and BOAD. Individual subprojects will require site specific assessments consistent with the approach of BOAD and the GCF including E&S and Gender due diligence and the agreed disclosure requirements (where applicable), and will develop project specific Environmental and Social Management Plans (ESMP). Only category C (I3) or B (I2) projects will be funded under this facility, consistent with the GCF accreditation of BOAD.

**All projects to be financed by the Programme fit in Category B or C according to both GCF and BOAD categorization of projects.** A project is classified as Category B if it may negatively impact environmental areas or human populations, while not being irreversible. For Category B projects, project developers will be required to develop all relevant environmental and social management frameworks needed to prevent, minimize, mitigate, or compensate for adverse impacts and to improve environmental performance. These environmental and social management frameworks can include, but are not limited to: Land Acquisition and/or Resettlement Action Plan (LARAP or RAP), Indigenous Peoples Plan (IPP) and Indigenous Peoples Planning Framework (IPPF). A project is classified as Category C if the probability of its negative impacts on the environment and human populations is considered minimal. For Category C projects, no environmental and social management framework will be required once the environmental and social screening has been conducted and approved. Guidance and indicative templates for the development of relevant environmental and social management frameworks can be found in the Programme's ESMF.

### Gender assessment

The energy markets are essential to the economies of the participating countries and women can in particular benefit from opportunities connected to the development of renewable energy sources. Women in the participating countries currently work primarily in services and agriculture, and to a much more limited extent in the power sector.

The Gender plan for the financing facility will ensure that all consultations with stakeholders will be designed in a gender responsive way and women will be equally consulted and participate in all discussions related to the subprojects. This approach will be incremental and would set a proper benchmark for other projects to follow.

The programme will be accompanied by a Gender Action Plan template for subprojects.

The ESMF for the program has been developed and submitted. All sub-projects will be assessed according to the relevant BOAD policies, and an environmental and social action plan (ESAP) will be developed during the implementation of each sub-project.

### F.4. Financial Management and Procurement

Financial resources from the GCF will be managed according to the general provisions of the AMA between the GCF and BOAD.

BOAD as part of its AE role will among others carry out the following:

- BOAD will be responsible for the overall oversight of the framework implementation and will report to GCF as per the terms to be agreed under the Accreditation Master Agreement (AMA);
- BOAD will report to GCF as per the terms to be agreed under the AMA and the Funded Activity Agreement (FAA).
- BOAD will prepare environmental and social impact analysis reports in line with the bank's environmental and social safeguards (including ensuring that the sub-project is within allowed environmental impact classification, i.e. only up to category B)
- BOAD will ensure that a robust climate rationale is in place for all projects.
- BOAD will ensure that an MRV framework will be established to ensure activities of the facility and other relevant stakeholders are in compliance with all applicable policies, procedures and other requirements of the GCF to enhance transparency. An independent party will be procured by BOAD for Monitoring, Reporting and Verification (MRV); BOAD will be ultimately responsible for reporting to the GCF.

BOAD as part of its EE role will be responsible for the implementation of the grant component. Procurement under the facility will be managed according to BOAD's procurement policies as detailed in BOAD's AMA. BOAD will hire qualified local and international consultants for the implementation of the technical assistance (component 2), based on its procurement policies as approved by the GCF. The consultants will work under the supervision of BOAD and will provide technical assistance

- to local private sector actors to build their capacities during the development phase of their projects (Sub-component 2.1);
- to BOAD staff to build their capacities during the development and implementation of the project portfolio (Sub-component 2.1);
- to national institutions to support them in the implementation of regulatory and policy regimes that provides clear and predictable rules for solar project development (Sub-component 2.2).

GCF and BOAD will sign a grant agreement for the implementation of these activities under Sub-components 2.1 and 2.2. The details in terms of financial management and flow of funds has been elaborated section in C.7.



## G.1. Risk Assessment Summary

It is considered that the programme might face risks in several different categories: macroeconomic, implementation, financial, environmental and social risks. The levels of these risks are considered moderate to low and expected to be mitigated by BOAD's operational tools and control mechanisms. BOAD will regularly evaluate the overall potential risks related to the implementation of the project and will provide recommendations for procedures to mitigate and address the risks.

## G.2. Risk Factors and Mitigation Measures

*Please describe financial, technical and operational, social and environmental and other risks that might prevent the project/programme objectives from being achieved. Also describe the proposed risk mitigation measures.*

### Selected Risk Factor 1

| Description                                                                                                                                                                             | Risk category | Level of impact                   | Probability of risk occurring |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----------------------------------|-------------------------------|
| Macroeconomic, regulatory and political risks can lead to a slowdown of the economy in the region and financial sector, increase regulatory / off-taker risks, and reduced investments. | Other         | Medium (5.1-20% of project value) | Medium                        |

### Mitigation Measure(s)

BOAD operations are covered at certain point against political risks given its political mandate as a regional development Bank and its close coordination with host Governments.  
BOAD will closely monitor the upcoming events on the markets and the potential impacts on the financial performance of the project developers who will have an obligation to regularly report on financial performance.  
BOAD will maintain a close relationship with host governments through policy dialogue activities that aim to identify policy and regulatory bottlenecks and create enabling environments.

### Selected Risk Factor 2

| Description              | Risk category | Level of impact            | Probability of risk occurring |
|--------------------------|---------------|----------------------------|-------------------------------|
| Credit and Currency risk | Financial     | Low (<5% of project value) | Low                           |

### Mitigation Measure(s)

Credit risk: The GCF will bear the credit risk of BOAD which is an investment grade financial institution rated by Moody's (Baa1) and Fitch (BBB). This reflects the solid financial position of BOAD which has one of the highest financial institution credit ratings in the region.

Foreign exchange fluctuations: The GCF's financing will be in EUR and repayments from BOAD to the GCF will also be EUR. The Franc CFA (XOF) is the common currency of the West African Economic and Monetary Union and is pegged to the EUR; therefore, currency risk is mitigated both for the GCF and the solar projects to be financed.

| Selected Risk Factor 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                           |                                   |                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-----------------------------------|-------------------------------|
| Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Risk category             | Level of impact                   | Probability of risk occurring |
| Implementation risk: Construction and operation risks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Technical and operational | Medium (5.1-20% of project value) | Medium                        |
| Mitigation Measure(s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                           |                                   |                               |
| <p>Project construction risk (completion and cost overrun) will be mitigated by a thorough due diligence process led by the BOAD technical and financial experts, and by ensuring robust Engineering, Procurement and Construction contracts.</p> <p>Project operation risk will be mitigated by a thorough due diligence process led by the BOAD technical and financial experts and by ensuring robust O&amp;M contracts. Resource risk (e.g. insufficient irradiation for solar PV projects) will be mitigated by ensuring high-standard technological studies in place.</p>                                                                                                                                                                                                                                                                                                                         |                           |                                   |                               |
| Selected Risk Factor 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                           |                                   |                               |
| Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Risk category             | Level of impact                   | Probability of risk occurring |
| Environmental and social Risks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Social and environmental  | Medium (5.1-20% of project value) | Medium                        |
| Mitigation Measure(s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                           |                                   |                               |
| <p>Environmental and social best practices will be applied by requiring all projects funded by the programme to follow BOAD environmental and social safeguard policies, standards and guidelines which are in line with GCF's ESS and gender standards.</p> <p>Risks will be mitigated through technical assistance provided to project developers to undertake technical feasibility studies, Environmental and Social Impact Assessments (ESIA) and environmental and social action plan (ESAP). These measures will identify potential environmental and social impacts and provide detailed mitigation measures. The project-level ESAP will be an integral part of the required agreement with the project developers. BOAD will only finance the projects classified under the category B or below, and apply its highest standards in the screening and managing of E&amp;S related issues.</p> |                           |                                   |                               |
| Selected Risk Factor 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                           |                                   |                               |
| Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Risk category             | Level of impact                   | Probability of risk occurring |
| Technological Risk: Investments are exposed to various levels of technological risk including but not limited to: equipment failure; supply chain failure such as delays or collapse of supply chain for technology or associated equipment; lack of qualified service technicians.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Technical and operational | Low (<5% of project value)        | Low                           |
| Mitigation Measure(s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                           |                                   |                               |
| The risk will be mitigated by the project pre-selecting a list of eligible technologies.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                           |                                   |                               |
| Other Potential Risks in the Horizon                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                           |                                   |                               |

*Please describe other potential issues which will be monitored as “emerging risks” during the life of the projects (i.e., issues that have not yet raised to the level of “risk factor” but which will need monitoring). This could include issues related to external stakeholders such as project beneficiaries or the pool of potential contractors.*

*\* Please expand this sub-section when needed to address all potential material and relevant risks.*

## H.1. Logic Framework.

Please specify the logic framework in accordance with the GCF's [Performance Measurement Framework](#) under the [Results Management Framework](#).

### H.1.1. Paradigm Shift Objectives and Impacts at the Fund level<sup>35</sup>

#### Paradigm shift objectives

|                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><i>Shift to low-emission sustainable development pathways</i></p> | <p>Under the programme, BOAD will address the market barriers by playing a catalytic role using a blended finance approach to provide long-term debt at affordable rates to increase the bankability of projects and attract private sector investments.</p> <p>The programme will contribute to a paradigm shift to low-emission development pathways through the mobilisation at scale of private climate finance. Transformational impacts will be achieved through the growth in commercially viable solar investments and the scaling up of financial products and services offered to the private sector in the region. This will essentially accelerate the GHGs emission reduction thereby contribute to national low-carbon development pathways.</p> <p>The programme will contribute to creating a replicable model for eventual scale-up of the Programme's success into the region as it is expected to showcase a viable model for all upcoming solar projects across the continent. Following the successful track-record in lending to these projects gained through the programme, it is expected that there will be a greater demand for financing solar projects and that local financial institutions will start using their own resources for climate lending.</p> <p>Furthermore, the programme will support the implementation of regulatory and policy regimes that provides clear and predictable rules for solar project development in the region. This will ensure that the programme will have a lasting transformative ("paradigm shifting") impact on the medium-large scale solar sector in the region.</p> |
|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

| Expected Result | Indicator | Means of Verification (MoV) | Baseline | Target                   |       | Assumptions |
|-----------------|-----------|-----------------------------|----------|--------------------------|-------|-------------|
|                 |           |                             |          | Mid-term (if applicable) | Final |             |

#### Fund-level impacts

|  |                                                                                                                                |                                                                                        |   |  |                                 |                                                                                                           |
|--|--------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|---|--|---------------------------------|-----------------------------------------------------------------------------------------------------------|
|  | <p><i>Tonnes of carbon dioxide equivalent (t CO<sub>2</sub>eq) reduced as a result of Fund-funded projects/ programmes</i></p> | <p>Program report (midterm and final)</p> <p>Regular reporting from BOAD to GCF as</p> | 0 |  | 4.8 million tCO <sub>2</sub> eq | <p>Successful completion of the plant construction</p> <p>Full disbursement of the approved financing</p> |
|--|--------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|---|--|---------------------------------|-----------------------------------------------------------------------------------------------------------|

<sup>35</sup> Information on the Fund's expected results and indicators can be found in its Performance Measurement Frameworks available at the following link (Please note that some indicators are under refinement):

[http://www.greenclimate.fund/documents/20182/239759/5.3\\_-\\_Performance\\_Measurement\\_Frameworks\\_PMF\\_.pdf/60941cef-7c87-475f-809e-4ebf1acbb3f4](http://www.greenclimate.fund/documents/20182/239759/5.3_-_Performance_Measurement_Frameworks_PMF_.pdf/60941cef-7c87-475f-809e-4ebf1acbb3f4)

|                                                                                                 |                                                                                                                                                                                          |                                    |   |  |                                 |                                                                                                                                       |
|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|---|--|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                 | <i>Financing leveraged through the programme</i>                                                                                                                                         | agreed in the AMA                  | 0 |  | Total funding EUR 128.1M        |                                                                                                                                       |
|                                                                                                 | <i>Cost per tCO<sub>2</sub>eq decreased for achieved projects</i>                                                                                                                        |                                    | 0 |  | EUR 12 tCO <sub>2</sub> eq      |                                                                                                                                       |
| <b>M1.0 Reduced emissions through increased low-emission energy access and power generation</b> | <i>Tonnes of carbon dioxide equivalent (t CO<sub>2</sub>eq) reduced as a result of Fund-funded projects/ programmes –gender-sensitive energy access power generation (sub-indicator)</i> | Program report (midterm and final) | 0 |  | 4.8 million tCO <sub>2</sub> eq | Information to be further disaggregated provided by project, country and type of technology in relation to the programme performance. |

### H.1.2. Outcomes, Outputs, Activities and Inputs at Project/Programme level

| Expected Result                                                               | Indicator                                                                                                         | Means of Verification (MoV)                                                                       | Baseline                                             | Target                   |                                                             | Assumptions                                                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|------------------------------------------------------|--------------------------|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                               |                                                                                                                   |                                                                                                   |                                                      | Mid-term (if applicable) | Final                                                       |                                                                                                                                                                                                                                               |
| Project/programme outcomes                                                    | Outcomes that contribute to Fund-level impacts                                                                    |                                                                                                   |                                                      |                          |                                                             |                                                                                                                                                                                                                                               |
| M6.0 Increased number of small, medium and large low-emission power suppliers | 6.1 Proportion of low-emission power supply in a jurisdiction or market                                           | Program report (midterm and final)<br><br>Regular reporting from BOAD to GCF as agreed in the AMA | Solar energy is 5% of the regional energy mix (2017) |                          | 19% solar in the regional energy mix of target countries    | The regulatory environment and financial profile for the solar project development is improving.<br><br>Information to be further disaggregated provided by project, country and type of technology in relation to the programme performance. |
|                                                                               | 6.2 Number of households, and individuals (males and females) with improved access to low-emission energy sources |                                                                                                   | 0                                                    |                          | At least 2.9m people indirectly benefit (700 000 household) |                                                                                                                                                                                                                                               |
|                                                                               | 6.3 MWs of low emission energy capacity installed                                                                 |                                                                                                   | 0                                                    |                          | 215 MW solar projects                                       |                                                                                                                                                                                                                                               |
| M5.0 Strengthened institutional and                                           | 5.1 Institutional and regulatory systems that                                                                     | Reports from the TA recipients;                                                                   | Scarce private investme                              |                          | One regulatory system that                                  | Close relationship with host                                                                                                                                                                                                                  |

|                                                                                                                                  |                                                                                                                                                                                                                                                            |                                                         |                                                            |                |                                                               |                                                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|------------------------------------------------------------|----------------|---------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| regulatory systems                                                                                                               | <i>improve incentives for low emission planning and development and their effective implementation</i>                                                                                                                                                     | regular reporting from BOAD to GCF as agreed in the AMA | nt in the solar sector due to lack of enabling environment |                | incentivises solar investment designed in each target country | governments through BOAD policy dialogue activities                                                                               |
| Project/programme outputs                                                                                                        | Outputs that contribute to outcomes                                                                                                                                                                                                                        |                                                         |                                                            |                |                                                               |                                                                                                                                   |
| Component 1: To scale up commercial and sustainable loan financing for solar projects (senior debt and standby loans component)  |                                                                                                                                                                                                                                                            |                                                         |                                                            |                |                                                               |                                                                                                                                   |
| 1.1 Solar projects financed and operational                                                                                      | Number of solar projects supported                                                                                                                                                                                                                         | Project monitoring and completion reports               | 0<br>(2019)                                                | 3<br>(2022 Q4) | 6<br>(2026 Q2)                                                | Commercial banks interests and capabilities for the solar financing                                                               |
| Component 2: To develop technical capacity and raise awareness of private sector actors - Technical assistance (Grant Component) |                                                                                                                                                                                                                                                            |                                                         |                                                            |                |                                                               |                                                                                                                                   |
| 2.1 Capacity built to BOAD and local project developers for investment preparation                                               | Within the local private sector companies, number of people trained, including number of women trained on project preparation phase<br><br>Within BOAD, number of people trained, including number of women, to manage and implement the project portfolio | Project monitoring and completion reports               | 0                                                          |                | 300, including 75 women<br><br>50, including 10 women         | No major external (economic, social, technological, etc.) issues that will have an adverse impact for solar on-grid interventions |
| 2.2 Capacity built to national institutions to address legal and regulatory barriers in the solar market                         | Number of people trained provided to national institutions to improve incentives for solar development projects and their effective implementation                                                                                                         |                                                         | 0                                                          |                | 300 including 25% women.                                      | Relevant institutions ready for the activities                                                                                    |
| 2.3 Awareness raised on the benefits of solar system technologies                                                                | Number of events organised to raise awareness on the benefits of solar system technologies disseminate experiences and good                                                                                                                                |                                                         | 0                                                          |                | 8                                                             |                                                                                                                                   |



|                                                                                                                       | practices with solar technologies                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |                                                                                                           |  |                                                                                                                                                                                                                               |  |
|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-----------------------------------------------------------------------------------------------------------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Activities                                                                                                            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  | Inputs                                                                                                    |  | Description                                                                                                                                                                                                                   |  |
| Component 1: To scale up commercial and sustainable financing for solar investments - Direct lending (loan component) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |                                                                                                           |  |                                                                                                                                                                                                                               |  |
| 1.1 Senior Loan Facility                                                                                              | Senior debt for medium to large scale solar projects                                                                                                                                                                                                                                                                                                                                                                                                                            |  | GCF senior debt financing and BOAD co-financing                                                           |  | Solar energy projects financed by the BOAD.                                                                                                                                                                                   |  |
| 1.2 Standby Loans Facility                                                                                            | Senior debt and standby loans (tenor extension instrument) for medium to large scale solar projects                                                                                                                                                                                                                                                                                                                                                                             |  | GCF standby loans to BOAD, to be on-lent to solar projects<br><br>Commercial banks: senior debt           |  | Tenor extension credit enhancement to solar projects financed by the BOAD.                                                                                                                                                    |  |
| Component 2: Technical assistance (grant component)                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |                                                                                                           |  |                                                                                                                                                                                                                               |  |
| 2.1 Building technical capacities to BOAD staff and local project developers                                          | Technical assistance will be provided to enhance project developers capacity in following sectors:<br><br>Technical and financial feasibility studies<br><br>Financial & commercial framework and risk mitigation strategy.<br><br>Environmental Impact Assessments (EIA) and environmental and social action plan (ESAP).                                                                                                                                                      |  | Trainings to BOAD and local project developers                                                            |  | Technical assistance consultant will be contracted by BOAD based on procurement process.<br><br>Procurement under the programme will be managed according to BOAD's procurement policies as detailed in BOAD's accreditation. |  |
| 2.2 Building the capacity of national institutions to address legal and regulatory barriers in the solar market       | TA under this sub-component will be focused on aspects of the enabling environment that have an immediate impact on stimulating solar private sector investment. Support can include the creation or revision of a simplified regulatory frameworks for Feed-in-tariffs for on-grid solar solutions, support on standard PPAs, support on IPP tenders and procurement process, on tariff structure, on a framework of financial and technical incentives and capacity building. |  | Trainings to national institutions                                                                        |  |                                                                                                                                                                                                                               |  |
| 2.3 Marketing and Awareness Raising                                                                                   | In order to create demand, the activities will support awareness raising activities on the benefits of solar system technologies, and broad dissemination of experiences and good practices.                                                                                                                                                                                                                                                                                    |  | Conduct awareness campaigns and promotional activities to potential project developers, commercial banks, |  |                                                                                                                                                                                                                               |  |

|  |                                                                                                                                                                                        |                                                                                                                             |  |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|--|
|  | Activities will include among other communications and outreach effort that promotes the objectives and good practices of the programme and the benefits of solar system technologies. | DFI's and other relevant stakeholders (e.g. donors, foundations, corporations, the media and public officials and agencies) |  |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|--|

## H.2. Arrangements for Monitoring, Reporting and Evaluation

*Besides the arrangements (e.g. semi-annual performance reports) laid out in AMA, please provide project/programme specific institutional setting and implementation arrangements for monitoring and reporting and evaluation. Please indicate how the interim/mid-term and final evaluations will be organized, including the timing.*

*Please provide methodologies for monitoring and reporting of the key outcomes of the project/programme.*

### **Third-party monitoring, reporting, verification and evaluation of implementation of loans**

An MRV framework will be established to ensure activities of the facility and other relevant stakeholders are in compliance with all applicable policies, procedures and other requirements of the GCF to enhance transparency. An independent party will be procured by BOAD for Monitoring, Reporting and Verification (MRV); BOAD will be ultimately responsible for reporting to the GCF.

BOAD is committed to the highest possible standards of monitoring, reporting and evaluation. BOAD's Monitoring and Evaluation policy was refined in 2013 in order to meet international standards. As a part of its monitoring and evaluation strategy, BOAD undertakes systematic monitoring, reporting and evaluation at all sates of the review cycle. It draws on a variety of tools - including logical framework, monitoring and evaluation matrix - to monitor and evaluate project development results. BOAD's monitoring and evaluation policy is comprised of 6 principles - (i) capitalization, (ii) accountability, (iii) partnership, (iv) credibility and independence (v) planning of monitoring and evaluation activities, (vi) dissemination.

Monitoring and Evaluation of the programme will be aligned to the BOAD Reporting Framework and the requirements of GCF as detailed GCF AMA. Under BOAD's policy, principle reports include: monitoring reports of development indicators for projects financed by the BOAD, performance reports of funded projects, and impact assessment reports of funded projects. With regards to this programme, the three main levels that quality assurance will occur: monitoring, reporting and evaluation.

### **Monitoring**

- **Project level monitoring:** Undertaken as per the project specific log frame that will be compiled for each individual project funded by BOAD. Project implementation will be monitored on an ongoing basis by the relevant BOAD project managers.
- **Programme level monitoring:** Information from projects will be collated into the annual project report and submitted to the GCF at the end of the calendar year. BOAD will compile the annual project report based on information received on individual projects.

### **Reporting**

- **Reporting – beneficiaries to BOAD:** Under a contract agreement, beneficiaries of the project will provide BOAD with (i) quarterly progress reports in a format consistent with BOAD's project performance reporting system; and (ii) consolidated annual reports including (a) progress achieved by output as measured through the indicator's

performance targets, (b) key implementation issues and solutions; (c) updated procurement plan and (d) updated implementation plan for next 12 months. This will also include financial information on the performance of each of project as well as impact data (capacity installed, electricity generated/saved etc.).

- **Reporting – BOAD to GCF:** GCF will receive bi-annual reports on individual projects. The programme log frame will be the backbone of this report and the report will be led by the relevant BOAD project manager and written in close collaboration with the beneficiary organization who has been provided a loan under the programme. Reporting to GCF will also include relevant information on the performance of the sub-loans and portfolio of sub-projects financed under the programme (including country and technology information and contribution to the overall expected results of the programme).

### Evaluation

The midterm and terminal evaluations will be undertaken by an independent consultant appointed by BOAD. BOAD will seek to enhance independence, including assessments, through transparency, largely through the use of experienced consultants through competitive bidding. The hired consultants will be invited to conduct their services and to deliver their opinions independently.

- **Independent interim evaluation:** A mid-term evaluation will be undertaken on the performance of the programme as per the indicators identified in the log frame. The mid-term evaluation will take place within 3.5 years of the programme being effective or at any time that the BOAD consider it necessary. The interim review serves to: (i) review institutional, administrative, organizational, technical, environmental, social, economic, and financial aspects of the programme based on the assumptions and risks included in the programme design; (ii) assess the need to restructure the programme's design and the effects of this on the immediate objectives (purpose) and long-term goals of the programme; and (iv) update the programme's design and monitoring framework if restructuring or reformulation is necessary or its immediate objectives will change.
- **Independent final evaluation:** At the end of the term of the programme, a final evaluation will be undertaken on the programme performance as per the log frame indicators and other qualitative information. The consultant will be tasked with evaluating the impact of each project (economic, environmental, etc.) and writing a report that will be shared with all relevant stakeholders (central and local governments, companies, NGOs, etc.) and could also be made public. It is worth noting that the evaluation function within BOAD refers to the same evaluation principles and criteria developed by the OECD Development Assistance Committee (DAC) and its evaluation protocols are similar. BOAD adheres to its principles for evaluation, which are impartiality and independence, credibility, usefulness, participation (of local partners) and coordination (with other donors). The evaluations aim to improve future projects and programmes by deriving lessons from experience and to find ways to make aid more effective.

In order to keep all the stakeholders informed of the projects financed under this programme, BOAD will use various means to communicate and disseminate the results, experience and best practices of the Programme as part of Component 2.3.

## I. Supporting Documents for Funding Proposal

- ☒ NDA No-objection Letter
- ☒ Feasibility Study
- ☒ Integrated Financial Model that provides sensitivity analysis of critical elements (xls format, if applicable)
- ☒ Confirmation letter or letter of commitment for co-financing commitment (If applicable)
- ☒ Project/Programme Confirmation/Term Sheet (including cost/budget breakdown, disbursement schedule, etc.) – *see the Accreditation Master Agreement, Annex I*
- ☒ Environmental and Social Impact Assessment (ESIA) or Environmental and Social Management Plan (If applicable)
- ☐ Appraisal Report or Due Diligence Report with recommendations (If applicable)
- ☐ Evaluation Report of the baseline project (If applicable)
- ☐ Map indicating the location of the project/programme
- ☒ Timetable of project/programme implementation
- ☒ Examples of Sub-projects in the pipeline
- ☒ Capacity Building Action Plan

*\* Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.*



N° 748/DGEC/MCVDD/SA

To: The Green Climate Fund ("GCF")

Cotonou November 21, 2018

**Re: Funding proposal for the GCF by Banque Ouest Africaine de Développement (BOAD) regarding "BOAD Climate Finance Facility to Scale Up Solar Energy Investments in Francophone West Africa LDCs"**

Dear Madam, Sir,

We refer to the program "BOAD Climate Finance Facility to Scale Up Solar Energy Investments in Francophone West Africa LDCs" as included in the funding proposal submitted by Banque Ouest Africaine de Développement (BOAD) to us on november 5, 2018.

The undersigned is the duly authorized representative of Ministry of Living Environment and Sustainable Development, the National Designated Authority of Benin.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the program as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Benin has no-objection to the program as included in the funding proposal;
- (b) The program as included in the funding proposal is in conformity with Benin's national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, the program as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the program as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the program.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Name: Mr. Martin Pepin AINA "[marnickson@yahoo.fr](mailto:marnickson@yahoo.fr)"

Title: General Director of Environment and Climate/ National authority Designed

**PREMIER MINISTERE**  
-----  
**SECRETARIAT GENERAL**  
-----

**BURKINA FASO**  
**Unité-Progrès-Justice**  
-----

**SECRETARIAT EXECUTIF DU FONDS VERT  
POUR LE CLIMAT AU BURKINA FASO**

Ouagadougou, le **27 DEC 2018**

**The Focal Point**

N° 2018-010 /PM/SG/SE-FVC/BF

**To**

**Green Climate Fund (GCF) Executive Director  
Songdo International Business District  
G-Tower, 175 Art Center daero  
Yeonsu gu, Incheon 406 840  
Republic of Korea**

**Objet : Funding proposal for the GCF by Banque Ouest  
Africaine de Développement (BOAD) regarding  
"BOAD Climate Finance Facility to Scale Up Solar  
Energy Investments in Francophone West Africa LDCs"**

**Dear Madam, Sir,**

We refer to the program "BOAD Climate Finance Facility to Scale Up Solar Energy Investments in Francophone West Africa LDCs" as included in the funding proposal submitted by Banque Ouest Africaine de Développement (BOAD) to us on november 5, 2018.

The undersigned is the duly authorized representative of the Prime Ministry, the National Designated Authority of Burkina Faso.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that :

- (a) The government of Burkina Faso has no-objection to the program as included in the funding proposal ;
- (b) The program as included in the funding proposal is in conformity with Burkina Faso's national priorities, strategies and plans ;



(c) In accordance with the GCF's environmental and social safeguards, the program as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the program as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the program.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards.



**Mamadou HONADIA**  
Chevalier de l'Ordre de l'Étalon

MINISTERE DE L'ENVIRONNEMENT,  
DE L'ASSAINISSEMENT ET DU  
DEVELOPPEMENT DURABLE

REPUBLIQUE DU MALI 79  
*Un Peuple - Un But - Une Foi*

Agence de l'Environnement et du  
Développement Durable (AEDD)

Bamako, le 31 DEC 2018



*The Director General of the Environment  
and Sustainable Development Agency*

To

Mr Javier Manzanarez, Executive Director i.a  
Green Climate Fund Songdo Business District, 175  
Art Center-daero Yeonsu-gu, Incheon 22004  
Republic of Korea

N° 2693 /MEADD/AEDD-DMRF

Re **BOAD** proposal for the GCF by Banque Ouest Africaine de Développement (BOAD) regarding  
"BOAD Climate Finance Facility to Scale Up Solar Energy Investments in Francophone West Africa  
LDCs"

Dear Madam, Sir,

We refer to the program "BOAD Climate Finance Facility to Scale Up Solar Energy Investments in Francophone West Africa LDCs" as included in the funding proposal submitted by Banque Ouest Africaine de Développement (BOAD) to us on november 5, 2018.

The undersigned is the duly authorized representative of Ministry of Environment, Sanitation and Sustainable Development, the National Designated Authority of Mali.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the program as included in the funding proposal.

By communicating our no-objection, it is implied that :

- (a) The government of Mali has no-objection to the program as included in the funding proposal;
- (b) The program as included in the funding proposal is in conformity with Mali's national priorities, strategies and plans ;
- (c) In accordance with the GCF's environmental and social safeguards, the program as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the program as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the program.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,



The Director general,

**Boureïma CAMARA**  
Chevalier de l'Ordre national

To: Mr Javier Manzanarez, Executive Director a.i.

Green Climate Fund  
Songdo Business District  
175 Art Center-daero  
Yeonsu-gu, Incheon 22004  
Republic of Korea

Niamey, 21 JAN 2019

**Re: Funding proposal for the GCF by Banque Ouest Africaine de Développement (BOAD) regarding "BOAD Climate Finance Facility to Scale Up Solar Energy Investments in Francophone West Africa LDCs"**

Dear Madam, Sir,

We refer to the program "BOAD Climate Finance Facility to Scale Up Solar Energy Investments in Francophone West Africa LDCs" as included in the funding proposal submitted by Banque Ouest Africaine de Développement (BOAD) to us on november 5, 2018.

The undersigned is the duly authorized representative of Prime Minister of Niger, the National Designated Authority of Niger.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the program as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Niger has no-objection to the program as included in the funding proposal;
- (b) The program as included in the funding proposal is in conformity with Niger's national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, the program as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the program as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the program.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

\_\_\_\_\_  
Name: Dr. Kamayé MAAZOU "[kamayemaazou@yahoo.fr](mailto:kamayemaazou@yahoo.fr)"

Title: FOCAL PONT / National authority Designed





**MINISTÈRE DE L'ENVIRONNEMENT  
ET DES RESSOURCES FORESTIERES**

**SECRETARIAT GENERAL**



**REPUBLIQUE TOGOLAISE**  
**Travail – Liberté - Patrie**

**DIRECTION DE L'ENVIRONNEMENT**

N° 01 /PF-FVC/2019

Lomé, 03 JAN 2019

To: The Green Climate Fund ("GCF")

Re: Funding proposal for the GCF by Banque Ouest Africaine de Développement (BOAD) regarding "BOAD Climate Finance Facility to Scale Up Solar Energy Investments in Francophone West Africa LDCs"

Dear Madam, /Sir,

We refer to the program "BOAD Climate Finance Facility to Scale Up Solar Energy Investments in Francophone West Africa LDCs" as included in the funding proposal submitted by BOAD to us on October 27, 2018.

The undersigned is the duly authorized representative of Directorate of Environment/Ministry of Environment and forestry resources of the National Designated Authority of Togo.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the program as included in the funding proposal.

By communicating our no-objection; it is implied that:

- (a) The government of Togo has no-objection to the program as included in the funding proposal;
- (b) The program as included in the funding proposal is in conformity with Togo's national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, the program as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the program as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the program.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

\_\_\_\_\_  
Name: Mr. AGRIGNAN Ezzo -Sam Abdou -Rassidou

Title: FOCAL POINT /National Authority Designed"





GOVERNO DA  
GUINÉ-BISSAU

SECRETARIA DE ESTADO DO AMBIENTE  
Direcção Geral do Ambiente

Bissau, October 29th, 2018

To: The Green Climate Fund ("GCF")

Re: Funding proposal for the GCF by Banque Ouest Africaine de Développement (BOAD) regarding "BOAD Climate Finance Facility to Scale Up Solar Energy Investments in Francophone West Africa LDCs"

Dear Madam, Sir,

We refer to the program "BOAD Climate Finance Facility to Scale Up Solar Energy Investments in Francophone West Africa LDCs" as included in the funding proposal submitted by Banque Ouest Africaine de Développement (BOAD) to us on November 5, 2018.

The undersigned is the duly authorized representative of Secretariat of State of Environment, the National Designated Authority of Guinea-Bissau.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the program as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Guinea-Bissau has no-objection to the program as included in the funding proposal;
- (b) The program as included in the funding proposal is in conformity with Guinea-Bissau's national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, the program as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the program as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the program.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Mr. Vitorio Luis Soares Cassama

GCF Focal Point



## Environmental and social report(s) disclosure

| Basic project/programme information                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Project/programme title                                                              | BOAD Climate Finance Facility to Scale Up Solar Energy Investments in Francophone West Africa LDCs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Accredited entity                                                                    | Banque Ouest Africaine de Développement (BOAD)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Environmental and social safeguards (ESS) category                                   | Intermediation 2 (I2)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Specific project location or target country/ies of the programme, as the case may be | Benin, Burkina Faso, Guinea-Bissau, Mali, Niger, and Togo                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Environmental and Social Impact Assessment (ESIA) (if applicable)                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Date of disclosure on accredited entity's website                                    | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Language(s) of disclosure                                                            | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Link to disclosure                                                                   | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Other link(s)                                                                        | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Environmental and Social Management Plan (ESMP) (if applicable)                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Date of disclosure on accredited entity's website                                    | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Language(s) of disclosure                                                            | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Link to disclosure                                                                   | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Other link(s)                                                                        | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Resettlement Action Plan (RAP) (if applicable)                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Date of disclosure on accredited entity's website                                    | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Any other relevant ESS reports and/or disclosures (if applicable)                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Description of report/disclosure                                                     | Environmental and Social Management Framework (ESMF)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Date of disclosure on accredited entity's website                                    | 2019-01-24                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Language(s) of disclosure                                                            | English, French and Portuguese                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Link to disclosure                                                                   | <p>English:<br/> <a href="https://www.boad.org/wp-content/uploads/2016/10/ESMF-Facility-to-Scale-Up-Solar-Energy-Investments-in-West-Africa-LDCs-English.pdf">https://www.boad.org/wp-content/uploads/2016/10/ESMF-Facility-to-Scale-Up-Solar-Energy-Investments-in-West-Africa-LDCs-English.pdf</a></p> <p>French:<br/> <a href="https://www.boad.org/wp-content/uploads/2016/10/CGES-Programme-Promotion-des-Investissements-Priv%C3%A9s-Secteur-de-lEnergie-Solaire-BOAD-FVC-FRENCH.pdf">https://www.boad.org/wp-content/uploads/2016/10/CGES-Programme-Promotion-des-Investissements-Priv%C3%A9s-Secteur-de-lEnergie-Solaire-BOAD-FVC-FRENCH.pdf</a></p> <p>Portuguese:<br/> <a href="https://www.boad.org/wp-content/uploads/2016/10/QGAS-Amplia%C3%A7%C3%A3o-Investimentos-Energia-Solar-em-PMA-da-%C3%81frica-Occidental--Portugu%C3%AAs.pdf">https://www.boad.org/wp-content/uploads/2016/10/QGAS-Amplia%C3%A7%C3%A3o-Investimentos-Energia-Solar-em-PMA-da-%C3%81frica-Occidental--Portugu%C3%AAs.pdf</a></p> |



|                                                               |                                                                                                                                                                                                                                                         |
|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                               | An Environmental and Social Management System (ESMS) consistent with the requirements for a category I-2 programme is contained in the ESMF.                                                                                                            |
| Other link(s)                                                 | <a href="https://www.boad.org/en/environmental-and-social-assessments/">https://www.boad.org/en/environmental-and-social-assessments/</a>                                                                                                               |
| <b>Disclosure in locations convenient to affected peoples</b> |                                                                                                                                                                                                                                                         |
| Date                                                          | 2019-01-24                                                                                                                                                                                                                                              |
| Place                                                         | BOAD Headquarters in Togo and resident missions in Benin, Burkina Faso, Guinea-Bissau, Mali and Niger. Please refer to the link below for further information:<br><br><a href="https://www.boad.org/en/contacts/">https://www.boad.org/en/contacts/</a> |

**Note: This form was prepared by the accredited entity stated above.**

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## Independent Technical Advisory Panel's review of FP105

|                         |                                                                                                    |
|-------------------------|----------------------------------------------------------------------------------------------------|
| Proposal name:          | BOAD Climate Finance Facility to Scale Up Solar Energy Investments in Francophone West Africa LDCs |
| Accredited entity:      | Banque Ouest Africaine de Développement (BOAD)                                                     |
| Project/programme size: | Medium                                                                                             |

### I. Assessment of the independent Technical Advisory Panel

#### 1.1 Impact potential *Scale: High*

##### 1.1.1 Brief description of the programme concept, goals and objectives

1. Slightly less than 50 per cent of the population in West Africa do not have access to electricity. The target countries for this programme are all in West Africa and belong to the subgroup collectively referred to as francophone West Africa. Six of the countries in this subgroup are least developed countries (LDCs) and are targeted by this programme, namely: Benin, Burkina Faso, Guinea-Bissau, Mali, the Niger and Togo. These countries are affected by poor access to electricity, poor energy security, and a high cost of electricity delivered to consumers, the common causes of which include:
  - (a) High dependence on fossil fuel, almost all of which is imported, for generating power;
  - (b) Insufficient availability of public funds to invest in the development of energy infrastructure needed;
  - (c) Inability to attract private capital to develop the necessary infrastructure for power generation; and
  - (d) High inefficiencies along the value chain of existing energy, and specifically electricity, infrastructure value chain.
2. As a group, the energy mix of the target countries is dominated by fossil fuel (76 per cent), followed by hydro (19 per cent) and solar (five per cent). In spite of the recent decline in the cost of solar electricity generation and the fact that many areas of the targeted countries are in the Sahel, which has high solar irradiation characteristics, solar photovoltaic (PV) power generation capacities in the six countries totalled only 70 megawatts (MW) at the end of 2017. Under their nationally determined contributions (NDCs) the targeted countries are expected to reach 1,192 MW by 2030. The actual capacity as of 2017 is therefore an underperformance, which if allowed to continue will mean the countries will not be able to meet that target.
3. A strategic plan is therefore called for that will put these six countries on a path to ensure that the NDC plan for solar peak capacity in 2030 is met. Evaluation of the circumstances on the ground in each of these countries, and especially the barriers to meeting the solar power generation capacity foreseen under the NDCs, showed that the strategic plan must catalyse private sector funding for the development of solar power to the grid. In the development of this programme, concessional funding from GCF was seen as key to achieving this. The West African Development Bank (BOAD) is expected to provide financial stimulus by making available funds from its own sources to blend with the concessional GCF funds. This will provide a blended finance approach to increasing solar investments (photovoltaic (PV) and concentrated solar

power (CSP)) through long-term debt financing to qualifying private sector companies with the capacity to invest in commercial scale grid-connected projects that, under the current market conditions, would not attract affordable financing at scale.

4. If successful, this blended finance approach is expected to deliver additional outcomes that will include:

- (a) Decarbonization of the energy mix of all the targeted countries;
- (b) Improvement in energy access in each of the countries through the scaling up of cost-competitive solar technologies; and
- (c) Development of a strategic plan that can be used to catalyse private sector funding of solar technologies in other LDCs, beyond those targeted under this programme.

5. It has been estimated that the total peak capacity of solar power projects to be delivered by this intervention will be about 215 MW. A summary of the expected funding is presented as follows:

- (a) The total funding that will be provided by both GCF and BOAD is estimated to be approximately EUR 122 million;
  - (b) This total fund will be structured as follows:
    - (i) A GCF loan of EUR 57 million as senior debt;
    - (ii) A GCF grant of EUR 4 million;
    - (iii) A BOAD loan of EUR 60 million as senior debt;
    - (iv) A BOAD grant of EUR 1 million; and
    - (v) A further EUR 128.1 million is expected to be leveraged by this blended GCF/BOAD loan as equity from project developers and other lenders (private and public sector banks) in the region; and
  - (c) Of the GCF long-term loan, EUR 45 million will be senior loan facilities available for the funding of the solar power projects; EUR 12 million (also senior loan) will be a standby loan facility for credit enhancement for tenor extension of the commercial loan available for the project from local financial institutions, unable to offer long-term and low interest loans to solar power projects. This credit enhancement facility for tenor extension is expected to incentivize the participation of local commercial banks. The GCF standby loan of EUR 12 million to BOAD is for on-lending to subprojects by facilitating the extension of commercial bank loans available at maturity to match the tenor of BOAD loans to such subprojects.
6. The structure of the programme as proposed in this funding proposal can be summarized as follows:
- (a) Component 1: to scale up commercial and sustainable financing for solar investments – direct lending (loan component):
    - (i) Subcomponent 1.1: senior loan facility; and
    - (ii) Subcomponent 1.2: standby loan facility (credit enhancement in the form of tenor extension); and
  - (b) Component 2: to develop technical capacity of private and public sector actors and raise awareness – technical assistance (grant component):
    - (i) Subcomponent 2.1: technical assistance to BOAD and local private sector actors for investment preparation;

- (ii) Subcomponent 2.2: technical assistance to national institutions to address legal and regulatory barriers in the solar market; and
- (iii) Subcomponent 2.3: marketing and awareness-raising activities.

### 1.1.2 Climate impact potential of the programme

7. The primary impact potential will be the reduction of greenhouse gas (GHG) emissions through the displacement of fossil fuel electricity generation by the solar power facilities implemented as a result of the programme.

8. It has been estimated that the programme will deliver about 215 MW of new solar power generating capacity, which will deliver approximately 339 gigawatt hours (GWh) of clean electricity annually thereby directly avoiding emissions of more than 190,000 tonnes of carbon dioxide equivalent (tCO<sub>2</sub>eq) per year (equivalent to 4.8 MtCO<sub>2</sub>eq over 25 years), that would be emitted by the current fossil fuel system.

9. The GHG emissions reduction (ER) methodology utilized in this calculation is summarized as follows:

- (a) The power supply systems in five of the six targeted countries, namely, Benin, Burkina Faso, Mali, the Niger and Togo, are connected to the West African Power Pool, while Guinea-Bissau's system is not. Therefore, each kilowatt hour (kWh) of electricity generated by solar power plants under this programme in the five countries will be deemed to displace a kWh of electricity on the baseline West African Power Pool network;
- (b) The GHG emissions reduction as a result of the delivery of electricity from the five targeted countries has been calculated using the grid emission factor from the Clean Development Mechanism methodology "Standardized baseline – Grid emission factor for West African Power Pool" (ASB0034: Grid emission factor for West African Power Pool (version 01.0)). This standardized baseline provides the values for grid emission factors (i.e. the CO<sub>2</sub> emission factors) for the West African Power Pool, the interconnected electricity system of nine countries: Benin, Burkina Faso, Côte d'Ivoire, Ghana, Mali, the Niger, Nigeria, Senegal and Togo. The grid emission factor used is 0.561 tCO<sub>2</sub>/MWh;
- (c) For the solar project in Guinea-Bissau, the country's emission factor, estimated as 0.765 tCO<sub>2</sub>/MWh was used to calculate the GHG ER implication of the solar power project implemented to the grid in that country; and
- (d) The total GHG emissions reduction of the entire programme was estimated as the sum of the ER from the solar projects in the five countries connected to the West African Power Pool and the ER achieved by the solar facilities connected to the stand-alone grid of Guinea-Bissau. Total ER in the six selected francophone countries was estimated to be 193,398 tCO<sub>2</sub>eq per annum.

10. In addition to the impact of GHG emissions reduction, it is expected that the programme will deliver clean electricity to an estimated 2.9 million people. This number is calculated using the per capita electricity consumption for the region (116 kWh/person-year<sup>1</sup>), and the total of 339 GWh of clean electricity expected to be delivered by the solar power projects created by the programme.

11. Given these impacts, the independent Technical Advisory Panel (TAP) assessed the scale of the impact potential as "high".

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<sup>1</sup> Available at <<https://www.se4all-africa.org/seforall-in-africa/country-data/mali/>>.

## 1.2 Paradigm shift potential

*Scale: High*

### 1.2.1 General overview

12. The projects covered by this programme will involve the development of grid-scale solar power projects. In terms of the technologies that will be implemented, these cannot be considered to be new as they now operate successfully in many parts of the world. However, although similar technology has been deployed in the target region in the recent past, its development has not taken place at a rate that is consistent with the vast potential of the primary resources in the region. Barriers to implementation of solar power technologies and delivery of power to the grid in the target countries are well documented in the funding proposal and include technical, financial, regulatory and policy issues. A key goal of the proposed programme is the reduction and/or elimination of these barriers. For example, a key component of the proposed intervention is the utilization of concessional financing from GCF to incentivize participation of the private sector (local investors and local financing institutions) in the implementation of grid-connected solar power facilities in the six target countries. Removing or reducing the barriers to these investments may represent a sound paradigm shift. The potential of this paradigm shift is discussed in the following subsections.

### 1.2.2 Potential for scaling up and replication

13. As stated in the previous section, the solar power technologies that will be employed by the programme are well developed and deployed internationally. Since the focus of the programme in this intervention is to put in place activities that will reduce or eliminate barriers that have prevented greater deployment of these technologies in the target countries, it can be safely concluded that effective mitigation of barriers will enhance the potential for replicating and scaling up activities in each country. The technical, regulatory, policy and market initiatives developed as a result of the programme implementation should be available to support scaling up or replication of the underlying projects. It is expected that the effects of the availability of concessional funding from GCF will include: more feasible solar power projects will be delivered because a major barrier, which has limited such projects in the status quo ante – high interest rate and short loan maturity – will be removed; and the introduction of better regulatory framework that will result will improve the skills of local project promoters, enabling local investors to replicate and scale-up solar power to the grid.

### 1.2.3 Potential for knowledge-sharing and learning

14. The potential for knowledge-sharing and learning of the proposed programme will be enhanced by the activities in component 2 of the proposed programme. This component has been designed to develop the technical capacity of private and public sector actors who will be involved in the different aspects of the project (implementation of infrastructure, enhancement of regulatory framework, and relevant policy issues and communications) and to raise awareness regarding the intervention among stakeholders in each of the target countries. Training activities under component 2 include: technical assistance (TA) to BOAD and local private sector actors for investment preparation; and TA to national institutions to address legal and regulatory barriers in the solar market. Activities in both these areas will generate knowledge enhancement and capacity-building, thus promoting learning and skills in each of the target countries. It is expected that approximately 300 people from the private sector will be trained. The transfer of knowledge will be ensured through the provision of operational manuals and guidelines and practical tools. TA to local project developers will be complemented by capacity-building for BOAD staff during the development and implementation of the project portfolio. BOAD experience in solar project investment and in mitigation and adaptation

projects in general is relatively new. BOAD staff expertise in several areas will be strengthened by the TA, which will be provided to remove the technical barriers faced by BOAD staff.

#### 1.2.4 **Contribution to the creation of an enabling environment**

15. The intervention will provide an enabling environment that can be measured by the extent to which the capacity of relevant stakeholders within each of the target countries is developed and strengthened. Stakeholders from BOAD and local private sector actors will acquire critical knowledge on how solar power projects are identified and developed as a result of this intervention. Furthermore, the TA activities foreseen under subcomponent 2.2 will provide public sector stakeholders with skill sets that will enhance their capability of understanding and handling technical, regulatory and policy issues related to solar projects. Thus, TAP is of the opinion that the proposed intervention will create an enabling environment for continued solar project development in francophone West Africa.

#### 1.2.5 **Contribution to regulatory framework and policies**

16. Subcomponent 2.2 of the proposed programme is specifically designed as a TA intervention aimed at building the capacity of national institutions to address legal and regulatory barriers in the solar market. It is expected that the regulatory and policy environment needed for development of grid-connected solar power projects will be achieved on completion of the programme.

17. The TAP therefore concluded that, based on the expected performance of the proposed programme, it will deliver a paradigm shift ensuring that the countries targeted will achieve an accelerated departure from the use of fossil fuel to deliver electricity to the grid, and will increase the utilization of low carbon solar power options (PV and CSP). The TAP has therefore scored the paradigm shift metric as “high”.

### 1.3 **Sustainable development potential**

*Scale: High*

18. The project contributes directly to the achievement of the following Sustainable Development Goals (SDGs):

- (a) SDG 1 – No poverty: end poverty, in all forms, everywhere;
- (b) SDG 5 – Gender equality: achieve gender equality and empower all women and girls;
- (c) SDG 7 – Affordable and clean energy: ensure access to affordable, reliable, sustainable and modern energy for all;
- (d) SDG 8 – Industry innovation and infrastructure: build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation;
- (e) SDG 10 – Reduced inequalities: reduce income inequality within and among countries;
- (f) SDG 11 – Sustainable cities and communities: make cities and human settlements inclusive, safe, resilient and sustainable;
- (g) SDG 12 – Responsible consumption and production: ensure sustainable consumption and production patterns;
- (h) SDG 13 – Climate action: take urgent action to combat climate change and its impacts by regulating emissions and promoting developments in renewable energy; and
- (i) SDG 17 – Partnerships for the goals: strengthen the means of implementation and revitalize the global partnership for sustainable development.



19. The major effort of this programme is expected to result in incentivizing the development of solar power to the grid in each of the six target countries through enabling good financing, regulatory and project development for the environment. The programme is expected to deliver environmental, economic and social co-benefits as well as gender-sensitive development impacts.

#### 1.3.1 **Economic and social co-benefits**

20. Renewable energy resources, most especially solar irradiation, are relatively abundant in each of the six target countries. Therefore, the opportunity to generate clean electricity at a relatively low cost can easily be created by the successful implementation of the proposed programme. This means that the pre-programme dependence on imported fossil fuel for electricity generation can be significantly reduced. Increased power generation with the solar facilities that will be connected to the country grids will reduce the need to import fossil fuel, reduce existing trade deficits caused by the import of refined oil products, and secure energy supply through diversified energy sources. Reduction of trade deficits in each of the target countries will improve their economies while also creating new job opportunities during the construction and operation of solar power plants. It has been estimated in the funding proposal reviewed that about 500 green jobs will be created during the construction phase ranging from highly skilled (field technicians, operators, corporate staff, etc.) to unskilled labour work (support and service), with an additional 200 jobs, approximately, for operations and maintenance. The increased, stable and affordable energy access that will be promoted by the proposed programme will bring about a positive impact to the sustainable growth of the region as a social co-benefit.

#### 1.3.2 **Environmental co-benefits**

21. Environmental co-benefits will also be delivered in each of the target nations due to the reduced use of fossil fuel for electricity generation; reducing the emission of pollutants, including GHGs, will deliver cleaner air. The stable supply of electricity will reduce dependence on charcoal and firewood for cooking, contributing to the reduction of deforestation and associated carbon emissions. This will have additional positive health impacts for the population.

#### 1.3.3 **Gender co-benefits**

22. Improved and more reliable access to electricity will produce strong co-benefits for gender-sensitive development. The initial design of the programme will take into account the different ways that women and men access energy, and the role that energy can play in increasing opportunities for women; technical support will be provided to promote gender equality throughout the programme's lifetime. The programme will add significant value in the target sector, where gender equality is poorly addressed. By mainstreaming gender throughout its lifetime, the programme will be in line with the GCF Gender Policy and Action Plan. Activities targeting women's participation and training as part of component 2 will ensure that the programme is gender sensitive and the benefits are shared among men and women. More information on gender mainstreaming in the programme is provided in the gender assessment and action plan included as annex 3 of the funding proposal.

23. Based on the assessment above, TAP assessed the sustainable development potential of the proposed programme as "high".

## 1.4 Needs of the recipient

*Scale: High*

### 1.4.1 Vulnerability of the countries and beneficiary groups

24. All target countries are classified as LDCs<sup>2</sup> and according to the Notre Dame-Global Adaptation Initiative Country Index, and they are all among the most vulnerable countries to climate change. Key challenges faced by all six countries include low access to electricity, a high cost for electricity and overdependence on imported fossil fuels for electricity generation, which will continue in the absence of this intervention. The inadequate availability of public funds to invest in energy infrastructure to meet the growing demand of electricity in each of the target countries, coupled with the difficulty of attracting private funds for investment in energy projects, means that the countries will continue to be vulnerable to these challenges. In the absence of this intervention, consumer groups will continue to face low access to electricity. Therefore, the proposed programme will address the needs of the beneficiary countries by improving access to clean and reliable energy for approximately 2.9 million people. The proposed programme will also create an enabling environment for increased financing of solar power provided to the grid and will deliver GHG emissions reduction as a result of the displacement of fossil-fuel-based generation, thus contributing to the achievement of the NDCs of the targeted countries.

### 1.4.2 Financial, economic, social and institutional needs

25. **Economic and social development level of the country and affected population.** Although all six Francophone countries targeted by this programme have experienced economic growth in recent years, with several averaging more than five per cent gross domestic product growth per annum, this has not translated to national poverty reduction. In most of these nations, more than half of the population still live below the national poverty line and a significant proportion of them do not have access to modern energy, including electricity. It is recognized in all the target countries that the prevailing poor access to electricity is one of the key barriers to better economic growth and social development. Therefore, improvement of access to electricity in both the rural and urban areas of the country has long been recognized as an important development strategy. Improved access cannot be achieved by relying on the status quo, which depends heavily and almost entirely on imported fossil fuel with a highly negative trade imbalance. The targeted countries are well endowed with solar energy resources, offering a credible alternative pathway to improved electricity access and hence of meeting the social and economic development needs in the region. To achieve these results, the barriers identified above – financial, technology, regulatory and other – must be properly mitigated. The proposed programme will initiate a systemic reduction of these barriers and therefore mobilize climate finance at scale, contributing to a paradigm shift and introducing a low-emission development pathway. The programme will accelerate private sector involvement in the solar sector, enabling market transformation and contributing to the economic and social needs of a large segment of the relevant populations. The funding structure of the proposed programme will also contribute to strengthening the institutional capacity required to sustain the development and supply of solar power to the grid in the targeted countries.

26. **Absence of alternative sources of financing.** A key barrier to a low carbon path to improving energy access has been the lack of funds (local equity, loans/debt – with medium to long maturity and lower interest rates) needed to implement clean power projects. BOAD, as an important regional financing institution, has a strategic objective to increase its portfolio of investments in the solar sector directly to private sector actors. This has not materialized as

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<sup>2</sup> United Nations, List of Least Developed Countries (as of March 2018), available at [https://www.un.org/development/desa/dpad/wp-content/uploads/sites/45/publication/ldc\\_list.pdf](https://www.un.org/development/desa/dpad/wp-content/uploads/sites/45/publication/ldc_list.pdf).

BOAD has been unable to mobilize funds at the level needed and at terms that will increase participation of the private sector in this endeavour. The proposed programme, which will enable BOAD to blend its resources with concessional debt financing from GCF, should contribute significantly to reducing the financial barriers around incentivizing low carbon energy provision. The availability of this blended finance option will also help to crowd in private sector equity and debt funding for solar power projects to the grid and thus provide a needed alternative fund source.

27. **Need for strengthening institutions and implementation capacities in the targeted countries.** Apart from the financial barrier discussed above, weaknesses are prevalent in institutional, regulatory, legal and policy frameworks, for example, tariff structures hindering the implementation of solar power facilities to the grid. The proposed GCF/BOAD programme grant funding will provide TA to relevant institutions in these countries to reduce or completely eliminate the negative impacts of regulatory and policy barriers. For example, as part of the grant funding, BOAD will carry out policy dialogue activities that will be informed by the activities of the proposed investments. The rationale of component 2.2 of this intervention is to remove the market barriers to the development of solar projects by supporting the implementation of regulatory and policy regimes that will provide clear and predictable rules for solar project development in the region. The TA under this programme is expected to stimulate the necessary conditions for a robust take-off and increase in private sector solar investment in the region. This will partly be achieved through institutional strengthening provided under the TA component of this intervention.

28. Given the various country needs that will be satisfied by the GCF/BOAD programme in the six targeted West African countries, TAP has concluded that the country need of the recipient countries and affected consumers can be rated as “high”.

## 1.5 Country ownership

*Scale: High*

29. Country ownership of the proposed GCF/BOAD intervention is discussed according to the following submetrics.

### 1.5.1 Alignment of the proposed programme with national climate strategies

30. A key document articulating the national climate strategies of targeted countries is the intended nationally determined contribution (INDC). All six of the targeted countries prepared the INDC and several have submitted the document, as required by international climate change agreements, to the United Nations Framework Convention on Climate Change secretariat in 2015/2016. All the countries' INDCs pointed to the need to implement solar power facilities to the grid, as well as off-grid, as a key strategy to achieve the national contribution to the global goal of GHG emissions reductions. In addition to the INDC documents, each target country also has in place national documents elaborating their climate change strategies. Measures spelt out in these documents, especially for the energy sector, included GHG emissions mitigation through the implementation of solar power systems, especially to the power grids, and as a means of increasing access to electricity across the various economic sectors. Therefore, it can be concluded that the proposed GCF/BOAD programme is aligned to country climate strategies.

### 1.5.2 Capacities of accredited and executing entities to successfully deliver this programme

31. BOAD will serve as both the accredited and executing entity for this project. Some of the relevant attributes of BOAD that point to the organization's capacity to deliver this programme are summarized as follows:

- (a) A key priority of BOAD, pre-dating the development this programme, is the intensification of its activities in francophone West Africa through increased provision of support to private sector development. This priority is in line with the planned intervention, which will enable the private sector to actively participate in development and implementation of solar power facilities. To maximize the programme impact, some of the activities of the TA component have been designed to improve the capacity of BOAD to develop, manage and implement the programme portfolio and to integrate climate change dimensions into project activities.
- (b) Over the years, BOAD has successfully used the resources that it mobilizes to invest in public and private sector projects and programmes aimed at building basic and modern infrastructure, improving rural livelihoods, generating energy, and climate change adaptation and mitigation. Key examples provided in documents reviewed by TAP include:
- (i) In West Africa, the promotion and financing of private sector actors by BOAD has taken the form of promoting financing vehicles and funding through commercial banks and financial institutions as well as direct loans to projects and the financing of studies;
  - (ii) BOAD has in recent times, partnered with entities, such as the Agence Française de Développement, German Development Agency, European Investment Bank, and African Development Bank, to set up specific vehicles including the Global Adaptation and Resilience Investment Fund (providing guarantees for private investments in West Africa) and Cauris Investissement, a venture capital company, that has greatly stimulated private sector investment in francophone West Africa;
  - (iii) Since 2013, BOAD has organized capacity-building activities and outreach with international financial institutions such as the World Bank, African Development Bank and United Nations Framework Convention on Climate Change secretariat to bring more awareness to the West African National Financial Institutions (NFIs) and commercial bankers in the financing of energy transition projects;
  - (iv) In June 2017 BOAD launched the creation of the West African network for climate finance. BOAD is also promoting the use of green bonds and green credit line mobilization throughout the region. Out of the 4,202.1 billion African Financial Community (CFA) franc (XAF) of total net commitments as at 30 September 2016, BOAD devoted about 24.3 per cent to the private sector, representing XAF 1,022.4 billion for 319 projects;
  - (v) BOAD private sector financing operations and the promotion of financial inclusion are carried out through the "Private Sector" window. The shareholding capital of BOAD for companies and financial institutions includes 82 transactions so far; and
  - (vi) Other initiatives set up by BOAD include: Regional Initiative for Sustainable Energy, set up to better overcome recurrent energy crises in many of the West African Economic and Monetary Union (WAEMU) member countries; the Energy Development Fund, established 25 September 2009, to contribute to improved access to energy in WAEMU member countries and to promote the development of sustainable energy under concessional conditions to finance projects eligible for the Regional Initiative for Sustainable Energy initiative; and the Infrastructure Development Fund, a profit-making fund for financing access to electricity in the WAEMU region.

- (c) BOAD is a GCF accredited entity and the above record of activities indicates its ability to act as executing entity for this programme. It is also important to stress that where the capacity of BOAD is deemed to be inadequate, especially its functions in programme execution, plans are in place to strengthen its capacity through the grant-funded TA.

### **1.5.3 Engagement with national designated authorities, civil society organizations and other relevant stakeholders**

32. BOAD has engaged a wide range of stakeholders in each of the targeted countries to ensure country ownership. These interactions included discussions and meetings with the national designated authorities of the targeted countries; and the successful collection of letters of no objection from all six countries (included as an annex to the funding proposal). BOAD is also continuing engagement with country stakeholders, including national institutions and authorities in the energy sector, local banks and major regional public banks, including the Bank for Investment and Development of the Economic Community of West African States, which has indicated support for the BOAD Climate Finance Facility and strong interest in co-financing climate projects.

33. Given the discussion above, TAP has concluded that country ownership is very strong as all targeted countries have shown interest in the programme. As such, the country ownership metric of this programme has been ranked as “high”.

## **1.6 Efficiency and effectiveness**

*Scale: Medium*

34. Efficiency and effectiveness of the proposed programme will be measured by the extent to which the intervention is able to release financing for implementation of solar power delivery to the grid, displacing fossil-fuel-based power generation and thereby achieving significant reductions of GHG emissions. The economic and financial soundness of the proposed programme are discussed as follows:

### **1.6.1 Cost effectiveness and efficiency**

35. Since public sector funding is not adequate to support delivery of solar power to the grid at the level needed, the effectiveness of the proposed intervention can be measured by the success and extent of private sector funding mobilization for solar power investment. According to the information included in the funding proposal, the concessional loan from GCF, which will be blended by BOAD with its own additional debt funding, will ensure that debt funding at better conditions (lower interest rate and longer tenure) will be available to solar projects in the countries targeted, acting as incentives for private sector actors. The concessionality received from GCF will enable BOAD to offer a final blended cost of capital, which will be below commercial rates. Furthermore, a key objective of the proposed programme is to crowd in private sector investment for the solar sector from commercial banks; this will be achieved through the provision of a tenor extension instrument from the concessional funding available from GCF. The efficiency of the programme will therefore be measured as follows:

- (a) Based on the way in which concessional funding will be made available, it has been estimated that the programme will leverage about EUR 128.1 million of private investment from project developers and local private and public banks; and
- (b) The programme is expected to deliver GHG emissions reductions totalling about 3.8 million tCO<sub>2</sub>eq over the 20-year lifespan of the programme, resulting in an abatement cost for GCF of about EUR 15 per tCO<sub>2</sub>eq.

### **1.6.2 Co-financing, leveraging and mobilization of long-term investment**



36. The programme has been designed to mobilize investment from GCF (debt and grant) as well as funds from BOAD (debt and grant) in such a way that public funding is not needed. The co-financing ratio for BOAD and GCF is therefore 1:1 as each entity will provide funding to the tune of EUR 61 million;

37. It is expected that the concessional funding from GCF, when blended with funds from BOAD, will also leverage a substantial amount of financing from project developers and private sector banks. Project developers are expected to provide equity investment amounting to EUR 50 million while the debt financing from local banks is expected to be about EUR 78.1 million. It is important to stress that the debt financing from local banks will only materialize as a result of the credit enhancement facility that will be made available by GCF in the form of tenor extension and credit enhancement for solar projects financed under the programme ("Standby Loan Facility"). The tenor extension instrument is intended to crowd in participation from interested and credible local commercial banks for solar projects financed under the programme ("Standby Loan Facility"). In total, the programme will achieve a leverage ratio of 1:3.

#### 1.6.3 **Financial viability of the programme**

38. Without the financial support of GCF for this programme, it is doubtful that delivery of solar power to the grid would materialize. The internal rate of return (IRR) for the programme with and without GCF funding shows the increased viability of this option as follows:

- (a) Without GCF: IRR of 11 per cent with a net present value of EUR 5.4 million will result. At such return levels, the private sector would not be willing to invest in this sector; and
- (b) With GCF: IRR of 18 per cent with a net present value of EUR 14.3 million will be recorded. This level of return is expected to secure private investment at scale.

39. The GCF/BOAD intervention will therefore deliver higher efficiency that will allow private sector investment to be released for the programme.

#### 1.6.4 **Application of best practices**

40. Equipment and technology will be selected from the best options available on the market. During the appraisal process of the projects to be funded under the programme, BOAD will support the application of best available technologies by requiring projects to demonstrate that they use solar technologies that are:

- (a) Effective in addressing climate-related goals;
- (b) Market-tested, technically performing and commercially viable; and
- (c) Contribute to market transformation in terms of demonstrating the viability and commercial application of new technologies.

41. Key components of the TA that will ensure the programme achieves high efficiency and effectiveness will include:

- (a) Support to project developers in selecting the technology suitable for the needs identified, to ensure viability of the technology and soundness of the investment;
- (b) Assessing the quality of equipment available in the market to help project developers to select a robust technology and avoid underperformance of the equipment;
- (c) Help for project developers in designing a monitoring system, which will be used to assess the equipment's performance and to identify any potential technical dysfunctions that may put at risk impacts (carbon savings) and efficiency (EUR/tCO<sub>2</sub>eq);



- (d) Encouraging project developers to engage maintenance contracts that will guarantee effective repair of the equipment and are therefore a complementary tool to ensure overall investment effectiveness; and
- (e) Securing continuous reviews of the selection of technologies and equipment to avoid underperforming technologies.

#### 1.6.5 **Key efficiency and effectiveness indicators**

42. The parameters utilized for the estimation of the indicators can be summarized as follows:

- (a) Total project financing = EUR 122 million;
    - (i) Of which GCF loan = EUR 57 million;
    - (ii) Of which GCF grant = EUR 4 million;
    - (iii) And BOAD loan = EUR 60 million; and
    - (iv) With BOAD grant = EUR 1 million; and
  - (b) Expected GHG emission reduction over 20-year lifetime = 3,867,964 tCO<sub>2</sub>eq.
43. Cost efficiency/effectiveness indicators for a 20-year project are estimated as follows:
- (a) Estimated cost per tCO<sub>2</sub>eq (for total project financing) = EUR 32/tCO<sub>2</sub>eq; and
  - (b) Estimated GCF cost per tCO<sub>2</sub>eq removed = EUR 15/tCO<sub>2</sub>eq.
44. Another set of efficiency/effectiveness indicator measures for this programme include:
- (a) Without GCF investment, the programme would have leveraged a 20-year IRR of 11 per cent compared to an IRR of 18 per cent with GCF investment;
  - (b) It has been estimated that without the GCF investment, the project has a negative net present value of about EUR 5.4 million compared to a positive net present value of about EUR 14.3 million with GCF funding; and
  - (c) It can therefore be concluded that the programme will have a better possibility of achieving a much higher efficiency/effectiveness ratio with GCF funding than without.
45. Based on the discussions presented above (item 17), TAP considered the efficiency and effectiveness of the programme can be “high”. However, the ability to achieve a high score on this metric is highly dependent on a successful capacity-building programme. The draft of the capacity-building plan, included by the accredited entity in an annex to the funding proposal, falls short with respect to its content; TAP feels that the programme design will not adequately deliver a high level of efficiency and effectiveness. TAP therefore scored this parameter as “medium”. TAP believes, however, that if continuous annual monitoring of the capacity-building element is incorporated into implementation of the programme, this parameter will eventually be rated as “high”.

## **II. Overall remarks from the independent Technical Advisory Panel**

46. This proposed programme meets GCF performance standards at a high level for: impact; paradigm shift; sustainable development potential; need of the recipient; country ownership; and achieved a medium score for efficiency/effectiveness.

47. The capacity-building plan included in the funding proposal of this submission is very broad and falls short with respect to its content. Since BOAD itself may lack the experience and capacity to produce a detailed plan immediately, TAP will recommend that the Secretariat request BOAD to include a report of capacity development and knowledge-sharing/transfer activities done, and a schedule for the next year in its annual performance report to GCF. With this recommendation, it will be possible to ensure that capacity development and knowledge-sharing/transfer activities which are very critical for the achievement of a high score on efficiency and effectiveness of the programme are being monitored in more detail.

48. With improved reporting in place, TAP feels confident to recommend this programme for approval by the Board.

## Reply to the Independent Technical Advisory Panel assessment findings (FP 105)

Proposal name: BOAD Climate Finance Facility to Scale Up Solar Energy Investments in Francophone West Africa LDCs

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Accredited entity: Banque Ouest Africaine de Développement (BOAD)

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Impact potential</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| AE concurs with ITAP assessment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Paradigm shift potential</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| AE concurs with ITAP assessment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Sustainable development potential</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| AE concurs with ITAP assessment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Needs of the recipient</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| AE concurs with ITAP assessment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Country ownership</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <p>AE concurs with ITAP assessment</p> <p>In Paragraph 15.b), the AE would like to add that it has recently participated in the financing of the largest private PV power plant in West Africa located in Kita, Mali. In January 2018, BOAD, the Emerging Africa Infrastructure Fund (EAIF), National Agricultural Development Bank of Mali (BNDA), Green Africa Power (GAP) and GuarantCo have secured a EUR 77 million credit facility to finance the construction and operation of a 50MW PV power plant. The financing will be used to develop the project, which will be built under a Build-Own-Operate and Transfer (BOOT) contract by local company Akuo Kita Solar SA, a subsidiary of French IPP Akuo Energy. Funding for the project was co-arranged by both BOAD and EAIF, with the 50MW installation standing as the first PV plant owned and operated by an independent producer in Mali.</p> <p>In paragraph 15. c), the AE would like to add that the African Guarantee Fund (AGF) and the ECOWAS Bank for Investment and Development (EBID) have confirmed their interest in participating in the programme. The Letters of Interests of AGF and EBID have been included in the annexes of the FP. Furthermore, Orabank has indicated its interest in co-financing solar projects through the proposed Facility.</p> |
| <b>Efficiency and effectiveness</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| AE concurs with ITAP assessment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Overall remarks from the independent Technical Advisory Panel:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| AE concurs with ITAP remarks and recommendation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

BOAD Climate Finance Facility to Scale Up Solar Energy  
Investments in Francophone West Africa LDCs

**Annex 3: Gender  
Assessment**

October 2018

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# 1. Introduction

Within the context of the six target countries, issues relating to gender equality remain an on-going challenge, even more so when it comes to energy access, poverty and affordability. While regional and national policy frameworks have existed for some time, the incorporation of women's needs and demands into energy development is absent or incomplete, and the sector remains largely male-dominated at the technical and decision-making level. While issues with regards to energy access and affordability impact the lives of both men and women, women are disadvantaged compared to their male counterparts and disproportionately suffer from low electricity access, energy insecurity and climate change. This is exacerbated in the context of women-headed households, where women miss out on the opportunity to contribute to their community because of their obligation to the household energy supply. Women in rural and isolated contexts incur even greater personal and social costs, spending several hours a day fetching wood for their cooking needs while having no or little access to electricity. Further, within the energy sector, a significant gender gap exists between men and women, with women not being seen as on par with men, resulting in fewer women technicians. The consequences of low or no electricity access, energy insecurity and unaffordability for women broadly include, but are not limited to:

- Limited capacity for professional development and fewer opportunities to be competitive in the job market, due to fewer numbers of girls enrolled in primary, secondary and tertiary education
- Inability to progress from traditional roles of women, particularly in the household where women are responsible for fetching wood for cooking
- Negative impacts on women's health: cooking fumes, inaccessibility of health care without electricity etc.
- Lower income and welfare compared to men and men-headed households
- Lower education levels
- Marginalisation of women from political and social life

The six countries targeted by the proposed programme show low to very low rankings in terms of both the Gender Inequality Index (GII) and the Human Development Index (HDI), as seen in Table 1 below. As of 2017, Mali scored the lowest of all target countries for Gender Inequality with a ranking of 157, while Togo scored the highest with 140, although this ranking is still among the lowest in the world. With regards to human development, all target countries are ranked very low, from 163 for Benin to 189 in Niger. These poor rankings show that social development inclusive of gender considerations will help advance human and economic development as a whole. Adversely, low social and gender development levels will hamper economic and human development. Indeed, as women represent around half of the total population in target countries (See Table 2), it is economic and political non-sense not to include them at every step of project development, and in particular in the context of energy interventions.

Table 1: Gender Inequality Index and Human Development Indicator by country

| Country       | Gender Inequality |         | Human Development |         |
|---------------|-------------------|---------|-------------------|---------|
|               | Index (GII)       | Ranking | Index (HDI)       | Ranking |
| Benin         | 0.611             | 146     | 0.515             | 163     |
| Burkina Faso  | 0.610             | 145     | 0.423             | 183     |
| Guinea-Bissau | N/A               | N/A     | 0.455             | 177     |
| Mali          | 0.678             | 157     | 0.427             | 182     |
| Niger         | 0.649             | 151     | 0.354             | 189     |
| Togo          | 0.567             | 140     | 0.503             | 165     |

Source: UNDP data, 2017



Data for women-headed households in the targeted countries is scarcely available, but the data that is available shows quite an important share of households with a woman head in Burkina Faso (11.8% in 2014), in Mali (5.3%) and a significant share of women-headed households in Togo (27.5% in 2014)<sup>1</sup>. Households with a woman head must be taken into account in energy interventions because they have different needs and challenges than men-headed households. For example, the woman head will be responsible for the household's energy supply as well as the income, as opposed to men-headed households where the man will be in charge of bringing an income to the household. This means that the lack of energy access and/or energy insecurity impacts women-headed households disproportionately compared to men-headed households, as less time is available for them to generate an income, which is further hampered in the event of energy/electricity deficiency. Women-headed households are therefore more at risk of poverty than men-headed households, with fewer means to alleviate their situation.

*Table 2: Total population disaggregated by sex*

| Country       | Total population | Population, male (% of total) | Population, female (% of total) |
|---------------|------------------|-------------------------------|---------------------------------|
| Benin         | 11,175,692       | 49.9                          | 50.1                            |
| Burkina Faso  | 19,193,382       | 49.9                          | 50.1                            |
| Guinea-Bissau | 1,861,283        | 49.2                          | 50.8                            |
| Mali          | 18,541,980       | 50.0                          | 50.0                            |
| Niger         | 21,477,348       | 50.1                          | 49.9                            |
| Togo          | 7,797,694        | 49.8                          | 50.2                            |

*Source: World Bank, 2017*

In the six target countries, gender inequalities evidently affect more women than men. For example, when analysing key indicators on gender inequality (See Table 3), the results show that women are socially, economically and politically more disadvantaged than men. Maternal mortality ratios are high in all target countries, ranging from 368 deaths per 100,000 live births in Togo to 549 in Guinea-Bissau. Across all gender inequality key indicators, Togo has the most favourable scores, while scores from Mali show the greatest gender disparities.

On average in all six target countries, one-quarter of seats in parliament are held by women, while as little as 10% of all women have some secondary education. The share of the male population with at least secondary education is around 20%, which is twice that of women but still low in light of the region's developmental challenges. Labour force participation rates also show important disparities, with the greatest difference recorded in Mali with female labour force participation showing a 21.7 point low compared to a male labour force participation of 82.5. These results derive from the traditional roles culturally assigned to women, who lack opportunities to go to school and later are responsible for providing the household with an energy supply for cooking. Similarly, adolescent birth rates over the period 2015-2020 represent around 120.8 births per 1,000 women aged 15-19 on average in the target countries. These rates adversely impact women's access to secondary and tertiary education, which explains the significant differences in the share of population with at least some secondary education. For example, in Togo, which scores the highest compared to other target countries, the share of men with at least some secondary education is 52.5%, double that of women with 26.3%.

---

<sup>1</sup> World Bank data

Table 3: Key indicators on gender inequality in target countries

|                                 | Maternal mortality ratio<br>(deaths per 100,000 live births) | Adolescent birth rate<br>(births per 1,000 women ages 15-19) | Share of seats in parliament<br>(% held by women) | Population with at least some secondary education<br>(% ages 25 and older) |                | Labour Force participation rate<br>(% ages 15 and older) |           |
|---------------------------------|--------------------------------------------------------------|--------------------------------------------------------------|---------------------------------------------------|----------------------------------------------------------------------------|----------------|----------------------------------------------------------|-----------|
|                                 | 2015                                                         | 2015-2020                                                    | 2017                                              | Female 2010-2017                                                           | Male 2010-2017 | Female 2017                                              | Male 2017 |
| <b>Benin</b>                    | 405                                                          | 86.1                                                         | 7.2                                               | 18.2                                                                       | 32.7           | 69.7                                                     | 73.1      |
| <b>Burkina Faso</b>             | 371                                                          | 104.3                                                        | 11                                                | 6                                                                          | 11.7           | 58.2                                                     | 75.2      |
| <b>Guinea-Bissau</b>            | 549                                                          | 84.5                                                         | 13.7                                              | N/A                                                                        | N/A            | 65.6                                                     | 78.1      |
| <b>Mali</b>                     | 587                                                          | 169.1                                                        | 8.8                                               | 7.3                                                                        | 16.4           | 60.8                                                     | 82.5      |
| <b>Niger</b>                    | 553                                                          | 192                                                          | 17                                                | 4.3                                                                        | 8.9            | 67.5                                                     | 90.7      |
| <b>Togo</b>                     | 368                                                          | 89.1                                                         | 17.6                                              | 26.3                                                                       | 52.5           | 75.8                                                     | 79.4      |
| <b>Average target countries</b> | 472                                                          | 120.8                                                        | 25.7                                              | 10.3                                                                       | 20.3           | 66.2                                                     | 79.8      |

Source: UNDP data

Therefore, gender equality remains an ongoing challenge in West Africa. In addition, as energy poverty affects men and women differently, mainstreaming gender in energy development is necessary to accommodate for the differences in the energy needs of men and women at the community and household level. For energy interventions to be effective, the programme must be tailored to the needs of the beneficiaries. Therefore, women need to be actively engaged in the processes of intervention to produce effective and inclusive results.

## 2. Strategic Context

### 2.1 Regional strategic context

Promoting gender equality and incorporating gender considerations has become a key objective of the ECOWAS region in recent years. The region, in cooperation with ECREEE and developmental partners, has drafted the ECOWAS Policy for Gender Mainstreaming in Energy Access (ECOWGEN), which was launched in 2013 and effectively implemented in 2015. This marks the recognition of women's different needs and challenges in terms of energy access, affordability and security and the region's commitment to reducing the gender gap both in general and specifically in the energy sector. The policy framework was conceived on the basis that the integration of a gender dimension into energy access challenges will enable the region to achieve both the SE4ALL and SDGs objectives by 2030. A specific gender policy framework for the WAEMU region is currently under development<sup>2</sup>. Fig. 1 below shows the 5 selected SDGs that are linked to gender equality and clean energy development in the target countries.

Figure 1: Selected SDGs linked to gender equality and clean energy development



BOAD also highlights the necessity to strengthen women economic empowerment as a key element of sustainable development (BOAD, 2012), with a word of caution. The Bank recognises that differences between women's groups must be carefully taken into account in the formulation and implementation of energy projects. Although women as whole face the same difficulties in terms of energy access and how society defines them, the extent of the impacts of such difficulties can vary across different women's groups based on factors such as social status, race, or the socio-cultural group women belong to. As BOAD puts it, "activities oriented toward a non-differentiated "women" category can serve the cause of some women and, at the same time, harming others" (BOAD, 2012, p. 19)

The ECOWAS Program on Gender Mainstreaming in Energy Access was accompanied by the ECOWAS Gender Strategy 2010-2020, developed in close cooperation with ECREEE. The strategy has the objective to promote favourable policy environments and to support institutional frameworks and resource mobilisation in order to more fully engage women in all areas of the energy access equation, including as energy suppliers, planners, financiers, educators and customers (ECOWAS, Situation Analysis Report on Energy and Gender Issues, 2015). The Strategy lays out five strategic objectives to be achieved by 2030, according to its vision of a world where men and women shall enjoy equal access to modern energy services that is easily available, affordable and contribute to high levels of standards of living and economic development (ECOWAS, 2015, p. 7). The table below lays out these five strategic objectives.

<sup>2</sup> According to <http://www.oecd.org/swac/topics/gender.htm>

Table 4: ECOWAS strategic objectives for Gender Mainstreaming in Energy Access

| Strategic objectives                                                                                                                                                                                                                                                                                               | Targets                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Achieve widespread understanding of energy and gender considerations at all levels of society                                                                                                                                                                                                                   | <ul style="list-style-type: none"> <li>100% of energy sector government employees will have received some relevant training by 2020 (and routinely thereafter)</li> <li>50% of citizens will have been exposed to some form of relevant public service announcement by 2020 growing to 90% by 2030</li> <li>At least 50 new scientific articles about gender and energy in West Africa published in peer-reviewed scientific journals by 2020, and 20 per year after that</li> </ul> |
| 2. Ensure that all energy policies, programmes and initiatives, including large energy infrastructures and investments, are non-discriminatory, gender-inclusive, gender-balanced and directed towards addressing inequalities, particularly energy poverty, which affects men and women in the region differently | <ul style="list-style-type: none"> <li>50% of energy policies by 2020 and 100% by 2030 will be gender-sensitive</li> <li>50% of energy projects, programmes, and initiatives with government participation will include gender dimensions in planning, implementation, analysis, and evaluation by 2020, rising to 100% in 2030</li> </ul>                                                                                                                                           |
| 3. Increase women's public sector participation in energy-related technical fields and decision-making positions                                                                                                                                                                                                   | <ul style="list-style-type: none"> <li>At least 25% women in the public sector energy workforce by 2020 and an equal (50-50) gender balance by 2030</li> </ul>                                                                                                                                                                                                                                                                                                                       |
| 4. Ensure that women and men have equal opportunities to enter and succeed in energy-related fields in the private sector                                                                                                                                                                                          | <ul style="list-style-type: none"> <li>At least 25% women participation in energy-related fields in the private sector by 2020 and an equal (50-50) gender balance by 2030, as determined through statistically rigorous random sampling</li> </ul>                                                                                                                                                                                                                                  |
| 5. Establish and maintain a gender responsive monitoring, accountability and review framework for objectives 1-4                                                                                                                                                                                                   | <ul style="list-style-type: none"> <li>100% percent compliance by 2017 in the monitoring, accountability and review framework</li> </ul>                                                                                                                                                                                                                                                                                                                                             |

Source: ECOWAS (2015) Policy for Gender Mainstreaming in Energy Access

The Framework constitutes a benchmark and guideline for member states to incorporate gender considerations and safeguards in energy Ministries and overall country policy frameworks. Although target countries have made progress in terms of energy access as a whole, much remains to be done to incorporate the gender dimension in national policy frameworks. Table 5 below provides an overview of energy and gender linkages in country policies in target countries.

Table 5: Energy and Gender linkages in national policy frameworks in target countries

|               | National Energy Policy | Renewable Energy Policy | Energy Efficiency Policy | Gender Dimension in Energy Policies | National Gender Policy | Energy Linkage in Gender Policy | Gender Focal Point or Unit in Ministry of Energy | Gender Focal Point or Unit at Agency Level | Gender audit of Energy Sector |
|---------------|------------------------|-------------------------|--------------------------|-------------------------------------|------------------------|---------------------------------|--------------------------------------------------|--------------------------------------------|-------------------------------|
| Benin         | ✓                      | ✗                       | ✗                        | ✗                                   | ✓                      | ✗                               | ✓                                                | ✗                                          | ✓                             |
| Burkina Faso  | ✓                      | ✗                       | ✗                        | ✓                                   | ✓                      | ✓                               | ✓                                                | ✗                                          | ✗                             |
| Guinea-Bissau | N/A                    | ✗                       | ✗                        | N/A                                 | ✓                      | N/A                             | N/A                                              | N/A                                        | N/A                           |
| Mali          | ✓                      | ✗                       | ✗                        | ✗                                   | ✓                      | ✗                               | ✗                                                | ✓                                          | ✗                             |
| Niger         | ✓                      | ✗                       | ✗                        | ✗                                   | ✓                      | ✗                               | ✓                                                | ✗                                          | ✗                             |
| Togo          | ✗                      | ✗                       | ✗                        | ✗                                   | ✓                      | ✓                               | ✗                                                | ✗                                          | ✓                             |

Source: ECOWAS (2015) Policy for Gender Mainstreaming in Energy Access

Other key initiatives supporting and complementing the ECOWAS Programme on Gender and Energy Access include: the Mano river Union (MRU) Framework Action Plan, the West African Clean Cooking Alliance (WACCA), the ECOWAS Federation on Business Women and Entrepreneurs (FEBWE), the Gender and Energy Program of the Africa Renewable Energy Access Program (AFREA) and the UNDP Multifunctional Platform Program. This array of frameworks and institutions show the region's commitment to bridging the gender gap in energy access, and fulfilling women's potential as energy suppliers, purchasers and producers. Ultimately, an energy programme or project which does not take into consideration the differences in gender roles as constructed by the society cannot effectively meet the needs of both men and women and will, ultimately, not lead to sustainable energy for all.

## 2.2 National strategic context

Inclusive participation in target countries remains an ongoing challenge. Gender inequality persists, as women are under-educated and generally have a lower socio-economic status, with inadequate access to basic social services and reduced economic opportunities compared to men (ECREEE, 2018). Female participation in education, particularly higher education, remains disproportionately low, as seen in Section 1. While gender discrimination is widespread, these issues tend to be more pronounced in rural areas. The following table summarises key aspects of gender inequality and inclusive participation in relation to the energy sector in each target country.

*Table 6: Gender and Energy linkages in the target countries*

| Country             | Gender and Energy linkages                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Benin</b>        | <p>The Government has adopted several policies and action plans to promote gender equality and has signed on to key international and regional framework agreements protecting women's rights. In 2007, the GoB adopted the National Gender Policy promoting gender equality, women's empowerment and improved social protection, including improvements to gender inequality in education and training. The National Policy for the Advancement of Women and Gender Equality (2009–2016) included specific targets for gender equality by 2025. Like many countries in the region, Benin has two parallel and overlapping judicial codes – one based on Western, mainly French, systems and one based on traditional systems. Under the Western system, women's rights are reasonably protected, and Benin is a signatory to most of the major international conventions on women's rights. Under the traditional code, however, while there is variation among ethnic groups, women often do not share the same freedoms as men.</p> <p>In the energy sector, efforts have been made to implement measures under the regional framework, ECOWAS Policy for Gender Mainstreaming in Energy Access, as well as at the national level. Gender mainstreaming in the country's energy policy requires capacity building of staff and the implementation of gender management systems at the institutional level to provide guidance on gender responsive leadership and decision making. As part of this process, the Government has established a gender focal point at the Ministry of Energy to promote inclusive participation for women in the energy sector.</p> <p>Under its newly developed off-grid regulatory framework (EHR), Benin has emphasised gender mainstreaming and social inclusion, encouraging project developers eligible to receive funding under the OCEF facility to take gender and social impact aspects into account in their project design. This includes an analysis of the project's implications for each target population group (men, women, youth, disadvantaged) at each stage of the project lifecycle along with corresponding corrective measures that will be implemented if the project does not benefit all groups fairly. Developers receiving assistance from OCEF are required to (i) commit to implementing certain gender and social measures; (ii) assess these issues in the project's area; and (iii) include indicators in the system operation monitoring plan.</p> |
| <b>Burkina Faso</b> | <p>Burkina Faso has adopted several policies to promote gender equality. In 2009, the Government established a National Gender Policy (Politique Nationale De Genre), a dedicated ministry – the Ministry of Women, National Solidarity and Family – as well as a National Council (Conseil National pour la promotion du Genre) to promote gender equality</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

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|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | <p>and develop several action plans and related programmes to advocate for gender inclusion. The existing national plan is the Strategy for Accelerated Growth and Sustainable Development (Stratégie de Croissance Accélérée et de Développement Durable, “SCADD”), which was adopted in 2010. The SCADD recognises the importance of gender in the country’s development and promotes gender mainstreaming and inclusive participation for women in the public and private sectors. As a result, FDE/ABER has considered gender aspects in all the projects implemented.</p> <p>In the energy sector, efforts have been made to implement measures under the regional framework, ECOWAS Policy for Gender Mainstreaming in Energy Access, as well as additional gender policies at the national level. The Government has included references and linkages to gender in its energy policies and has established a gender focal point at the Ministry of Energy. Despite these efforts, gender mainstreaming in the country’s energy policy remains insufficient and requires capacity building of staff and the implementation of gender management systems at the institutional level to provide guidance on gender responsive leadership and decision making. Across the energy sector, there are very few women-led businesses and women professionals in senior-level positions. In the off-grid space, NGOs like the Women Environmental Programme are working to promote gender equality.</p>                                         |
| <b>Guinea-Bissau</b> | <p>Under the Guinea-Bissau Constitution, men and women are equal before the law. The Government has even adopted a National Gender Policy (Política Nacional para a Promoção da Igualdade e Equidade de Género 2012–2015) to promote gender equality and economic opportunities for women. The Strategic and Operational National Action Plan “Terra Ranka” within the Strategy for Poverty Reduction Document (DENARP II, 2013) also includes gender equality measures. Among other initiatives, the Government has also established an Institute for Women and Children, IMC (Instituto da Mulher e Criança), the Ministry of Women, Family and Social Cohesion, and the Peace Building Fund’s “Gender Promotion Initiative.” Unfortunately, gender inequality persists throughout the nation especially in rural areas where customary and religious practices tend to supersede state policies and laws.</p> <p>In the energy sector, efforts have been made to implement measures under the regional framework, ECOWAS Policy for Gender Mainstreaming in Energy Access, as well as at the national level. Gender mainstreaming in the country’s energy policy requires capacity building of staff and the implementation of gender management systems at the institutional level to provide guidance on gender responsive leadership and decision making. As part of this process, the Government has established a gender focal point at the Ministry of Energy to promote inclusive participation for women in the energy sector.</p> |
| <b>Mali</b>          | <p>The Government has adopted policies and action plans to promote gender equality. Mali has signed on to several key international frameworks and regional agreements related to gender equality. The GoM has established a Ministry for the Advancement of Women, Children and the Family as well as a National Gender Policy that endorses gender equality and the improvement of rights for women. The National Policy also includes gender-mainstreaming provisions and actions to be taken by 15 government ministries, including those involved in the management of public employment.</p> <p>The Government has also made attempts to address gender equality in the energy sector. The National Director of Energy is a woman; moreover, AMADER has created a joint Gender and Energy Program in partnership with the UN and the World Bank called ‘SHER’ (Hybrid Rural Electrification Systems), aimed at increasing access to productive energy for women in rural areas. The Government has also established a gender focal point at the Ministry of Energy and has taken additional measures under the regional framework, ECOWAS Policy for Gender Mainstreaming in Energy Access, to further improve inclusive participation for women in the energy sector.</p>                                                                                                                                                                                                                                                              |
| <b>Niger</b>         | <p>Niger has adopted several policies and action plans to promote gender equality and has signed on to key international and regional framework agreements protecting women’s rights. According to available data, Niger has ratified both the Convention on the</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |



|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             | <p>Elimination of All Forms of Discrimination against Women (CEDAW) and the Optional Protocol on violence against women (2004). They signed but have not ratified the Protocol to the African Charter on Human and Peoples' Rights on the Rights of Women in Africa. In 2008, the Ministry of the Advancement of Women and Child Protection (le Ministère de la Promotion de la Femme et de la Protection de l'Enfant) developed and implemented a National Gender Policy that promoted rights and economic opportunities for women.</p> <p>In the energy sector, efforts have been made to implement measures under the regional framework, ECOWAS Policy for Gender Mainstreaming in Energy Access, as well as at the national level. Gender mainstreaming in the country's energy policy requires capacity building of staff and the implementation of gender management systems at the institutional level to provide guidance on gender responsive leadership and decision making. As part of this process, the Government has established a gender focal point at the Ministry of Energy to promote inclusive participation for women in the energy sector. The ECOWAS Federation on Business Women and Entrepreneurs (ECOWAS/FEBWE) is an initiative open to national federations/associations of business women and entrepreneurs from within the Community as well. Other initiatives such as the WAPP Master Plan includes provisions to address gender disparities in Niger's energy sector. The Comité Permanent Inter-Etats de Lutte contre la Sécheresse dans le Sahel (CILSS) includes four high level objectives pertaining to gender equality.</p> |
| <b>Togo</b> | <p>Togo has adopted several policies and action plans to promote gender equality and has signed on to key international and regional framework agreements protecting women's rights. These include ratification of the Convention on the Elimination of All Forms of Discrimination against Women and the Protocol to the African Charter on Human and Peoples' Rights on the Rights of Women in Africa. A General Directorate for the Advancement of Women has existed since 1977, a Ministry for the Promotion of Women was established in 2010, and a National Policy on Gender Equality and its corresponding Action Plan were adopted in 2011. Despite these policy initiatives, the level of participation among women remains very low, especially in rural areas where customary and religious practices tend to supersede state policies and laws.</p> <p>In the energy sector, efforts have been made to implement measures under the regional framework, ECOWAS Policy for Gender Mainstreaming in Energy Access, as well as at the national level. Gender mainstreaming in the country's energy policy requires capacity building of staff and the implementation of gender management systems at the institutional level to provide guidance on gender responsive leadership and decision making. As part of this process, the Government has undertaken a Gender Audit of its energy policy and established energy linkages in its gender policy. The GoT has also established a gender focal point at the Ministry of Energy to promote inclusive participation for women in the energy sector.</p>                                                  |

*Source: ECREEE (2018) Regional Off-Grid Electrification Project, Regional Off-Grid Solar Market Assessment & Private Sector Support Facility Design, Country Reports, August.*

## 3. Objectives of the Programme

### 3.1 Gender and the energy sector

As access to affordable and clean energy is a prerequisite for economic empowerment and poverty reduction, gender inequalities limit economic opportunities for women and subsequently have significant negative impacts on their families and communities. Women's and men's needs and priorities for energy and access to energy resources often differ based on their roles and responsibilities in societies. This results in gender-differentiated impacts at the individual, household, community and project levels. Women and men enjoy different access and use of energy and benefits that derive from energy services. As explained above, women and girls tend to be disadvantaged compared to men and boys, and this can be exacerbated by poverty, intrinsic social and cultural norms, bias, discriminatory or blind policy frameworks, as well as political conflict and fragility. As a result, women and girls have little or no access to and control over how energy is accessed and/or used, their educational and professional opportunities and their own maternal and reproductive health. In 2018, ECREEE partnered with AfDB to launch a regional workshop to advance the participation of women in the renewable energy sector. The programme intends to address the lack of inclusion of women in the energy value chain as only 2% of energy sector entrepreneurs in West Africa today are women (ECREEE, 2018). The joint initiative ultimately seeks to develop a pipeline of investment-ready, women-owned energy businesses across the region.

Regarding women's employment in the energy sector, in the target countries and overall in West Africa, women suffer from gender occupational segregation, which means that women who are employed are often assigned to roles in the sales and service sector in roles covering retail, administration and HR, with lower salaries and fewer benefits than men. They are unable to integrate into more competitive and higher-paid jobs in the energy and construction sectors. The fact that women are disadvantaged compared to men with regards to economic opportunity and resource use results in them not having due access to energy and a lack of opportunities in the energy sector. Overall, males and females are not encouraged in equal measures to pursue studies, and eventually careers, in the energy sector. Specific constraints with regards to women's employment in the energy sector include but are not limited to:

- Insufficiently resources initiatives to promote female education in STEM fields
- Little targeted marketing of energy sector jobs and business opportunities for women
- Too few resources dedicated to gender sensitive adaptation of vocational, management and entrepreneurial training and outreach
- Gender equal capital flows to female entrepreneurs in the energy sector are too small and too concentrated in just a few sub-sectors
- Few incentives for the private sector to create a gender-balanced workforce (ECOWAS, 2015, p. 20)

At the consumer level, the challenges for gender equality are twofold: one is awareness about product availability, costs, benefits and maintenance; the other is ability to pay. The lack of female agency in households poses difficulties for the adoption of improved and cleaner energy in some settings. Gender equality at the consumer level requires that women be fully recognised as consumers, be economically empowered, and have equal agency. Specific constraints include:

- Many supply side agents still lack market intelligence about demand composition and gender aware sales techniques
- Remaining inefficiencies in the application of end-user finance and targeted subsidies, along with continued market impediments, affecting affordability
- Incomplete progress on achieving gender equality in household budgets, including earning power and agency in decision making

### 3.2 Gender and energy access

Improved access to modern, sustainable energy services can significantly reduce the time and labour burdens women are traditionally subject to, as well as improve the health conditions and opportunities for enterprise and capacity building in communities<sup>3</sup>. The provision of lighting in the household, for example, provides an opportunity for extended and/or flexible working hours, and therefore increases the time available

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<sup>3</sup> UNIDO (2014) Guide on Gender Mainstreaming

for income-generating activities, particularly for women who are time-constrained due to their household obligations (UNIDO, 2014). Similarly, labour-saving technologies such as power water wells, drip irrigation systems for agricultural production and post-production, such as grinding and milling, can result in significant time saved for women and enhance productivity. With electricity access, both men and women can engage in other income-generating activities, while boys and girls are free of household chores and can attend school. Without energy access, rural women and girls have to spend long hours performing subsistence tasks, such as collecting wood for cooking and heating, which constrains them from accessing decent employment, educational opportunities and livelihood-improving options. This also limits their opportunities for social and political interaction and involvement outside the household sphere (UNIDO, 2014).

Improving gender equality and social inclusion is therefore critical to fulfil the development impacts of energy projects. Tables 6 and 7 below lay out the various linkages between energy access, health and education.

*Table 6: Selected energy access and health linkages*

| <b>Energy services and end uses</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Health outcomes</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Incubation facilities for infants</li> <li>• Cold chain for vaccine, medicine, and blood bank refrigeration</li> <li>• Laboratory and diagnostic equipment</li> <li>• Powering medical equipment</li> <li>• Lighting surgery and examination rooms</li> <li>• Electricity for prevention and cure</li> <li>• Indoor lighting for examination, surgery, reading, ambient light and outdoor, including security lighting</li> <li>• Clean water supply</li> <li>• Water heating</li> <li>• Sterilisation</li> <li>• Energy services for health clinics and for housing doctors, nurses and clinic staff</li> </ul> | <ul style="list-style-type: none"> <li>• Reduced child and adult mortality and morbidity</li> <li>• Improved maternal health</li> <li>• Reliable access to clean/safe drinking water</li> <li>• Enhanced communications (especially for emergencies)</li> <li>• Technologies and education for health data and information</li> <li>• Combat HIV, malaria and other diseases</li> <li>• Improved life expectancy</li> <li>• Improved health service delivery (staff retention, motivation and training, lab services and end uses)</li> <li>• Evening health training for local communities</li> <li>• Reduced fertility rates</li> <li>• Night birthing and safety</li> <li>• Cleaner living conditions at home</li> </ul> |

*Source: ECREEE (2015) Situation Analysis Report on Energy and Gender Issues in ECOWAS*

*Table 7: Selected energy and education linkages*

| <b>Energy services and end uses</b>                                                                                                                                                                                                                                                                                                                                                     | <b>Education outcomes</b>                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Electricity for lighting</li> <li>• Computers</li> <li>• Radios/TV</li> <li>• Telecommunications</li> <li>• Thermal energy for cooking (students and staff)</li> <li>• Water pumping</li> <li>• Energy efficient lighting and appliances</li> <li>• Electricity and modern fuel services both for schools and for teachers' housing</li> </ul> | <ul style="list-style-type: none"> <li>• Poverty reduction</li> <li>• Income generation</li> <li>• Improved quality of education</li> <li>• Improved grades</li> <li>• Enhanced school enrolment/retention rates</li> <li>• Improved ability to attract and retain teachers and staff</li> <li>• Improved sanitation</li> <li>• Gender/balance equity</li> <li>• Facilities for evening classes and adult education</li> </ul> |

*Source: ECREEE (2015) Situation Analysis Report on Energy and Gender Issues in ECOWAS*

Lastly, improved electricity access, especially in rural and remote areas, can enhance security at the individual and community level. One example is street lighting, or lack of same, which results in increased violence (physical and/or sexual) against women and girls. Illuminated night markets, streets and other evening activities will have the potential to enhance women's participation in social contexts outside of the household while improving their security. Telecommunications in the event of political and social unrest or natural disasters can also be used to send alerts and information.

### 3.3 Benefits and challenges of on-grid renewable energy development

#### *Benefits*

The proposed programme will result in increased solar capacity in the target countries and improved affordability of energy services, which will enable energy access to a larger number of people, including women. Additionally, jobs in the operating, maintenance and support areas will be created during the construction and operation of solar power plants. Women's and children's health will be improved as a result of the use of cleaner and sustainable energy in the household, for cooking and lighting needs. Complementing Tables 6 and 7 above, the benefits of improved energy/electricity access through the development of on-grid renewable energy include, but are not limited to:

- Improved energy access will disproportionately benefit women as they are traditionally responsible for household tasks i.e. collecting wood
- Reduced gap in employment opportunities for men and women
- Positive impact on local economic dynamics due to the "construction boom", with increased number of employment opportunities for women
- Exposure to different standards of development, capacity building and skill strengthening
- Long-term change at the community level and spillover benefits due to the presence of the project for a long period of time
- Community development projects derived from increased clean energy access that alleviate poverty and increase choices for women
- Improved learning conditions for children and adults: extended reading hours, to enable children to do their homework
- Affordable electricity for a larger number of households, including women-headed households
- Positive impact on health thanks to the electrification of local health centres, and eliminating toxic fumes at the household level
- Overall availability of electricity that will benefit all members of the community, including women
- Street and public space lighting will result in reductions of violence against women and crime rates
- Possibility of training in solar energy for both men and women

#### *Challenges*

There are, however, potential negative effects that could occur during the development and construction of the projects, which could worsen the existing gender disparities and inequalities. This could be caused by the fact that access to benefits generated by the project i.e. compensation, employment, access to land and water etc, is subject to existing social relationships and hierarchies. For instance, as large-scale solar projects require an important land area, this could lead to conflict of interests over ownership and land use, governed by norms which typically limit women's access.

Indeed, existing gender disparities in energy poverty can be aggravated at the country, regional and local level, if energy access is expanded in a gender-blind manner. The benefits of on-grid expansion could apply disproportionately to men, thus reinforcing gender inequalities.

The project must therefore pursue gender gains in its investment activities and look at opportunities for female entrepreneurship, improvements in the representation of women in the labour force, and energy access for the poorest population groups – particularly women-headed households – in the target countries.

Other negative impacts could include:

- Lack of engagement of women that would result in no genuine engagement in the project development process
- Employment opportunities benefitting a selected few, excluding women to a large extent, thus increasing women's dependency on men as household income providers
- Inadequate, non-reflective representation in stakeholder consultations and contribution to local development plans
- Increase in physical and sexual violence against women around construction sites
- Risk of human trafficking
- Inequitable participation between men and women in decision-making and benefits flowing from the newly operating renewable services.

For each of the projects, stakeholders will ensure that thorough gender assessments will be carried out in order to establish a baseline and identify challenges and opportunities specific to the concerned communities. These will help avoid and/or minimise the potential negative impacts and enhance the positive impacts of the project with regards to women and women's groups.

## 4. Applicability of the Gender Assessment at the project level

The Gender strategy of the programme consists of this document (the Gender Assessment) and the accompanying Gender Action Plan. The Gender Strategy is consistent with international standards on gender equality and reflects the safeguards set out in BOAD's Gender Policy<sup>4</sup>. BOAD's Gender policy is designed with the dual objective of promoting the mainstreaming of gender issues in the operations of the Bank and to support the action undertaken by the member countries of the Bank for the achievement of equality between women and men. This dual objective entails:

- The integration of gender analysis in all projects, projects and policies of the Bank;
- The strengthening of cooperation relationships between men and women;
- The strengthening of women empowerment economically: a key element of sustainable development;
- The taking into account of the differences between women groups in the interventions of the Bank and
- Targeted interventions in the implementation of the strategy for the integration of gender issues.

The different needs and priorities of women and men in each individual project will also be assessed and taken into account within project-level Gender Assessment and Action Plans. The present Gender Assessment acts as an overview of the current gender situation in targeted countries, while the Programme's Gender Action Plan constitutes a safeguards benchmark laying out procedures, timings, and aspects which must be incorporated and mainstreamed in project design. Each project funded as part of the Programme will comply with the safeguards laid out in the present Gender Assessment and will follow the procedures specified in both the Gender Assessment and the Gender Action Plan, in addition to that of BOAD's Gender policy and national safeguards. Annual Monitoring and Evaluation requirements will ensure the sound application and mainstreaming of both frameworks at project-level.

At a project level, thorough Gender Assessments will be carried out using inputs from stakeholder consultations in order to incorporate and ensure the mainstreaming of needs and priorities of men, women and youth impacted by the project. As a result of stakeholder consultations and the assessment of gender-specific needs and priorities, a project-level Gender Action Plan will be developed to specify the specific interventions and/or activities that will be implemented in the project to address the identified vulnerabilities and enhance the mainstreaming of gender issues at every step of project planning and implementation.

### 4.1 Guidance on conducting stakeholder consultations, gender assessment and developing gender action plans

The general purpose of a Stakeholder Engagement Plan (SEP) at project-level is to ensure the involvement of men and women in discussions about the project's activities, objectives and measures, in order to assess and analyse their respective roles and responsibilities as main actors and beneficiaries of the project. Stakeholder consultations must be carried out early on at the project design stage. The SEP constitutes a benchmark framework that sets out all relevant Grievance Response Mechanism and resettlement procedures, along with a project-level stakeholders matrix presenting all stakeholders and their affiliated roles, responsibilities, risk potential and level of impact by the project.

Based on the mapping of relevant stakeholders at project-level, and using the present Gender Assessment, field-based data collection and documentation must be conducted to define the specific risks and gender mainstreaming opportunities at project-level. Below is a non-exhaustive list of questions that can be asked to ensure knowledge gaps are appropriately bridged:

- How women, men and youth are differently affected by the project?
- Are project developers taking into account the different roles and responsibilities of women, men and youth?
- Are indicators in the Risk Mitigation Framework disaggregated by sex?

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<sup>4</sup> West African Development Bank (2012) Policy of the West African Development Bank in terms of Gender

- Are plans and procedures for resettlement gender sensitive?
- Are women included both as organisers and beneficiaries of the awareness-raising, training and capacity-building activities?
- Are men and women equally able to participate in decision-making processes?
- Are consultations and stakeholder's involvement gender sensitive?

To efficiently bridge those gaps, the hiring of a local gender expert and reaching out to local women groups and organisations is highly recommended. The affiliated costs must be mainstreamed in the project-level implementation budget.

The project-level Gender Action Plan sets out the targets, inputs, outputs and outcomes of each of the project's activities by component. It is derived from the project-level logical framework and must include all relevant indicators, targets, timelines and costs disaggregated by gender.

## 4.2 Designing the Gender Action Plan at project-level

As previously mentioned, the programme-level Gender Action Plan constitutes a framework of reference for designing monitoring and evaluation targets and milestones in order to determine true progress in mainstreaming gender considerations at the project-level. The design of the project-level Gender Action Plan must be done jointly with the monitoring and evaluation framework. The table below outlines five indicative key steps to be followed for the development of project-level gender action plans:

|                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Provide a detailed breakdown of project activities to be monitored and evaluated for gender considerations                                     | To design a gender action plan, the first step is to identify and design the phases of the intervention applying a gender-sensitive lens. The project team will have to ensure that all project activities include consultations with women's groups and other relevant stakeholders, that data collection is disaggregated by gender and that all pre-assessments incorporate relevant gender issues and considerations                                                       |
| 2. Identify relevant indicators, data sources and tools to be used and/or develop, decide on the frequency of data collection                     | The focus of this step is for the project team and all relevant stakeholders to agree on the specific indicators to be included in the project-level results framework, the different data sources and the tools that would need to be developed/used to collect data as well as the frequency of data collection. A gender-sensitive The GAP would be developed to include indicators, outputs, outcomes and impacts of the project, aligned with the project-level log frame |
| 3. Decide on evaluation methods                                                                                                                   | Evaluations methods should be based on project's priority areas and the data collected. A combination of quantitative and qualitative should be used, since some key dimensions cannot be captures using only quantitative data. Such qualitative data will be collected through stakeholder consultations.                                                                                                                                                                    |
| 4. Develop effective communication channels for monitoring data, and identify resources, constraints and opportunities with regards to monitoring | Monitoring and reporting requirements must be defined at project design phase to ensure data flows directly from the field to the project team. These requirements include the definition of responsibilities pertaining to data collection, documentation, checking and reporting.                                                                                                                                                                                            |
| 5. Compile and combine the data gathered                                                                                                          | The Monitoring and Evaluation framework and the Gender Action Plan outline the gender aggregated data to be used to determine the projects' impacts on men and women. Both frameworks should be integrated in the workplan and revised/updated if the projects' activities or objectives change. Progress should be reviewed regularly to make the necessary adjustments.                                                                                                      |



## Gender Action Plan

| Activities                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Targets                                                                                                                              | Timeline and Monitoring                                                                                                                | Responsible                                        | Costs                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|--------------------------------------------------------|
| <b>Impact Statement:</b> Enhanced energy access and power generation for both women and men in target countries through the implementation of an enabling regulatory and financial environment for solar energy development, including in poor and female-headed households<br><b>Outcome Statement:</b> The project will contribute to a paradigm shift through the scale up of investments for solar project development through improved regulatory environments and financial profiles |                                                                                                                                      |                                                                                                                                        |                                                    |                                                        |
| <b>Component 1: To scale up commercial and sustainable financing for solar investments – Direct lending</b>                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                      |                                                                                                                                        |                                                    |                                                        |
| <b>Outcome 1:</b> Commercial banks interests and capabilities for solar financing                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                      |                                                                                                                                        |                                                    |                                                        |
| <b>Output 1.1:</b> Project design incorporates findings from gender analyses/impact assessments at the project-level                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                      |                                                                                                                                        |                                                    |                                                        |
| 1.1 % of solar projects having conducted gender analysis/impact assessment and developed a GAP by the end of project design phase                                                                                                                                                                                                                                                                                                                                                          | % of projects reaching end of project design phase with a complete Gender Assessment and Action Plan<br>Mid-term: 50%<br>Final: 100% | Monitoring:<br>Yearly basis<br><br>Timeline:<br>baseline information to be collected at the start of project design phase, year 1 to 2 | BOAD<br><br>Project developers in target countries | Mainstreamed in implementation budget at project-level |
| <b>Output 1.2:</b> Projects have produced a dedicated gender action plan based on project-level Gender Assessments                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                      |                                                                                                                                        |                                                    |                                                        |

| Activities                                                                                                                                                                                                                                                                                                                                                                                                | Targets                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Timeline and Monitoring                                                                        | Responsible                                               | Costs                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------|
| <p>1.2 Number of solar projects having produced a project-based gender action plan that includes:</p> <p>Share (in %) of women in total population benefitting from improved energy access</p> <p>% of women employed in project implementation</p> <p>% of women included in stakeholder consultations between the project and local communities</p> <p>Number of consultation meetings led by women</p> | <p>Number of projects<br/>Mid-term: 3 projects<br/>Final: 6 projects</p> <p>50% of those benefitting from improved energy access in communities around projects are women</p> <p>25% of people employed for roles created as a result of the construction and operation of solar power plants are women</p> <p>Among which 50% of roles will be at the technical level, 50% at management level</p> <p>25% of staff employed in project implementation are women</p> <p>50% of people included in stakeholder consultations are women</p> <p>50% of consultation meetings are led by women</p> | <p>Monitoring:<br/>Yearly basis</p> <p>Timeline: 3 projects by 2024<br/>6 projects by 2031</p> | <p>BOAD</p> <p>Project developers in target countries</p> | <p>Mainstreamed in implementation budget at project-level</p> |
| <b>Component 2: To develop technical capacity and raise awareness of private sector actors – Technical Assistance</b>                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                |                                                           |                                                               |
| <b>Outcome 2:</b> Project implementation and monitoring ensures gender inclusion and equality (outputs 2.1 – 2.4)                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                |                                                           |                                                               |
| <b>Output 2.1:</b> Strengthened capacity of private sector actors for investment preparation                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                |                                                           |                                                               |

| Activities                                                                                                                     | Targets                                                                                                                                                                                                                                                                                                                                                                                                 | Timeline and Monitoring                                              | Responsible  | Costs                                                                   |
|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|--------------|-------------------------------------------------------------------------|
| 2.1 Ensure inclusion of 1 to 4 ratio of women to men in training and capacity building activities for private sector actors    | <p>50% of hired training agents are women</p> <p>25% of capacity building activities participants are women</p> <p>Training programmes include awareness raising on gender dimensions and accommodate women's schedules</p>                                                                                                                                                                             | <p>Monitoring:<br/>Yearly basis</p> <p>Timeline:<br/>Year 1 to 5</p> | BOAD         | <p>No additional costs expected</p> <p>EUR 10,000 (indicative)/year</p> |
| <b>Output 2.2:</b> Strengthened capacity of national institutions to address legal and regulatory barriers in the solar market |                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                      |              |                                                                         |
| 2.2 Ensure inclusion of an equal share of women in capacity building and training activities for national institutions         | <p>50% of hired training agents are women</p> <p>50% of capacity building activities participants are women</p> <p>Training programmes include awareness raising on gender dimensions and accommodates women's schedules</p> <p>Training programmes ensure incorporation of gender safeguards and dimensions into revised regulatory frameworks for on-grid solar solutions, and tariffs structures</p> | <p>Monitoring:<br/>Yearly basis</p> <p>Timeline:<br/>Year 1 to 5</p> | BOAD<br>NDAs | <p>No additional costs expected</p> <p>EUR 10,000 (indicative)/year</p> |
| <b>Output 2.3:</b> Awareness raised on the benefits of solar system technologies                                               |                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                      |              |                                                                         |
| 2.3 Ensure communication and outreach efforts include gender considerations                                                    | <p>50% agents in communication and outreach campaigns are women</p> <p>50% of awareness raising materials shows women as energy suppliers, purchasers</p>                                                                                                                                                                                                                                               | <p>Monitoring:<br/>Yearly basis</p>                                  | NDAs<br>BOAD | N/A                                                                     |